Beliefs, Aggregate Risk, and the U.S. Housing Boom*

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Abstract

This paper presents a one-step solution method to endogenize beliefs about future prices in general equilibrium models with incomplete markets and aggregate risk. Applying this method to the U.S. housing boom of the 2000s, adaptive learning about house prices is triggered by loosening credit conditions. Adaptive learning induces partial updating of beliefs, which is key for 1) matching both the time-path and volatility of house prices and 2) generating beliefs that are increasingly optimistic and co-move with house prices, as observed in existing and novel empirical evidence. Because model-generated beliefs are sensitive to policy interventions, how beliefs are formed has direct implications for the effectiveness of prudential policies.

Keywords: housing boom; aggregate risk; heterogeneous agents; adaptive learning
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1 Introduction

Record house prices in the 2000s preceded one of the worst recessions in modern U.S. history. Understanding the drivers of this housing boom—particularly the role of loose credit conditions and optimistic beliefs—is central to researchers and policymakers seeking to safeguard against another crisis.[1] While looser credit conditions allow households to overcome mortgage financing constraints and purchase otherwise unattainable homes, the resulting effect on house prices depends on household heterogeneity, particularly the distribution of constrained households. Likewise, the effect of beliefs on house prices depends on belief formation and departing from full-information rational expectations has been shown to capture empirical features of beliefs like correlation with past price changes and partial updating.[2]

Existing models of the U.S. housing boom typically prioritize either heterogeneity via a sophisticated housing sector (Kaplan et al., 2020) or endogenous belief formation (Chodorow-Reich et al., 2023), but not both because of computational tractability. To allow for both, this paper contributes a one-step method that solves for a Krusell and Smith (1998) approximate equilibrium by forecasting house prices directly with a rule that is updated via adaptive learning and disciplined with survey evidence. Whereas Moll (2026) and Moll and Ryzhik (2026) extol the one-step method's computational efficiency and amplification dynamics in expositional models, the viability of these advantages in a large-scale quantitative model has been an open question. This paper answers this methodological question by contributing a foundational realization of these promised advantages in a quantitative model with incomplete markets and aggregate risk.[3] In doing so, it addresses the economic questions of 1) why beliefs became more optimistic during the housing boom and 2) how belief formation matters for the effectiveness of prudential policies.

The U.S. housing boom is an ideal application of the one-step method because of the

[1] See Favilukis et al. (2017), Greenwald (2026), Arslan et al. (2026), Lind (2021), Johnson (2025), Landvoigt (2016), Kermani (2012), Justiniano et al. (2019), Mian and Sufi (2017) and Di Maggio and Kermani (2017) for the role of loose credit conditions in the housing boom. See Kaplan et al. (2020), Burnside et al. (2016), Piazzesi and Schneider (2009), Gelain and Lansing (2014), Adelino et al. (2018), Nathanson and Zwick (2018), Glaeser and Nathanson (2017), Albanesi et al. (2022), and Foote et al. (2012, 2018) for the role of optimistic beliefs. Duca et al. (2021) provide a cross-country synthesis of the drivers of housing cycles. Low interest rates have also been studied with mixed success at generating the U.S. housing boom of the 2000s. Even though Adam et al. (2017), Garriga et al. (2019), Jordà et al. (2015), Chodorow-Reich et al. (2023) and Diamond and Landvoigt (2022) successfully link low interest rates to high house prices, Dokko et al. (2011) and Glaeser et al. (2013) do not find a similar connection.

[2] Kuchler et al. (2023) give an overview. See Glaeser and Nathanson (2017), Adam et al. (2025), Kindermann et al. (2024), Armona et al. (2019), Granziera and Kozicki (2015), Gelain et al. (2018), and others.

[3] Giusto (2014) and Hoffman (2016) are the first to use adaptive learning with incomplete markets and aggregate risk, the former in the Krusell and Smith (1998) model and the latter to study the interactions of beliefs and income fluctuations in the housing boom. Broer et al. (2022, 2023), Porapakkram and Young (2007), and Kübler and Scheidegger (2025) explore aggregate risk under information frictions more generally.

central role of heterogeneity and beliefs along with important non-linearities and amplification dynamics. I model the housing boom as an aggregate state where the exact effect of fundamentals—particularly looser credit conditions—on house prices is unknown. Given today's ongoing debate about the degree to which credit conditions affect house prices, it is unlikely that everyone in the early 2000s had a clear understanding. And how could they have known if expectations about future house prices were in line with fundamentals when innovations in mortgage finance that allowed for looser credit conditions had never before been experienced on a nation-wide scale?

When credit conditions loosen, agents form beliefs to forecast future house prices and do so via adaptive learning. Among the many forms of subjective beliefs, previously studied forms of adaptive learning provide a good framework to endogenize beliefs in housing models because they have been shown to capture realistic co-movements of empirical house prices and beliefs (Kuang, 2014; Kindermann et al., 2024; Adam et al., 2025), while also maintaining tractability and convergence in a state-of-the-art general equilibrium model. In this paper's model specifically, partial updating of beliefs about future house prices is a key mechanism of adaptive learning that generates optimistic beliefs in the boom. Partial updating prevents agents from fully internalizing the increase in demand for housing services when credit conditions loosen, which results in house prices rising by more than expected. Agents interpret this positive forecast error as a signal that their beliefs were not optimistic enough, revise their beliefs upward, but only partially, and continue to do so absent a new signal suggesting that their beliefs were too optimistic. House prices rise because optimism about future prices pulls forward demand and leads to more actual housing purchases.

With the contribution of rich heterogeneity and endogenous belief formation, the model matches 75 percent of the empirical aggregate house price increase from 1997 to 2007 and replicates 89 percent of observed volatility. While the increase in house prices is similar to that in Kaplan et al. (2020), generating any volatility—let alone matching that of the data—represents a significant advancement in the study of boom-bust episodes, as noted by Piazzesi and Schneider (2016).

I validate the model's persistently positive forecast errors—the key mechanism for generating optimistic beliefs—via a novel proxy for 2000s house price beliefs using the University of Michigan Survey of Consumers.[4] Although Kuchler et al. (2023, Table 2) document limited availability of empirical house price expectations prior to 2007, I develop the proxy by exploiting the tight correlation of a question on house price beliefs that is only available

[4]Surveys by Case and Shiller (1988, 2004), Case et al. (2012), and Shiller and Thompson (2022) document high expectations for house price growth during local booms. For the 2000s boom specifically, Piazzesi and Schneider (2009), Ben-David et al. (2024), and Soo (2018) all document optimism rising with house prices.

starting in 2007 with a question on selling conditions going back to 1992 to construct the empirical proxy. In applications of learning to housing in later periods when expectations data is available, Adam et al. (2025) and Kindermann et al. (2024) also find forecast error persistence like that generated by the model and observed in the 2000s proxy. By contrast, the exogenous beliefs framework of Kaplan et al. (2020) falls short of reproducing these empirical features.

In the model, credit conditions have only a modest direct effect on house prices because their loosening only affects a fraction of households—those that are younger and have lower income—who purchase homes similar in size to their rentals.[5] These mechanisms include an influx of foreign borrowers (Favilukis et al., 2017), market segmentation of renters and homeowners (Greenwald and Guren, 2025), types of credit conditions (Greenwald, 2026; Justiniano et al., 2019), credit condition endogeneity (Diamond and Landvoigt, 2022), and feedback between households and the financial sector (Arslan et al., 2026). However, credit conditions play two important roles. First, they allow the model to match the boom dynamics of homeownership, leverage, and foreclosures. Second, they increase the demand for housing services, which, even when small, is sufficient to trigger the initial forecast errors that drive endogenously optimistic beliefs.

Finally, I show that belief formation is critical for assessing policy effectiveness in the management of housing booms and that the house price response is non-linear with respect to the size and timing of policy interventions. I find that macroprudential interventions, such as raising interest rates or tightening credit conditions are more effective at dampening or eliminating booming house prices when beliefs are endogenous, which aligns with Chodorow-Reich et al. (2023), Igan and Kang (2011) and Kuang et al. (2026). The paper builds on the theoretical contribution of Fontanier (2025) by providing a quantitative assessment. Because the one-step method endogenizes beliefs while preserving nonlinearities, this paper finds that the house price responses scale nonlinearly with respect to the size and timing of prudential policy interventions. Implications of these nonlinearities include 1) limited downside risk of high interest rates deepening a contraction in house prices and 2) tightening credit conditions earlier in a boom is better than later.

2 Related Literature

By endogenizing beliefs with incomplete markets and aggregate risk, this paper provides a tractable one-step innovation to the Krusell and Smith (1998) solution method. This method

[5]Constrained households are the closest parallel to subprime borrowers who have been shown to account for the dynamics of the housing boom (Mian and Sufi, 2009, 2017; Griffin et al., 2021). While credit conditions have been shown to have a larger effect on house prices in other work, these mechanisms may still fall short of matching house price statistics other than the level increase as in this paper.

unifies credit and belief-based explanations of the 2000s housing boom, which are typically assessed in isolation (Favilukis et al., 2017; Kaplan et al., 2020) or together, but in less rich environments (Johnson, 2025; Cox and Ludvigson, 2021; Dong et al., 2022; Landvoigt, 2016).[6] This unification is important for understanding why beliefs about future house prices became optimistic in the 2000s.[7]

Other structural models with belief formation via adaptive learning have successfully accounted for housing market dynamics via limited knowledge about the supply of credit and demand for assets, house prices, the permanence of financial innovations, or local fundamentals.[8] These papers typically abstract from rich household heterogeneity present in this paper, which may overstate the direct effect of looser credit conditions on higher house prices. This paper's learning about market clearing prices instead of quantities has microfoundations in Adam and Marcet (2011) and Bastianello and Fontanier (2024, 2026).

While this paper and others rely on homogeneous beliefs, other work with heterogeneous beliefs finds that optimists tend to push up house prices in housing booms.[9] Homogeneous beliefs have the advantage of a clean interpretation of mechanisms and establish computational efficiencies for the one-step solution method that are foundational for incorporating the richer heterogeneous beliefs into state-of-the-art quantitative models.

By assuming that shifts in aggregate credit conditions trigger adaptive learning about house price forecasts, this paper takes the stance that beliefs respond to fundamentals. Even though this direction of causality is confirmed by Mian and Sufi (2009, 2017) but contested by Adelino et al. (2018) and Foote et al. (2012), it allows for a clean evaluation of macroprudential policies, such as adjusting borrowing limits or raising interest rates as studied by Allen and Greenwald (2022) and Graham and Sharma (2024), respectively.

This paper's finding that macroprudential policies can dampen and prevent housing booms aligns with macroprudential policies under behavioral biases more generally (Fontanier, 2025) and previous studies of interest rate policy under learning.[10] Complementing the rich general equilibrium dynamics of house prices and interest rates shown in Kinnerud (2025), this paper's quantitative assessment shows that house price responses scale nonlinearly with the timing and size of policy interventions. This insight is possible because the one-step

[6]Hoffman (2016) uses a related one-step method to study the interaction of beliefs and income.

[7]Howard and Liebersohn (2023) and Chodorow-Reich et al. (2023) propose regional heterogeneity as a complementary fundamentals-based explanation for why beliefs shifted. Sun (2025) combines regional and household heterogeneity in one framework.

[8]See Caines (2020); Adam et al. (2012); Kuang (2014), Boz and Mendoza (2014); and Chodorow-Reich et al. (2023); Kindermann et al. (2024), respectively.

[9]See Piazzesi and Schneider (2009); Burnside et al. (2016); Glaeser and Nathanson (2017); Mian and Sufi (2022); Kindermann et al. (2024); Dong et al. (2022).

[10]See Adam and Woodford (2021); Adam et al. (2025); Caines and Winkler (2021).

solution method allows for subjective beliefs and also preserves nonlinearities.

3 Model

Sections 3.1-3.6 describe the environment and sectors of the model, which are designed to nest those of Kaplan et al. (2020) (see Appendix E for a comparison). This nesting structure is key for the main and counterfactual macroprudential policy comparisons of exogenous and endogenous beliefs in section 5. Section 3.7 introduces endogenous beliefs by detailing why this state-of-the-art model needs a rule for forecasting house prices and the implementation of adaptive learning about the coefficients of this rule. Because agents only learn about these coefficients and no other model prices or quantities, this discussion is relegated to the end of the section.

3.1 Environment

Time is discrete and the economy is populated by a continuum of measure one finitely lived households who are heterogeneous in ages and idiosyncratic income endowments. These households trade goods and services with a lending sector, a rental housing sector, housing construction firms, and final goods firms. Lowercase letters denote individual household quantities and uppercase letters denote aggregates.

Preferences: Households are active in economic life for $j = 1, \ldots, J$ periods where they work from $j = 1$ to J^{ret-1} and retire at J^{ret} until they exit the economy with certainty at J. Households have preferences over final goods consumption and housing expenditures $\{c_j, s_j\}_{j=1}^{J}$ with final goods consumption as the numeraire. All j subscripts have been dropped unless needed. The utility function is given as:

$$u_j(c_j, s_j) = e_j \frac{[(1-\phi)c_j^{1-\gamma} + \phi {s_j}^{1-\gamma}]^{\frac{1-\sigma}{1-\gamma}} - 1}{1-\sigma}$$

Expected lifetime utility at time t for $0 < \beta < 1$ is given as:

$$\mathbb{E}_t \left[\sum_{j=1}^{J} \beta^{j-1} u_j(c_j, s_j) + \beta^J v(\flat) \right]$$

Where the warm-glow bequest motive at the end of life J follows the functional form of De Nardi (2004) which modifies that of Carroll (2002):

$$v(\flat) = \psi \frac{(\flat + \underline{\flat})^{1-\sigma} - 1}{1-\sigma} \tag{1}$$

The bequest motive prevents households from counterfactually ramping up debt and drawing

down housing when they exit the economy. The strength of the bequest motive is regulated by $\psi \geq 0$ and the extent to which bequests are luxuries is given by $\underline{\flat} \geq 0$.

Income endowments: While working, households receive an income endowment comprised of an aggregate stochastic endowment $\Theta(Z)$, a deterministic life-cycle profile that varies by age χ_j, and an idiosyncratic stochastic endowment $\epsilon_j(z)$ that follows a Markov process.

$$\log y = \log \Theta(Z) + \chi_j + \epsilon_j(z), \quad \text{when} \quad j < J^{ret}$$

The transition matrix for earnings $\epsilon_{j+1}(z') \sim \Upsilon_{j+1|j}(\epsilon_j(z))$ is age-dependent which helps account for rising income volatility throughout working life.

In retirement, households receive a fraction ρ_{SS} of their last working period income with aggregate income averaged across states $\bar{\Theta}$. Heterogeneity in retirement income helps preserve the wealth distribution of retired agents.

$$y = \rho_{SS}(\log \bar{\Theta} + \chi_{J^{ret}-1} + \epsilon_{J^{ret-1}}), \quad \text{when} \quad J^{ret} \leq j \leq J \tag{2}$$

Income tax follows the functional form of Heathcote et al. (2017) where τ_y^0 sets the average level of taxation and τ_y^1 sets the degree of tax progressivity. To capture the tax benefits of homeowning, households can deduct ϱ fraction of the interest paid on their mortgage $r_m m$ for the first \$1,000,000 of mortgage debt, $\bar{m}$.

$$\mathcal{T}(y, m) = y - \tau_y^0(y - \varrho r_m min\{m, \bar{m}\})^{1-\tau_y^1} \tag{3}$$

If a household does not have a mortgage, the expression simplifies to, $\mathcal{T}(y) = y - \tau_y^0(y)^{1-\tau_y^1}$.

Housing: Households can either rent or own houses. To capture the lumpy dynamics of housing over the life cycle—as described by Chambers et al. (2009)—housing units are indivisible and available in discrete fixed sizes $\mathcal{H} \in \{h^0, \ldots, h^N\}$ for homeowners and $\tilde{\mathcal{H}} \in \{\tilde{h}^0, \ldots, \tilde{h}^N\}$ for renters. If housing units were instead a continuum of sizes households would make a counterfactually large number of adjustments to square footage over the life cycle as they could upsize or downsize by a fraction of a room.

Markets for rental and owner-occupied housing are both frictionless and competitive where the law of one price holds and selling is instantaneous.[11] Rental units cost $\rho(\mu, Z)$ and owner-occupied units cost $p(\mu, Z)$. Renting generates housing services equal to the size of the housing unit, $s = \tilde{h}$ while homeownership offers additional utility $s = \omega h$ for $\omega \geq 1$. Homeowners pay per-period taxes and maintenance costs $(\delta_h + \tau_h)p(\mu, Z)h$ where maintenance

[11] See Hedlund (2016) for the role of search frictions such as time delays in housing market dynamics.

offsets the depreciation of the housing unit. When selling, homeowners pay a transaction cost $\kappa p(\mu, Z)h$ that is linear in housing value. Renters do not pay these costs and can adjust the size of their housing unit without transaction costs.

Liquid savings: Households can save in one-period bonds b at the price $q_b = 1/(1+r_b)$, the reciprocal of the exogenous risk-free rate. Unsecured borrowing is prohibited and households cannot trade among themselves. Households transact bonds with risk-neutral non-modeled foreign agents with deep pockets. Assuming an exogenous financial sector implies that the model is a small open economy.[12]

Mortgages: Homeowners finance housing purchases with multi-period defaultable mortgages subject to a fixed origination cost $\kappa^m(Z)$. These mortgages are amortized over the life of the household at interest rate $r^m = (1+\iota)r_b$ which is equal to the risk-free rate scaled by an intermediation wedge.[13] Default risk is thus priced by the mortgage pricing function $q_j(\mathbf{x}', y; Z, \mu)$ rather than the interest rate. This function depends on variables that predict future repayment which include the age j of borrowing households, individual state variables $\mathbf{x}' = \{b', h', m'\}$, individual income y, and aggregate states Z and μ.

When a household aged j obtains a mortgage with principal balance m' they receive $q_j(\mathbf{x}', y; Z, \mu)m'$ from the lender where $q_j(\mathbf{x}', y; Z, \mu) < 1$. The down payment at origination is thus $p(\mu, Z)h' - q_j(\mathbf{x}', y; Z, \mu)m'$. Households make $J - j$ mortgage payments until the mortgage is fully repaid, $\pi_j^{min}(m) \leq m(1+r_m) - m'$. The minimum payment is determined by constant amortization:

$$\pi_j^{min}(m) = \left[\frac{r_m(1+r_m)^{J-j+1}}{(1+r_m)^{(J-j+1)} - 1}\right] m \tag{4}$$

A loan-to-value (LTV) constraint limits mortgage size to a fraction of the value of housing collateral $m' \leq \theta^{LTV}(Z)p(\mu, Z)h'$ and the payment-to-income constraint (PTI) caps the minimum mortgage payment at a fraction of household income $\pi_j^{min}(m') \leq \theta^{PTI}(Z)y$. Because mortgages are multi-period debt, households are only subject to these constraints in the period when mortgages are originated.

HELOCs: Home equity lines of credit (HELOCs) allow homeowners to borrow one-period,

[12]This assumption follows Arslan et al. (2026), Kaplan et al. (2020), Greenwald and Guren (2025), Chodorow-Reich et al. (2023).

[13]Allowing homeowners to choose the length of mortgages with borrower-specific interest rates would add more realism at the cost of more state variables. Tractability is assured by 1) mortgages being due at the deterministic end of economic life and by 2) amortization at the common mortgage interest rate r^m.

non-defaultable debt up to a fraction of the value of housing collateral $-b' \leq \theta^{HELOC} p(\mu, Z)h$ with an interest rate equal to that on mortgages r_m.

Shocks: Z indexes an aggregate state where income $\Theta(Z)$ fluctuates via a two-state Markov chain with transition matrix $Z' \sim \Gamma_Z(Z)$ and credit conditions
$\mathcal{C}(Z) = \{\theta^{LTV}(Z), \theta^{PTI}(Z), \kappa^m(Z), \zeta^m(Z)\}$ loosen via a one-time unanticipated shock as in Favilukis et al. (2017). In the housing boom, income and credit conditions attain their high and loose values, respectively. Because agents expect permanently looser credit conditions in the housing boom, the return of credit conditions to their tight state in the housing bust is also a one-time unanticipated shock.

Individual income $\epsilon_j(z)$ follows an AR(1) process with persistence ρ and an age-dependent standard deviation σ_{ε_j} resulting in an age-dependent transition matrix $\epsilon_{j+1}(z') \sim \Upsilon_{j+1|j}(\epsilon_j(z))$

$$\epsilon_j(z) = \rho\epsilon_{j-1}(z_{-1}) + \varepsilon_j(z), \quad \varepsilon_j(z) \sim \mathcal{N}(0, \sigma^2_{\varepsilon_j}) \tag{5}$$

Aggregate state space: The distribution of individual household states μ is a necessary state variable for agents to correctly forecast next period prices under incomplete markets and aggregate risk. $\Gamma_\mu(\mu; Z, Z')$ is the equilibrium law of motion of the measure of agents such that $\mu' = \Gamma_\mu(\mu; Z, Z')$. The aggregate state space of the economy is thus the distribution over individual states and the aggregate shock $\{\mu, Z\}$. Let $\mathcal{X}^h = \mathcal{B} \times \mathcal{H} \times \mathcal{M} \times \mathcal{E} \times \mathcal{J}$ denote the set of individual states for homeowners and $\mathcal{X}^r = \mathcal{B} \times \mathcal{E} \times \mathcal{J}$ that of renters with measure $\int_{\mathcal{X}^h} \mu^h d\mu^h + \int_{\mathcal{X}^r} \mu^r d\mu^r = 1$.

Beliefs: This paper introduces a solution method where agents form boundedly rational expectations with respect to fully known shocks $\{Z, \epsilon_j(z)\}$ and only form beliefs about house prices $p(\mu, Z)$, which are a subset of endogenous prices that are not pinned down by modeling assumptions. Section 3.7 discusses implementation in more detail and Appendix B.2 provides an algorithm.

3.2 Households' Problem

See Appendix A.1 for the full recursive households' problems.

Renters are endowed with savings $b > 0$ and income y. They must choose to stay a renter or purchase a house and become a homeowner solving the problem:

$$V^r_j(b, y; \mu, Z) = \max\{V^{rent}_j(b, y; \mu, Z), V^{own}_j(b, y; \mu, Z)\} \tag{6}$$

Newly originated mortgages are subject to loan-to-value and payment-to-income constraints:

$$m' \leq \theta^{LTV}(Z) p_h(\mu, Z) h' \tag{7}$$

$$\pi_j^{min}(m') \leq \theta^{PTI}(Z) y \tag{8}$$

Relative to renting, homeowning offers households the benefits of extra utility, tax deductible mortgage interest payments, and collateral for borrowing liquid assets in the form of HE-LOCs. Renting has the advantage of shielding households from capital gains fluctuations due to movements in house prices. When model simulated house prices match the empirical standard deviation as is the case under endogenous beliefs, renting becomes relatively more advantageous due to more volatile house prices.

Homeowners have four options: sell their house to purchase a new house or become a renter, default and become a renter, stay in their house and pay their existing mortgage, or stay in their house and refinance a new mortgage.

$$V_j^h(\mathbf{x}, y; \mu, Z) = \max\{V_j^r(b^n, y; \mu, Z), V_j^{default}(b, y; \mu, Z), V_j^{stay,pay}(\mathbf{x}, y; \mu, Z), V_j^{stay,refi}(\mathbf{x}, y; \mu, Z)\} \tag{9}$$

When households sell houses, they use the proceeds to pay off their existing mortgage. The difference is transformed into liquid financial assets b^n and the household solves the renters' problem (6) where they purchase or rent. When households default, they incur a utility penalty and become renters in the smallest size house. Because of the McClements scale on utility, households want larger houses as family size increases so the penalty of living in a one-room flat with a large family is sufficient for households to avoid default. When households stay in their house and pay their mortgage, they stay in the same structure and make at least the minimum payment on their existing mortgage. And finally, when households stay and refinance, they stay in the same structure but take out a new mortgage. Because mortgages have a fixed origination cost, households will only refinance when the terms are sufficiently favorable.

3.3 Lending Sector

The lending sector originates mortgages m' to households aged j at price $q_j(\mathbf{x}', y; \mu, Z) \leq 1$. Although lenders are competitive, uninsurable aggregate risk may induce profits and losses along the equilibrium path. Lenders are owned by non-modeled foreign agents with deep pockets who receive these profits and losses as net exports. The mortgage market clears loan-by-loan with the pricing of mortgage points depending on individual

future default probabilities and collateral of foreclosed homes.

$$q_j(\mathbf{x}', y; \mu, Z) = -\zeta(Z) + \mathbb{E}_{Z',\epsilon'} \begin{cases} 1 & \text{sell/refi} \\ \dfrac{(1-\delta_h^d-\tau_h-\kappa_h)p'(\mu',Z')h'}{(1+r_m)m'} & \text{default} \\ \dfrac{(1+r_m)m'-m''+q_{j+1}(\mathbf{x}'',y';\mu',Z')m''}{(1+r_m)m'} & \text{pay} \end{cases} \tag{10}$$

If a homeowner sells or refinances, they repay the full balance of their mortgage so that the lender receives the principal plus interest less the intermediation cost $\zeta(Z)$. If the household defaults, the lender forecloses and sells the house to recover the market value of the house as a fraction of the original mortgage. If the homeowner pays their existing mortgage, the lender values the contract as the next period mortgage payment $(1+r_m)m' - m''$ plus the continuation value of the contract $q_{j+1}(\mathbf{x}'', y'; \mu', Z')m''$.

3.4 Final Goods and Construction Firms

See Appendix A.2 for the full recursive problems. The competitive final goods sector has a linear constant returns to scale technology $Y = \Theta(Z)N_c$ with inelastic labor supply N_c. Profit maximization delivers an aggregate wage equal to aggregate productivity:

$$w(\mu, Z) = \Theta(Z) \tag{11}$$

A competitive construction sector produces houses with technology $H_h = (\Theta N_h)^\alpha \bar{L}^{1-\alpha}$ where N_h is labor services and $\bar{L}$ is the amount of newly available buildable land. Using the equilibrium wage in equation (11) housing supply is:

$$H_h = [\alpha p(\mu, Z)]^{\frac{\alpha}{1-\alpha}} \bar{L} \tag{12}$$

The profit maximization of the construction sector thus pins down a single price for housing $p(\mu, Z)$ via aggregate housing supply H_h and the aggregate productivity shock $\Theta(Z)$.

3.5 Rental Sector

The rental rate is pinned down by a user-cost formula that is a function of current and future house prices along with a per-period operating cost Ξ as shown in Appendix A.2. The competitive rental sector frictionlessly buys and sells housing units to convert them into rental housing which incurs the same depreciation and taxes as owner-occupied housing. Appendix A.2 discusses how the landlord heterogeneity of Greenwald and Guren (2025) can

improve the quantitative importance of credit conditions on the price-rent ratio.

$$\rho(\mu, Z) = \Xi + p(\mu, Z) - (1 - \delta_h - \tau_h)\mathbb{E}_{Z',\epsilon'|Z,\epsilon}[q_b p'(\mu', Z')] \tag{13}$$

3.6 Government

The sum of income tax revenues $\mathcal{T}(y, m)$ less mortgage interest rate deductions, property tax revenues $p(\mu, Z)\tau_h \int_{\mathcal{X}^h} h d\mu^h$, and land permit revenues $p(\mu, Z)H_h - w(\mu, Z)N_h$ net of pension outlays $\int_{\mathcal{X}} y_{ret} d\mu_{\mathcal{J}^{ret}}$ are always positive and spent on government services G that are not valued by households and thus discarded.

3.7 Computation of Equilibrium with Learning

See Appendix A.3 for the definition of the recursive competitive equilibrium and Appendix B.2 for the detailed computational algorithms, goodness of fit, and a discussion of the standard Krusell and Smith (1998) method. Because the model features substantial nonlinearities in the form of constraints and discrete choices, a global solution method maintains these, which is important for quantifying the effectiveness of macroprudential policy interventions as discussed in section 5.1.

Price determination with incomplete markets and aggregate risk requires agents to solve a computationally intractable problem: track a potentially infinite dimensional equilibrium law of motion for the distribution, $\mu' = \Gamma_\mu(\mu; Z, Z')$.[14] Because agents ultimately care about the evolution of prices, they can track prices directly for each combination of current and future aggregate states, $\mathcal{Z} \in \{Z, Z'\}$, to solve for a Krusell and Smith (1998) approximate equilibrium.[15] In the standard Krusell and Smith (1998) solution method, prices $p(Z)$ are a state variable and the forecasting coefficients $\boldsymbol{a}_{\mathcal{Z}} = (a^0_{\mathcal{Z}}, a^1_{\mathcal{Z}})$ pin down future prices $p'(p(Z); \mathcal{Z})$. Because $\boldsymbol{a}_{\mathcal{Z}}$ maps each grid point $p(Z)$ to a future price $p'(p(Z); \mathcal{Z})$ for each $\mathcal{Z}$, no additional grids are needed. This implicitly assumes that agents know the forecasting coefficients in all aggregate states of the economy, including ones like the loose credit conditions of the housing boom that may only occur once a generation.

$$\log p'(p(Z); \mathcal{Z}) = a^0_{\mathcal{Z}} + a^1_{\mathcal{Z}} \log p(Z) \quad \iff \quad \mu' = \Gamma_\mu(\mu; \mathcal{Z}) \tag{14}$$

This paper's one-step method with endogenous beliefs relaxes this assumption and allows for limited knowledge of these forecasting coefficients once the economy enters the aggregate

[14] Ahn et al. (2018) offer an alternative to Krusell and Smith (1998) that linearizes around aggregate steady states. However, Krusell and Smith (1998) is more robust than linearization when endogenous beliefs cause substantial deviations in the wealth distribution from steady state.

[15] A single housing price $p(\mu, Z)$ is the only price determined via market clearing. The lending sector, rental sector, final goods firms, and construction firms are all perfectly competitive with linear objective functions. The number of prices is thus reduced from four to one.

state with loose credit conditions. Agents know the form of the rule, but not the parameter values.[16] Given that the effect of credit conditions on house prices is still debated today (Greenwald and Guren, 2025), it is unlikely that everyone in the early 2000s had a clear understanding. To that end, imperfect knowledge of the mapping between credit conditions and house prices is implemented by altering the standard Krusell and Smith (1998) solution method by 1) expanding the state space of the individual households' problems to include beliefs and 2) allowing beliefs to vary over time in the simulation of the ergodic distribution. As discussed by Moll (2026), this approximate equilibrium does not require any additional moments such as means or variances and allows for tractable computation with aggregate nonlinearities.

First, beliefs are endogenized in the household's problem by making the forecasting coefficients $\boldsymbol{a}_{\mathcal{Z}}$ in equation (14) state variables.[17] No functional forms of beliefs are yet needed in this step. Consequently, the households' problem can be solved with standard methods such as value function iteration (as used in this paper). Because endogenizing beliefs requires as many grids as there are future aggregate states, with two values for aggregate income $\Theta(Z)$ evolving via a Markov chain and aggregate credit conditions $\mathcal{C}(Z)$ shifting between two values via an unanticipated shock, there are two future states $\#Z' = 2$, which requires additional grids $\{\boldsymbol{a}^{loose}_{Z_{high},Z'_{high}}, \boldsymbol{a}^{loose}_{Z_{high},Z'_{low}}\}$ for beliefs in the low and high income states. Although there are four possible aggregate realizations, agents only expect two at a time because changes in credit conditions are unanticipated.[18] With as many states as there are future aggregate states (two in this setting), beliefs can be state dependent such that there are different degrees of optimism or pessimism depending on whether income is high or low, $\boldsymbol{a}^{loose}_{Z_{high},Z'_{high}} \neq \boldsymbol{a}^{loose}_{Z_{high},Z'_{low}}$. This state-dependence aligns with the evidence of Roschitsch and Twieling (2025) who find a larger degree of pessimism in the bust than optimism in the boom. If a single belief grid, $\boldsymbol{a}^{loose}_{Z_{high},Z'} = \boldsymbol{a}^{loose}_{Z_{high},Z'_{high}} = \boldsymbol{a}^{loose}_{Z_{high},Z'_{low}}$, were used instead, then there would be no state-dependency and beliefs would either be optimistic or pessimistic in both income expansions and contractions. Appendix B.1 provides an example.

For any point on the $p(Z)$ grid, the households' problem is solved for all combinations of belief grids $\{\boldsymbol{a}^{loose}_{Z_{high},Z'_{high}}, \boldsymbol{a}^{loose}_{Z_{high},Z'_{low}}\}$. If each coefficient grid has N values, the households'

[16] Although additional statistics, lags, or forms of a forecasting rule may be used, Krusell and Smith (1998) find that more complexity only incrementally improves accuracy. Moreover, Pancrazia and Pietrunti (2019) find that simple forecasting rules are best at fitting the paths of house prices throughout the 2000s.

[17] In practice, it is equivalent to assign grids for beliefs or future prices, $\{\boldsymbol{a}^{high}_{Z_{high},Z'_{high},t}, \boldsymbol{a}^{high}_{Z_{high},Z'_{low},t}\}$ or $\{p'_{high,high}, p'_{high,low}\}$. This is explained in more detail in Appendix B.1.

[18] If credit conditions were also Markov, then there would be four additional state variables ($\#Z' = 4$), one for each combination of tight/loose credit conditions and low/high income. Allowing credit conditions to evolve via an unanticipated shock so that there are only two additional grids instead of four simplifies the computation.

problem is solved $N^{\#Z'}$ times, so N^2 times in this setting for two future aggregate states. Adding two state variables to an already computationally intensive problem is possible because the belief grids $\{\boldsymbol{a}^{loose}_{Z_{high},Z'_{high}}, \boldsymbol{a}^{loose}_{Z_{high},Z'_{low}}\}$ are fully parallelizable. That is, agents do not internalize that their beliefs are time-varying and instead expect them to be in place next period, which makes the problem solved at each point on the coefficient grids independent of the next point.

Second, how endogenous beliefs vary over time is specified in the simulation of the ergodic distribution. For the advantages of tractability and realism, this paper proposes step-size recursive stochastic gradient adaptive learning à la Evans et al. (2010), Evans and Honkapohja (1998), and Mele et al. (2020), but with three differences conducive to its embedding in a general equilibrium framework with aggregate risk.[19] First, agents learn about prices rather than quantities. Second, agents only learn about a subset of prices—house prices—rather than the entirety of endogenous variables or prices as in the other papers. Finally, this paper circumvents a potential simultaneity problem by using a lagged forecast error e_{t-1} so that agents only form time t beliefs with information available at time $t-1$ as in Carvalho et al. (2023) and as discussed by Evans and Honkapohja (1999, Chapters 8, 10). Appendix F.6 shows simulations of versions using least-squares learning or a simplified Bayesian learning. Because forecast errors are not weighted by the inverse of second moments $R^{-1}_{\mathcal{Z},t}$, past house price volatility does not enter the updating rule. Let $\boldsymbol{x}_{t-2} = (1, \log p_{t-2})'$.

$$\boldsymbol{a}_{\mathcal{Z},t} = \boldsymbol{a}_{\mathcal{Z},t-1} + g_t \boldsymbol{x}_{t-2} \underbrace{(\log p_{t-1} - \boldsymbol{x}'_{t-2}\boldsymbol{a}_{\mathcal{Z},t-1})}_{e_{t-1}} \tag{15}$$

In this adaptive learning rule, coefficients $\boldsymbol{a}_{\mathcal{Z},t}$ are a combination of last period's coefficients $\boldsymbol{a}_{\mathcal{Z},t-1}$ and weighted lagged forecast errors e_{t-1}.[20] How these time-varying coefficients $\boldsymbol{a}_{\mathcal{Z},t}$ differ from their near-rational known counterparts $\boldsymbol{a}_{\mathcal{Z}}$ depends on the evolution of the house price forecast error, which, in turn, depends on the gain parameter g_t governing the speed of updating and the vector of initial coefficients $\boldsymbol{a}_{\mathcal{Z},0}$, both of which are discussed in more detail in section 4.

While there are many ways to endogenize beliefs about house prices, this adaptive learning rule allows for three key features: 1) comparison to the exogenous beliefs of Kaplan et

[19] Mele et al. (2020) highlight that stochastic gradient learning keeps the state space small by setting $\boldsymbol{R}_{\mathcal{Z},t} = 1$ such that the forecast error is not weighted by inverse of the matrix of second moments, $\boldsymbol{R}_{\mathcal{Z},t} = \boldsymbol{R}_{\mathcal{Z},t-1} + g_t(\boldsymbol{x}_{t-2}\boldsymbol{x}'_{t-2} - \boldsymbol{R}_{\mathcal{Z},t-1})$, $\boldsymbol{a}_{\mathcal{Z},t} = \boldsymbol{a}_{\mathcal{Z},t-1} + g_t \boldsymbol{R}_{\mathcal{Z},t}\boldsymbol{x}_{t-2}(\log p_{t-1} - \boldsymbol{x}'_{t-2}\boldsymbol{a}_{\mathcal{Z},t-1})$ as in more traditional adaptive learning. Consequently, when $\boldsymbol{R}_{\mathcal{Z},t} = 1$, stochastic gradient learning only weights the forecast error e_{t-1} by $g_t\boldsymbol{x}_{t-2}$ so that the coefficients are always updated in the direction of the forecast error. When $\boldsymbol{R}_{\mathcal{Z},t} \neq 1$, the updating could be in a different direction which could complicate interpretation.

[20] There are several alternative forms of equation (15) and Appendix F.2 shows that these have little material difference on the main housing boom simulations.

al. (2020) for the prudential policy exercises, 2) realism, and 3) convergence and tractability in general equilibrium. First, the model adheres as closely as possible to that of Kaplan et al. (2020) to allow for comparisons between the two solution methods and macroprudential counterfactual simulations as in section 5. Second, learning is a realistic model for housing expectations as its partial updating has been shown to match features of house price and belief data (Kuang, 2014; Adam et al., 2025; Kindermann et al., 2024) such as co-movements in beliefs and house prices as well as house price volatility. Third, learning relies on agents updating their imperfect knowledge with new information to find true values, which is important for ensuring convergence in a general equilibrium setting. In other forms of subjective beliefs that rely on behaviorial biases, beliefs may stray far from true values, which could compromise convergence.

Finally, the convergence properties of learning allow the one-step method to recover the near-rational fixed-point Krusell and Smith (1998) coefficients. Appendix F.9 shows that if the model is solved at a reasonable guess for coefficients, one could potentially use this one-step method to solve the model in one hour instead of the 10 or more hours needed for Krusell and Smith (1998) fixed point iteration. Applications of the one-step method are not limited to house prices—Belda et al. (2026) use a variant method in a model with stock prices, beliefs, and aggregate risk.

4 Parameterization and Calibration

4.1 Parameters

The model's parameters are discussed in detail in Appendix C.1 and are set to resemble the U.S. economy in the late 1990s with the cross-sectional moments from the 1998 Survey of Consumer Finances via Kaplan et al. (2020). Table 1 lists targeted moments from the model's stochastic steady state which is characterized by tight credit conditions and fluctuations in income. See Appendix C.1 for a discussion of the fit including improving the right tail of housing to net worth.

Housing Preferences: The preference for housing relative to goods consumption $\phi = 0.13$ pins down housing demand so that the average share of housing expenditure to total expenditures from the model is 0.16, as in the data. In Kaplan et al. (2020) this parameter is not fixed and instead follows a stochastic process where it can take on a low value $\phi_{low} = 0.12$ and a high value $\phi_{high} = 0.20$.

Aggregate shocks: The economy has aggregate shocks to income $\Theta(Z)$ and credit conditions $\mathcal{C}(Z) = \{\theta^{LTV}(Z), \theta^{PTI}(Z), \kappa^m(Z), \zeta^m(Z)\}$. Aggregate income follows a two-state

Moment	Parameter	Empirical Value	Model Value
Agg. net worth/annual agg. labor income	β	5.5	4.9
Median ratio of net worth to labor income	β	1.2	1.2
Median net worth: age 75/age 50	ψ	1.55	1.48
% of bequests in bottom 1/2 of wealth dist.	$\underline{\flat}$	0	0
Housing/total cons. expenditures	ϕ	0.16	0.16
Aggregate home-ownership rate	ω	0.66	0.68
Foreclosure rate	ξ	0.005	0.0002
P10 housing/total net worth of owners	$\min \mathcal{H}$	0.11	0.13
P50 housing/total net worth of owners	$\#\mathcal{H}$	0.5	0.33
P90 housing/total net worth of owners	gap $\mathcal{H}$	0.95	0.74
Average sized owned house/rented house	$\min \tilde{\mathcal{H}}$	1.5	2
Average earnings of owners to renters	$\#\tilde{\mathcal{H}}$	2.1	2.8
Annual fraction of houses sold	κ_h	0.1	0.09
Homeownership rate of < 35 y.o.	Ξ	0.39	0.34
Employment in construction sector	$\bar{L}$	0.05	0.04

Table 1: Targeted moments in calibration corresponding to model parameters.

Markov chain with values $\{\Theta(Z_{low}), \Theta(Z_{high})\}$ following a discrete approximation of an AR(1) process estimated from a linearly de-trended series of total U.S. labor productivity. The transition probabilities $\pi^{\Theta}_{Z,Z'}$ are similarly obtained from the Markov chain approximation.

Aggregate credit conditions loosen via a one-time unanticipated shift to characterize the onset of the housing boom in the 1990s. They are represented by perfectly correlated movements in the loan-to-value constraint $\theta^{LTV}(Z)$, payment-to-income constraint $\theta^{PTI}(Z)$, the fixed mortgage origination cost $\kappa^m(Z)$, and the mortgage intermediation wedge $\zeta(Z)$. Although higher loan-to-value constraints are the most common representation of looser credit conditions in macro housing models, other innovations in mortgage finance also played a role in extending relatively more credit to households and are hence included.[21] Dokko et al. (2019) find that 60 percent of all mortgages contained at least one non-traditional feature by 2005 and Greenwald (2026) and Ma and Zubairy (2020) show that payment-to-income constraints $\theta^{PTI}(Z)$ are quantitatively more important than loan-to-value constraints $\theta^{LTV}(Z)$ in accounting for mortgage debt throughout the 2000s.

The loan-to-value constraint is equal to 0.95 in the low state which is slightly higher than Duca et al.'s (2011) estimates for cumulative loan-to-value ratios of first-time home buyers in the 1990s. Following Kaplan et al. (2020), the value 0.95 replicates the 90th percentile of the

[21] Loan-to-value constraints also make the households' problems well defined as explained by Engelhardt (1996). Along with Kiyotaki et al. (2011), Engelhardt (1996) notes that loan-to-value constraints are a reduced-form representation of contract enforcement frictions in credit markets. Because lenders fear that borrowers might not repay mortgages, they can distribute some of the downside risk by requiring the house to be pledged as collateral and forcing borrowers to hold equity via loan-to-value constraints.

Interpretation	Parameter	Value
Aggregate income $\Theta(\mathbf{Z})$		
Aggregate income	$\{\Theta(Z_{high}), \Theta(Z_{low})\}$	$\{1.035, 0.965\}$
Transition probability	$\pi^{\Theta}_{h,h} = \pi^{\Theta}_{l,l}$	0.9
Aggregate credit conditions $\mathcal{C}(Z)$		
Loan-to-value ratio	$\{\theta^{LTV}(Z^{loose}), \theta^{LTV}(Z^{tight}))\}$	$\{1.1, 0.95\}$
Payment-to-income ratio	$\{\theta^{PTI}(Z^{loose}), \theta^{PTI}(Z^{tight})\}$	$\{0.5, 0.25\}$
Fixed origination cost	$\{\kappa^{m}(Z^{loose}), \kappa^{m}(Z^{tight})\}$	$\{\$1{,}200, \$2{,}000\}$
Proportional origination cost	$\{\zeta(Z^{loose}), \zeta(Z^{tight})\}$	$\{0.006, 0.010\}$
Beliefs/learning		
Constant gain	$g_t = g$	0.508
Initial learning coefficients	$\boldsymbol{a}_0 = \boldsymbol{a}^{tight}_{Z_{high}, Z'_{high}}$	{-0.102,0.838}
	$\boldsymbol{a}_0 = \boldsymbol{a}^{tight}_{Z_{high}, Z'_{low}}$	{-0.111,0.838}

Table 2: Parameters for aggregate shocks to income and credit conditions along with those for endogenous beliefs characterized via adaptive learning. The model period is two years and values are not annualized.

loan-to-value distribution observed in the pre-boom data. The high state value of 1.1 targets a 15 percentage point rise in combined loan-to-value ratios from the late 1990s to 2006. The payment-to-income constraint values are $\theta^{PTI}(Z_{high}) = 0.5$ and $\theta^{PTI}(Z_{low}) = 0.25$, which are slightly lower than the values of Greenwald (2026) to account for the lifetime amortization of mortgages being longer than the typical 30 years. Falling financial intermediation and funding costs are represented by a decrease in the fixed mortgage origination costs $\kappa^m(Z)$ from \$2,000 to \$1,200 as described by Favara and Imbs (2015) and a decrease in the mortgage intermediation wedge $\zeta^m(Z)$ from 100 basis points to 60 basis points as described by Arslan et al. (2026).

Beliefs: This paper's contribution of endogenous beliefs via adaptive learning introduces three parameters that affect the formation of house price expectations.

First, the learning gain g_t assigns the speed at which agents update their beliefs with incoming information. Following Marcet and Nicolini (2003), Milani (2014), and Carvalho et al. (2023), I use a mixed gain that switches between a constant gain $g_t = g$ and a decreasing gain $g_t = g_{t-1}/(g_{t-1} + 1)$ in order to, "restrict the learning mechanism to produce good forecasts within the model," as described by Marcet and Nicolini (2003). While these mixed gain frameworks typically estimate a threshold for which the gain switches, I instead tie the switch to underlying aggregate states, which tracks the degree of knowledge agents have about house price forecasting coefficients. The constant gain in the boom state means

that agents update their beliefs more aggressively with more recent house price observations, which is consistent with the interpretation that agents have never seen an aggregate loosening of credit conditions before. By contrast, agents have experience in the non-boom states and thus the decreasing gain allows their beliefs to more closely track the sample mean of house price observations based on realized experiences. While the choice of a state-dependent mixed gain can resolve some of the economic tension between the weight given to new information in different states of the economy, it is further justified via empirical forecast errors.[22]

$$g_t = \begin{cases} g & \text{if} \quad Z_t \in \{Z_{high}^{loose}, Z_{low}^{loose}\} \\ \dfrac{g_{t-1}}{1+g_{t-1}} & \text{otherwise} \end{cases}$$

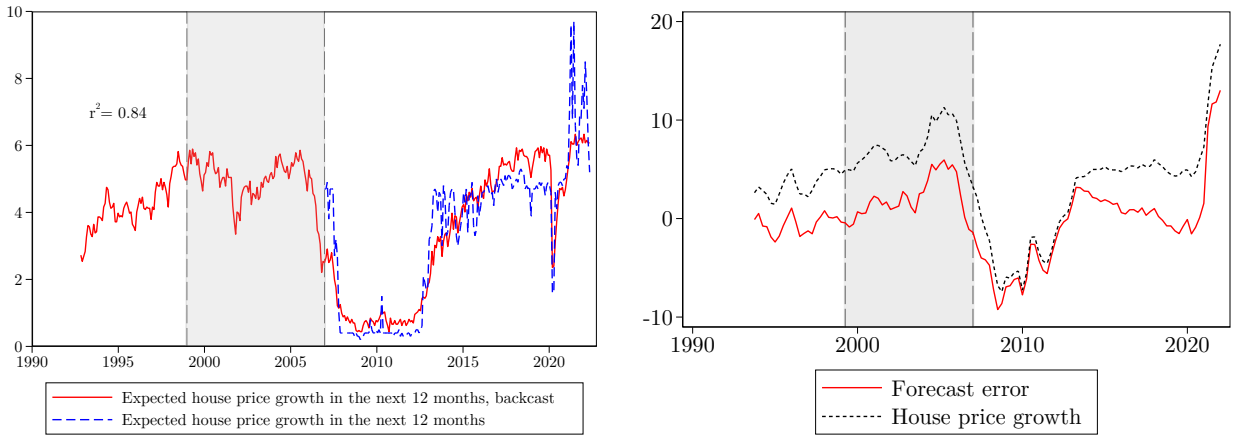
The data on forecast errors used is an empirical proxy constructed from the University of Michigan Survey of Consumers detailed in Appendix D and shown in Figure 1. Even though series of expected house prices are only available at the end of the housing boom in 2007, the "next 12 months" expectations series has a 0.92 correlation with a series on selling conditions that dates back to 1992. As a result, backcasting the shorter expectations series from the longer selling conditions series can deliver a proxy of house price expectations throughout the 2000s. The forecast errors used to calibrate the constant learning gain $g_t = g$ are large and have a steady upward path following that of house prices as shown in Panel 1b for the proxy constructed from the 75th percentile of expectations. This evidence aligns with that of Adam et al. (2025) as shown in Appendix D.3 which compares their exercises to those of this paper's proxy as well as proxies constructed from the mean and median.[23] Finally, Appendix D.2 discusses why the 75th percentile is the main calibration instead of the mean or median, but shows that housing boom results are broadly similar under any of these calibrations.

Following Caines (2020), the constant learning gain $g_t = g$ is set to minimize the mean squared errors of house price forecasts from the model and data. The period targeted is that of the housing boom in this paper: 1999:Q1 to 2007:Q4, which yields a mean-squared forecast error of 28.66. The calibration yields $g = 0.3$ (0.508 non-annualized) to set the model's mean squared error of house price forecasts to that of the data, as shown in Figure

[22]Although decreasing gain learning can potentially guarantee convergence to a rational expectations equilibrium, the Krusell and Smith (1998) approximate equilibrium solved for in this paper may not necessarily be a true rational expectations equilibrium, and convergence may only be to an ergodic distribution. Nonetheless, the decreasing gain in some states may help ensure that coefficients indeed converge, as shown in Figures 42 and 43 in Appendix F.9, as a constant gain may be more prone to oscillations.

[23]Mian and Sufi (2022) show somewhat contradictory evidence that all but optimistic marginal home buyers thought it was a bad time to buy a house in the later part of the boom using 1) disaggregated data from the University of Michigan Survey of Consumers and 2) survey data from Case et al. (2012). However, Appendix F.5 confirms their evidence and shows that model-generated endogenous beliefs would have eventually become endogenously pessimistic had credit conditions stayed looser for longer.

4. The non-annualized value of $g = 0.508$ implies that 4 years of data are retained by agents and this short history can be justified from the fact that the housing boom state is a new state of the economy agents have never before experienced.



(a) Expectations of house price growth, %

(b) House price growth and forecast errors

Figure 1: 12-month percent change in expected house price growth (75th percentile from the University of Michigan Survey of Consumers) and 4-quarter percent change in house price growth with forecast errors calculated as house price growth less the proxy. Shaded bar between the dashed vertical lines marks the U.S. housing boom (1999 to 2007). Appendix D details the construction of the proxy and compares it to its counterparts constructed from the mean and median.

With an annualized value of about $g = 0.3$ (0.508 non-annualized), this paper's constant gain is on the larger end of values typically found elsewhere in the literature. Milani (2014) notes that when forecast errors are large, agents may be concerned that the economy is experiencing a structural break and assign a large weight to incoming information and hence a high value for the constant gain. This interpretation is consistent with agents having limited knowledge of the evolution of future house prices in the boom state they have never visited before.

Moreover, Caines (2020), Adam et al. (2012), Adam et al. (2025) use annualized gains of about 0.03 to 0.06 in their applications of adaptive learning to housing. These smaller values may arise from 1) agents learning about house price growth instead of house price forecasting coefficients and 2) differences in house price expectations used in calibration. First, Figure 35 in Appendix F.6 shows that a lower gain of $g = 0.15$ (0.28 non-annualized) based on learning about house price growth instead of levels can generate a similar housing boom while targeting the same mean-squared forecast error in the boom period. Second, both Caines (2020) and Adam et al. (2025) use post-boom house price expectations from the Case-Shiller house price futures or the Michigan Survey. Roschitsch and Twieling (2025) show that expectations can differ in the boom and post-boom periods.

On the other hand, Marcet and Nicolini (2003) estimate an annualized constant gain ranging from 0 to 0.5 in boom-like settings, Milani (2014) estimates constant gains ranging from 0.21 to 0.32, and Carvalho et al. (2023) close to 0.45 at an annualized frequency. Taken together, this evidence suggests that an annualized constant gain of about 0.3 is high but still within an established range of values.

Given that a constant gain can reproduce a similar housing boom to one under a mixed-gain (Figure 32 in Appendix F.4), what does switching to a decreasing gain in the non-boom states accomplish? The answer lies in reproducing an empirically consistent forecast error. Appendix Panel 32g in Appendix F.4 shows that non-mixed learning gains tend towards too much pessimism in bust and too much optimism in the recovery. By contrast, the forecast error under the mixed gain tracks its empirical counterpart almost exactly, while targeting the mean-squared forecast error of the boom via $g = 0.508$ and separately the path of the forecast error in the bust via a decreasing gain with a decay parameter $\tau = 5$.

The decreasing gain $g_t = g_{t-1}/(g_{t-1}+1)$ only affects the non-boom states and is initialized at $g_t = g/(1+\tau \times g) = 0.14$ and decreases thereafter. This initialization follows the convention of Marcet and Nicolini (2003) and Milani (2014) plus a decay parameter $\tau = 5$ that takes into account the number of periods since the start of the housing boom and thus accumulated experience of realizations since learning began. This implies that agents retain about 14 years of data when forming their post-boom beliefs. Figure 34 in Appendix F.5 shows that simulations are broadly similar under various or no decay parameters and discusses that the mean-squared forecast error is notably worse without a decay parameter.[24]

The second and third learning parameters are contained in the vector of lagged initial learning coefficients $\boldsymbol{a}_0$. These parameters determine the initial forecast error and there is no definitive convention, as noted by Lubik and Matthes (2016). Using the known pre-boom coefficients for tight aggregate credit conditions and the high income state $\boldsymbol{a}^{tight}_{Z_{high},Z'_{high}}$ as initial beliefs for the boom state leads to an initial forecast error of 2.36 percent which is in the 2.35 to 4.1 range of empirical values detailed in Table 15 in Appendix D.2. Appendix F.3 shows that the results are similar if the constant coefficient a^0_0 is adjusted so that the initial forecast error attains an upper value in this range, while leaving the slope coefficient unchanged at $a^1_0 = a^{1,tight}_{Z_{high},Z'_{high}}$. Furthermore, results are similar if the slope coefficient is adjusted and the constant coefficient is unchanged. As a sensitivity check, Appendix F.3 also shows that the housing boom dynamics are broadly similar if either coefficient is adjusted so that it is below the empirical range.

[24]Appendix F.5 discusses why it is difficult to distinguish mean-squared forecast errors in the bust. However, the fit for $\tau = 1$ is much worse than any $\tau \in [3, 7]$. Because $\tau = 5$ can be justified via the observations since the start of the housing boom, this is retained as the main calibration.

4.2 Calibration

See Appendix C.2 for an in-depth discussion of the cross-sectional distribution of households in the model's stochastic steady state which assumes known forecasting coefficients, tight aggregate credit conditions, and fluctuations in aggregate income.

Parameter	Empirical Value	Model Value
Fraction of homeowners with a mortgage	0.66	0.65
Fraction of homeowners with a HELOC	0.06	0.02
Aggregate mortgage debt/housing value	0.42	0.48
P10 LTV ratio for mortgages	0.15	0.02
P50 LTV ratio for mortgages	0.57	0.5
P90 LTV ratio for mortgages	0.92	0.85
Share of NW held by bottom quintile	0	0
Share of NW held by middle quintile	0.05	0.09
Share of NW held by top quintile	0.81	0.66
Share of NW held by top 10 percent	0.7	0.43
Share of NW held by top 1 percent	0.46	0.06
P10 house value/earnings	0.9	0.94
P50 house value/earnings	2.1	1.8
P90 house value/earnings	5.5	4.1
Weight on new information	[0.25,0.33]	0.35
Cyclicality of expected house price growth, $\hat{c}$	$[0.036, 0.112]$	0.064
Cyclicality of actual house price growth, $\hat{c}$	$[-0.525, -0.055]$	-0.24

Table 3: Untargeted moments in calibration.

Note: Weight on new information is $1/(1+\hat{b}^{CG})$ from the regressions of house price forecast errors on forecast error revisions as described in Appendix D.3 at a yearly frequency. The cyclicality moments are also at a yearly frequency and described in Appendix D.3, and are regressions of expected and actual house price growth on the lagged price-rent ratio.

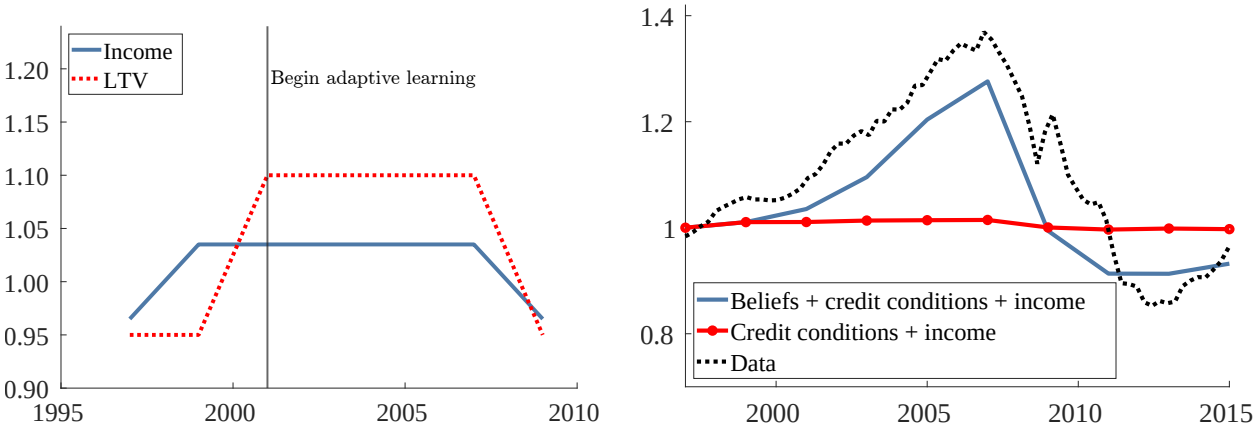
Table 3 shows a comparison of untargeted moments from the model to their empirical counterparts. The model matches the distributions of homeownership and mortgage debt shown in the top panel and the bottom 90th percentile of the wealth distribution shown in the bottom panel. As is common in models with heterogeneous households, the top 10th percentile of the model's wealth distribution is far below that of the data because wealthy households can only hold housing or liquid financial instruments and thus accumulate a counterfactually high share.[25] Appendix C.3 discusses the assumption on the partial seg-

[25] Although allowing for risky assets with heterogeneous returns is a solution proposed by Cioffi (2021) and Xavier (2021), it is beyond the scope of this paper as it would complicate how a shift in beliefs affects house prices. Relatedly, while adding two types of discount factors could better capture the right tail of the wealth distribution, solving a Krusell and Smith (1998) with two discount factors could lead to multiple equilibria, convergence issues, or numerical inaccuracy where an AR(1) forecasting rule for house prices may not be enough moments to pin down the evolution of house prices.

mentation of housing unit sizes by renters and homeowners. The bottom panel of the table shows how the model captures untargeted statistics on subjective beliefs studied by Adam et al. (2025) (details in Appendix D.3), thereby validating that the learning process used in this model can capture features of the data.

5 Housing Boom Simulations

The housing boom is initiated by a sequence of aggregate shocks to income and credit conditions, as shown in Panel 2a. In 1999, aggregate income (solid blue lines) transitions from the low state to the high state and aggregate credit conditions (dotted red lines) unexpectedly loosen in 2001 which is when adaptive learning about house price forecasting coefficients begins.[26] After remaining in this boom state with high income and loose credit conditions for 4 periods (8 years), aggregate income transitions back to the low state and aggregate credit conditions unexpectedly tighten back to their pre-boom values. The housing boom is thus a period where agents do not anticipate looser credit conditions, but then expect them to stay in place permanently.[27]



(a) Aggregate shocks (b) House prices, 1997=1

Figure 2: Aggregate shocks and house prices. Aggregate shocks correspond to values in Table 2 for 1997-2009. House prices from the data and the model solved under endogenous beliefs. See Appendix G for sources and definitions of the data series.

Panel 2b shows that optimistic beliefs, along with shocks to income and credit conditions (solid blue lines), can match the boom in U.S. house prices in the 2000s. Aggregate house prices peak 35 percent and 28 percent above their pre-boom values in the data and the model, respectively. The path of house prices generated by the model closely tracks that of

[26] Appendix F.5 shows that the results are similar if adaptive learning instead begins at the start of the model simulation instead of in 2001.

[27] Appendix E shows credit conditions evolving via a one-time shock or a Markov process has little effect on the housing boom simulations under exogenous beliefs. A one-time shock is therefore assumed under both exogenous and endogenous beliefs.

the data, which results in the model matching additional features of aggregate house price data such as 96 percent of the autocorrelation and 89 percent of the standard deviation. Key to the model generating any volatility in house prices—let alone successfully matching the data—is partial updating of endogenous beliefs via adaptive learning. Given that agents can only partially internalize the increase in aggregate demand when credit conditions loosen, house prices come in higher than expected, which results in a positive forecast error that is interpreted as signal that beliefs were too low and results in even more optimistic beliefs.

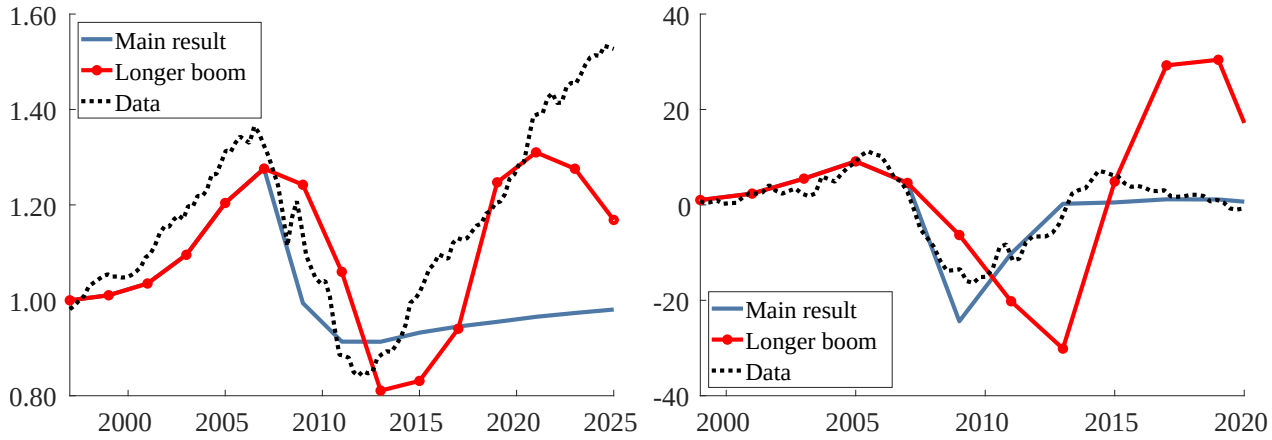
By contrast, Panel 2b shows that shocks to income and credit conditions without endogenous beliefs (red lines with circles) result in only an incremental increase in house prices and almost no volatility. Because demand for housing services—and hence house prices—is generally determined by income, the 7 percentage point increase in aggregate income shown in Panel 2a is not large enough to generate the nearly 35 percentage point increase in house prices shown in Panel 5a. Relatedly, the increase in housing demand from looser credit conditions is limited to only a fraction of households due to heterogeneous incomes and ages. Because these constrained households tend to be younger and buy the same size of house they were renting when credit conditions loosen in this framework, overall demand for housing services only increases slightly, which results in almost no increase in house prices.[28] As will be discussed and is shown by Ma and Zubairy (2020), looser credit conditions are important for accounting for homeownership dynamics across the age distribution. Figure 28 in Appendix F.1 shows additional simulations such as those with just beliefs and no shift in credit conditions.[29] Figure 29 in Appendix F.1 shows that shifting only one, instead of all four, credit conditions can also result in a similar boom in house prices. However, shifts in all credit conditions are necessary for matching the dynamics of other model quantities such as homeownership, leverage, and foreclosure.

Because the bust in house prices shown in Figure 2 is attributed to the contraction of aggregate states, Figure 3 shows a counterfactual simulation where the housing boom state persists for another twenty years until 2029 and results in a boom-bust-rebound cycle. This endogenous bust and rebound can be explained via the path of forecast errors shown in Panel 3b. In both the main simulation and the counterfactual, the forecast error is lower in 2007 than in 2005. However, because forecast errors affect beliefs with a lag, 2009 is when the drop in the forecast error first enters belief formation, which only occurs in the counterfactual

[28] Greenwald and Guren (2025) show that this is not a general result and frameworks that incorporate endogenous investment in the housing stock along with landlord heterogeneity can generate higher house prices from looser credit conditions.

[29] While the simulation with endogenous beliefs but no shift in credit conditions can somewhat match the increase in house prices, it delivers no increase in the homeownership rate and a drop in leverage, which suggests that credit conditions are important for pinning down these dynamics in the housing boom.

where there is no shift in aggregate states back to their pre-boom values. Mirroring the optimism that generates the boom, the pessimism that generates the bust arises because agents interpret a decrease in the forecast error as signal that their beliefs were too optimistic and revise their beliefs downward. Due to partial updating, these downward revisions are slow and give rise to persistently negative forecast errors, persistently pessimistic beliefs, and thus a drop in house prices. In this counterfactual, the forecast error overshoots zero and turns negative because agents interpret the forecast error as a signal that their optimistic beliefs need correction. Figure 33 in Appendix F.5 shows simulations of additional prices and quantities.



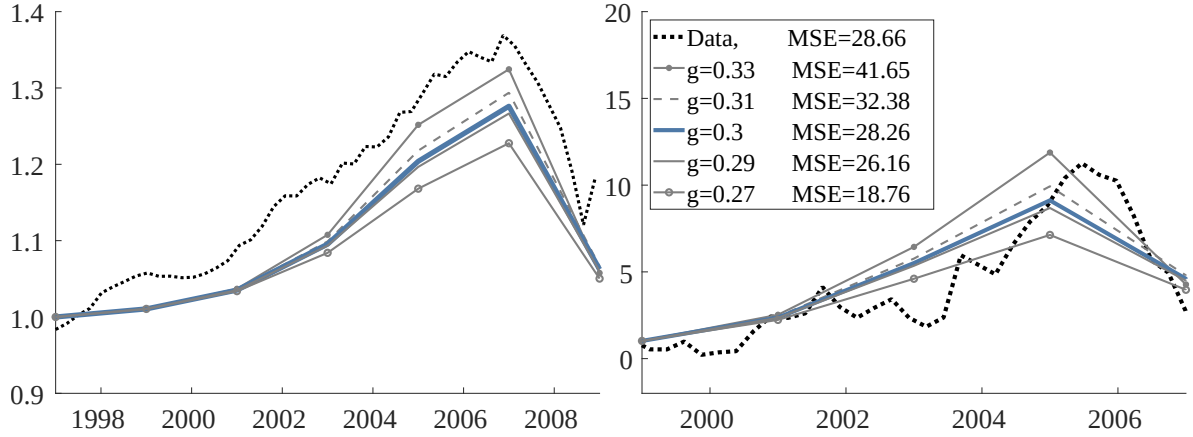
(a) House prices, 1997=1 (b) House price forecast error, percentage points

Figure 3: House prices and forecast errors from the data and the model solved under endogenous beliefs. The model's forecast error is $\log p_t - \mathbb{E}_{t-1}[a^0_{\mathcal{Z},t-1} + a^1_{\mathcal{Z},t-1} \log p_{t-1}]$. See Appendix G for sources and definitions of the data series and Appendix D for details on the construction of the proxy for empirical forecast errors from University of Michigan Survey of Consumers.

How closely model-generated house prices match the data depends on the constant gain parameter g as shown in Panel 4a. Although house prices steadily rise under all values of the constant gain g shown, a smaller constant gain results in a more muted housing boom while a larger constant gain results in one that is more pronounced. There is a positive correlation between the constant gain and house price growth because the constant gain determines the speed at which agents incorporate new information into their beliefs—a larger value of the constant gain g thus updates beliefs more aggressively.[30] While Figure 4 shows that model results are sensitive to values of the constant gain g, Appendix F shows that the housing boom simulations in the main result are robust to differences in the learning rule. These include a variety of forecast error formulations (Appendix F.2); a variety of initial forecast

[30] Appendix F.9 shows that beliefs do eventually converge to an ergodic distribution in a longer simulation with a counterfactual recurrence of the housing boom state.

errors (Appendix F.3); different constructions of the learning gain such as a constant gain, decreasing gain, or a reversed mixed gain (Appendix F.4); and different learning processes such as the Bayesian learning of Adam et al. (2025) or least-squares learning (Appendix F.6). Taken together, these findings imply that many learning models relying on partial updating of beliefs that prevent agents from fully internalizing shifts in demand for housing services can generate the housing boom.



(a) House prices, 1997=1 (b) House price forecast errors, percent

Figure 4: House prices and forecast errors from the data and model solved under endogenous beliefs. The model's forecast error is $\log p_t - \mathbb{E}_{t-1}[a^0_{\mathcal{Z},t-1} + a^1_{\mathcal{Z},t-1} \log p_{t-1}]$. The parameter g is the learning gain and is annualized in the legend. See Appendix G for sources and definitions of the data series and Appendix D for details on the construction of the proxy for empirical forecast errors from University of Michigan Survey of Consumers.

Empirical house price forecast errors constructed from the proxy of the University of Michigan Survey of Consumers are persistently positive throughout the 1999 to 2007 period, as shown in Panel 4b via the dotted black line. Under all values of the constant gain g shown, forecast errors are positive and match the upward path observed in the data, but with less volatility due to their observation at a lower two-year frequency.[31] Agents underpredict house prices because their initial beliefs do not fully internalize the increase in housing demand from looser credit conditions. Since beliefs are a combination of past values and recent forecast errors, partial updates of large positive forecast errors can result in persistently underpredicted house prices and positive forecast errors. At 28.26, the mean squared error for the annualized constant gain $g = 0.2986$ (g=0.508 at a two-year frequency) is the closest fit to the data and hence used in the main results.[32] Appendix D.2 shows that results are

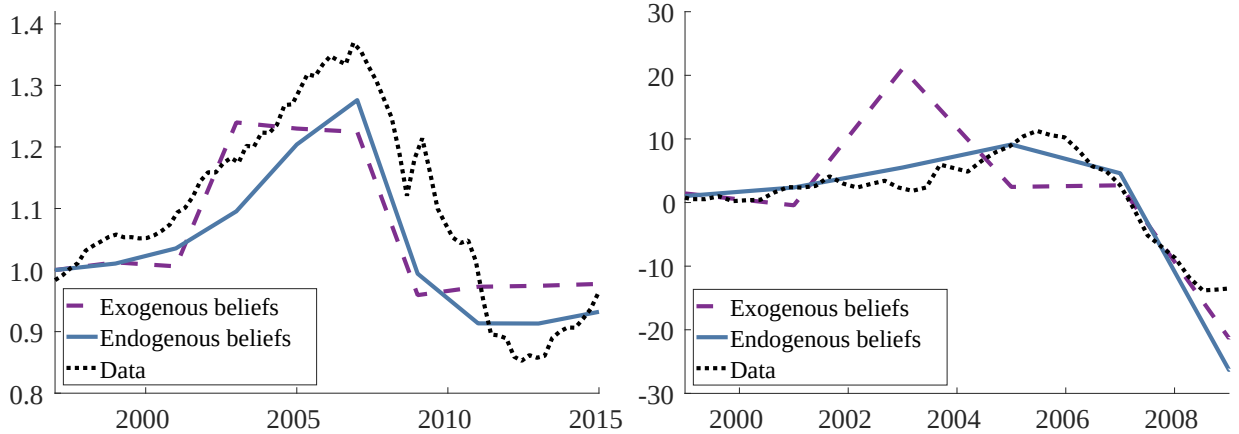
[31]Because it is unclear if it is better to aggregate via end of period observations or averages, I do not transform the quarterly data to a 2-year frequency.

[32]Because the data used to calibrate the constant gain g is only an empirical proxy there could be concerns about measurement noise that biases results. However, Appendix F.4 shows that the housing boom simulation results are similar if a decreasing learning gain that does not rely on data constructed from an empirical

similar if the model is calibrated to the mean or median proxy instead of the 75th percentile.

This paper's finding that beliefs can account for the 2000s housing boom has also been shown by Kaplan et al. (2020), but under exogenous instead of endogenous beliefs. I next compare how model prices, forecast errors, and quantities evolve under different types of belief formation and find many similarities. Despite these similarities, section 5.1 shows that belief formation matters for the effectiveness of policy interventions.

Under either the endogenous beliefs of this paper or the exogenous beliefs of Kaplan et al. (2020), the housing boom is initiated by the same sequence of aggregate shocks to income and credit conditions shown in Panel 2a. Under exogenous beliefs, there is no adaptive learning but a third shock to housing preferences that occurs in 2003 in the period following the shift in credit conditions in 2001. While the housing preference parameter ϕ is fixed under endogenous beliefs it evolves according to a three-state Markov process under exogenous beliefs. In the first state, it is at its low value and has zero probability of shifting to its high value. In the second state, it also has its low value but has a probability of 0.85 to shift to its high value. This second state where a shift to the high value is likely is the exogenous shift in beliefs that accounts for higher house prices observed in Panel 5a.



(a) House prices, 1997=1

(b) House price forecast error, percentage points

Figure 5: House prices and forecast errors from the data and the model solved under endogenous beliefs and exogenous beliefs. The model's forecast error is $\log p_t - \mathbb{E}_{t-1}[a^0_{\mathcal{Z},t-1} + a^1_{\mathcal{Z},t-1} \log p_{t-1}]$. See Appendix G for sources and definitions of the data series and Appendix D for details on the construction of the proxy for empirical forecast errors from University of Michigan Survey of Consumers.

Model-generated house prices under exogenous beliefs (dashed purple lines) peak just

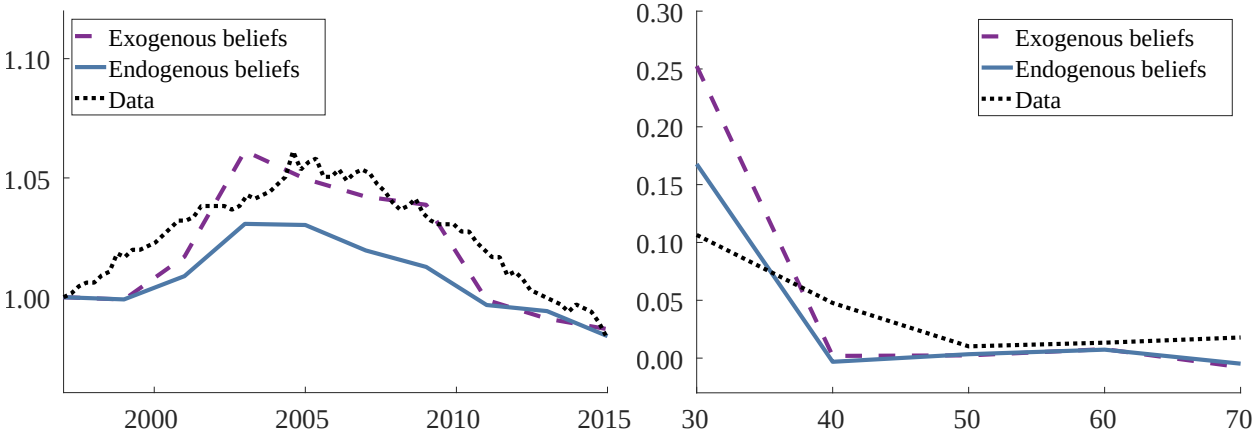
proxy is used instead. Appendix F.4 also includes a simulation that mirrors the findings of Roschitsch and Twieling (2025)—a larger gain in the bust and a smaller gain in the boom, which suggests a role for over pessimism as well as over optimism. This is achieved by switching the pattern of the switching gain so that it is constant in the bust and recursive in the boom.

	St. dev., σ_{p_t}	$\frac{\sigma_{p_t}}{\sigma_{data}}$	Autocorr., $\rho(p_t, p_{t-1})$	$\frac{\rho(p_t, p_{t-1})}{\rho_{data}}$
Exogenous beliefs, 2003-2007	0.020	16%	0.325	71%
Exogenous beliefs, 1999-2007	0.152	115%	0.225	39%
Endogenous beliefs, 1999-2007	0.113	89%	0.440	96%
Data, 1999-2007	0.127	100%	0.458	100%

Table 4: Biennial house prices statistics from the data and model for the U.S. housing boom.

under the 28 percent peak of their counterparts generated under endogenous beliefs (solid blue lines), as shown in Panel 5a. However, instead of a steadily rising path like that under endogenous beliefs, exogenous beliefs generate a counterfactual jump in a single period. As a result, Table 4 shows that exogenous beliefs severely undershoot or overshoot the autocorrelation and standard deviation of aggregate house prices, while the fit under endogenous beliefs is 96 percent and 89 percent, respectively. The house price forecast errors shown in Panel 5b provide external empirical validation: while forecast errors are positive and persistent under endogenous beliefs as in the data, they display a counterfactual sign change and are negative in all periods except for 2003 under exogenous beliefs. Table 18 in Appendix E shows that the better match of the autocorrelation of house prices under endogenous beliefs holds in an unconditional simulation as well as in the housing boom.

Given that endogenous beliefs generate house prices that match the path, autocorrelation, and standard deviation of the house price data while also matching empirical forecast errors, this paper next investigates how endogenous beliefs affect the homeownership rate which is a key determinant of housing demand and hence house prices.



(a) Homeownership rate, 1997=1

(b) Log change in homeownership by age

Figure 6: The homeownership rate from the data and the model solved under endogenous and exogenous beliefs. See Appendix G for sources and definitions of the data series.

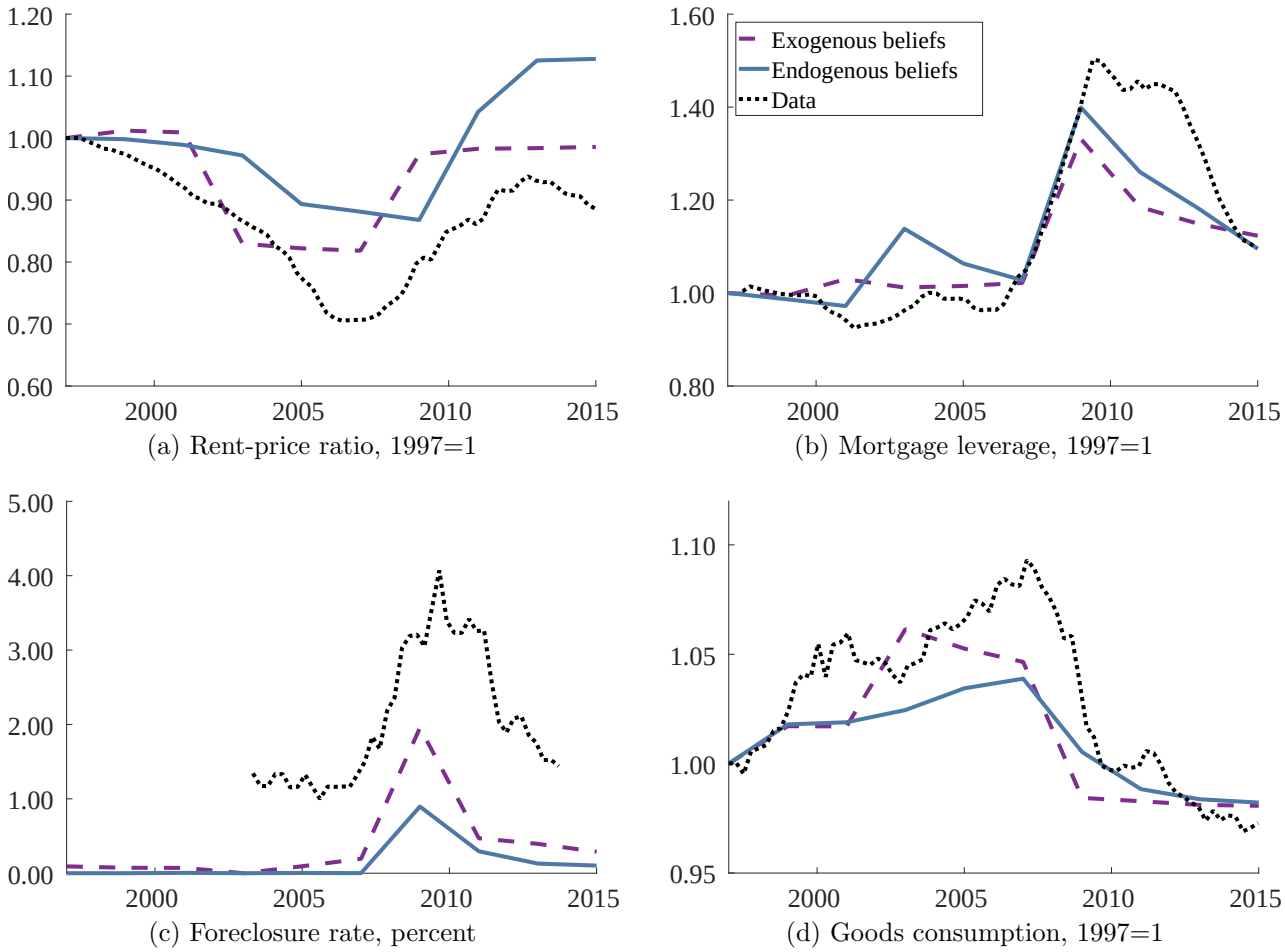
Figure 6 shows that the homeownership rate largely matches its empirical counterpart

over time and in the cross section. Panel 6a shows that the aggregate homeownership rate under endogenous beliefs captures the gradual rise and decline of the path and about half of the peak observed in the data throughout the boom. Although there is a more pronounced initial jump in the homeownership rate under exogenous beliefs, the simulations are quite similar with either type of belief formation. The increase in the homeownership rate in both the model and the data is mostly driven by households under the age of 40 purchasing homes as shown in Panel 6b. The size of this increase in homeownership among these young households in Panel 6b is reflected in the aggregate homeownership in Panel 6a. Looser credit conditions allow a larger fraction of young households to overcome payment-to-income and loan-to-value constraints earlier in life than under tighter credit conditions. New homeowners tend to purchase the same sized housing unit they were previously renting, which results in slightly more aggregate demand for housing services and only accounts for a small fraction of the increase in house prices. Higher house prices can mostly be attributed to unconstrained households upsizing their housing units in response to wealth effects from higher expected future house prices.

Figure 7 shows that other prices and quantities also match the data under either exogenous or endogenous beliefs. Matching the volatility of house prices under endogenous beliefs leads to the rent-price ratio and aggregate consumption better matching the path observed in the data which is useful for understanding the consumption response to house prices.

Panel 7a shows that the decline in the rent-price ratio under either exogenous or endogenous beliefs matches about half of the decline observed in the data. Although most models of housing succeed at generating an increase in prices and homeownership, it is often at the expense of counterfactually rising rents. The rental rate in equation (13) shows that high expected future house prices—like those from optimistic beliefs—are key to rents decreasing and thus a drop in the rent-price ratio. When model-generated house prices more closely track their counterparts in the data, as under endogenous beliefs, the model-generated rent-price ratio also better tracks its empirical counterpart.

Greenwald and Guren (2025) show that the tenure supply elasticity—the elasticity of house prices to credit supply shocks divided by the elasticity of homeownership to credit supply shocks—is an empirically important moment for understanding the effects of credit conditions on house prices. Dividing the peak percentage point increase in the price-rent ratio (27 p.p.) by that of the homeownership rate (3.1 p.p.) under the endogenous beliefs proposed by this paper delivers a ratio of 8.9 which is close to that observed in the data of 11 (70 p.p. peak in the price-rent ratio divided by a 6 p.p. peak in the homeownership rate). While the ratio under endogenous beliefs is closer to the no segmentation case of Greenwald and Guren (2025), it is in line with their empirical estimates of at least 3.8



(a) Rent-price ratio, 1997=1

(b) Mortgage leverage, 1997=1

(c) Foreclosure rate, percent

(d) Goods consumption, 1997=1

Figure 7: Housing boom prices and quantities from the data and the model solved under endogenous and exogenous beliefs. See Appendix G for sources and definitions of the data series.

Panel 7b shows that modeled generated mortgage leverage remains near its pre-boom values throughout most of the boom and then rises in the bust, similar to what is observed in the data. The only shortcoming of endogenous beliefs is a counterfactual one-time jump during the boom that is not present in their exogenous beliefs counterparts.

The model's foreclosure rate shown in Panel 7c rises in 2009 when income contracts and credit conditions tighten in the bust. The rise in the foreclosure rate is smoother under endogenous beliefs relative to exogenous beliefs reflecting a smoother path of house prices rather than a one-time jump under exogenous beliefs.

Finally, Panel 7d shows that optimistic beliefs and higher house prices generate a path of consumption that matches about half of the peak in the data with a path that gradually rises throughout the boom, as observed empirically. In line with a housing wealth effect, the paths of consumption under either exogenous or endogenous beliefs show a strong correlation with

the respective paths of house prices.[33] The model under endogenous beliefs can thus better match the data's steady rise in the path of goods consumption due to its corresponding steady rise in house prices.

5.1 Policy counterfactuals

Although model-generated house prices under endogenous beliefs are validated by external empirical evidence and match many features of house prices beyond the level increase, other prices and quantities from the model are quite similar under either type of belief formation. This section addresses the implications of belief formation by showing that endogenous beliefs solved via the one-step method are more sensitive to prudential policy interventions than their exogenous counterparts solved via Krusell and Smith (1998) fixed-point iteration. While this finding echoes that of Fontanier (2025), the one-step method contributes a quantitative assessment that reveals substantial nonlinearities such that the responses of house prices do not scale linearly with the size or timing of interventions. By combining endogenous beliefs with a global solution method, these nonlinearities are preserved and found to be quantitatively important.

The sensitivity of beliefs to prudential policy interventions stems from two counterfactual scenarios during the housing boom, one where interest rates increase and another where credit conditions tighten during the housing boom. Using monetary policy to "lean against" rising house prices has been studied as a prudential policy for housing boom management because higher policy rates push up mortgage rates, thereby dampening demand for housing services. However, because higher interest rates also 1) dampen output along with house prices and 2) may make homeownership less affordable, researchers have cautioned that lowering house prices may be too costly.[34] On the other hand, researchers have also shown that leaning against house prices may be optimal when knowledge is limited (Caines and Winkler, 2021; Adam and Woodford, 2021).

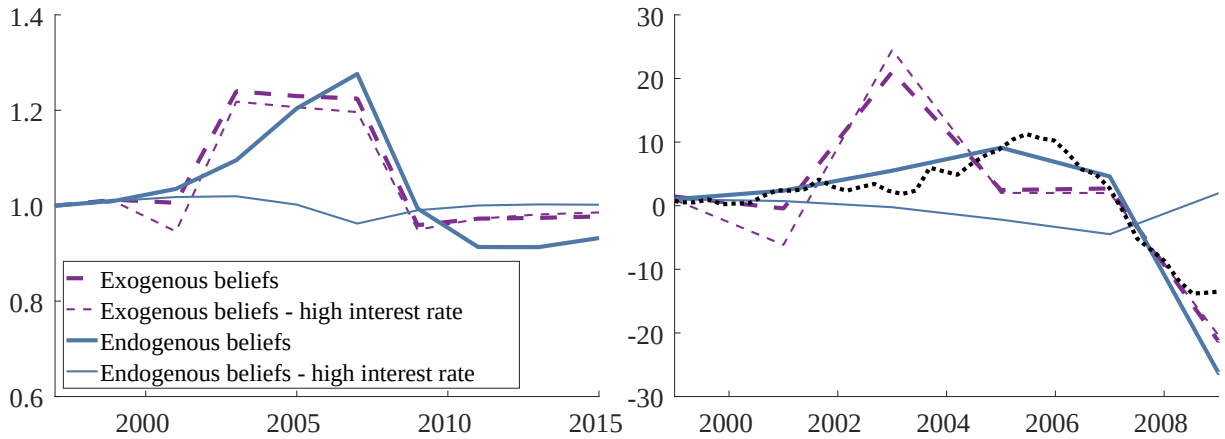
The higher interest rate scenario relies on the same aggregate shocks as the main housing boom simulation shown in Panel 2a, along with an additional unexpected shock to the risk-free rate that occurs in 2001 when credit conditions loosen. More specifically, the risk-free rate r_b rises 50 basis points from 6 percent to 6.5 percent on a two-year basis resulting in the mortgage rate $r^m = (1 + \iota)r_b$ rising from 8 percent to 8.65 percent on a two-year basis.

Figure 8 shows that house prices under endogenous beliefs are much more sensitive to a 50 basis point increase in the risk-free rate than their exogenous counterparts. In fact, the

[33]The size of the housing wealth effect remains debated. While Berger et al. (2018) find large consumption responses to house price movements, Guren et al. (2020) note that responses in the 2000s are smaller than those of the 1980s.

[34]See Benati (2021), Lambertini et al. (2013), Ehrenbergerova et al. (2021), Svensson (2017).

thin solid blue line in Panel 8a shows a bust instead of a boom. House prices drop because of pessimistic beliefs—the decrease in demand for housing services from higher interest rates leads to a negative forecast error that persists due to the slow updating of beliefs under adaptive learning, as shown in Panel 8b. By contrast, under exogenous beliefs house prices and forecast errors are little different in the scenario with higher interest rates (thin dashed purple lines) relative to the main simulation (thick dashed purple lines). Because exogenous beliefs rely on households pulling forward higher expected demand for housing services that never materializes, house prices will only be substantially lower if the dampening effects of higher interest rates can offset these amplifying effects of beliefs. The counterfactual shows that this is unlikely with only a 50 basis point increase in interest rates.



(a) House prices, 1997=1 (b) House price forecast errors, percentage points

Figure 8: House prices and forecast errors with counterfactually higher interest rates. The model's forecast error is $\log p_t - \mathbb{E}_{t-1}[a^0_{\mathcal{Z},t-1} + a^1_{\mathcal{Z},t-1} \log p_{t-1}]$.

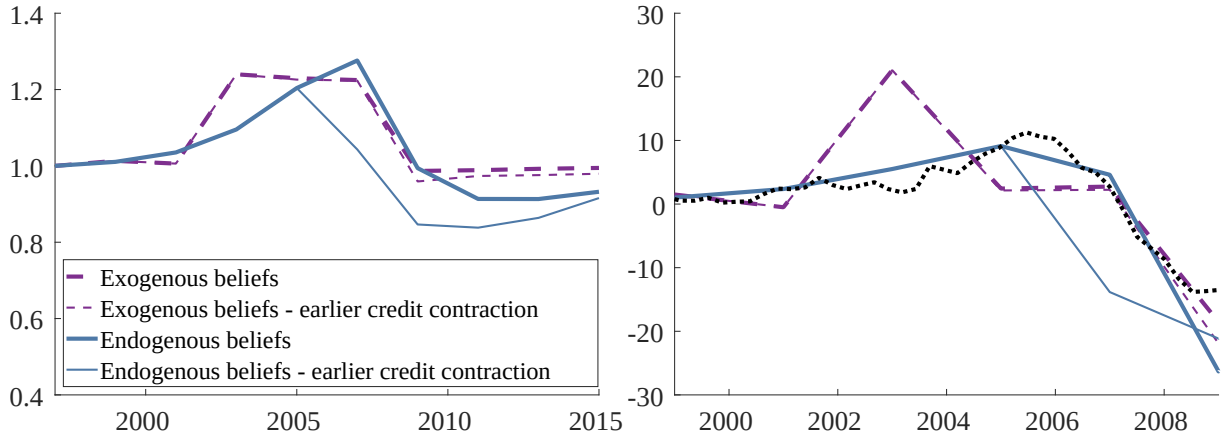
Raising interest rates to lean against high house prices is one of many prudential policies in housing boom management. Statutory borrowing limits on loan-to-value ratios and payment-to-income ratios are more widely studied as these policies directly limit leverage, thereby reducing demand for housing services and decreasing house prices.[35] However, statutory borrowing limits may be costly for younger and less well-off borrowers who remain renters for longer (Gete and Reher, 2016; Carozzi, 2019).

The statutory tighter credit conditions scenario is the same as the main housing boom simulation shown previously, but with credit conditions tightening in 2005 instead of 2009. This scenario can be interpreted as one where innovations in mortgage finance allow for looser credit conditions, but the government reacts by imposing statutory limits at pre-boom values.

Figure 9, like Figure 8, shows that house prices are more sensitive to macroprudential policy under endogenous beliefs than under exogenous beliefs. Under endogenous beliefs,

[35]See Kelly et al. (2018); Fuster and Zafar (2016); Bolliger et al. (2025).

house prices peak as credit conditions tighten, as shown via the thin solid blue line in Panel 9a. When house prices are high and credit conditions tighten, households that are younger and with relatively lower income can no longer afford to become homeowners thereby dampening demand for housing services. The decrease in house prices pushes down the house price forecast error, as shown in Panel 9b, which then dampens house prices. By contrast, under exogenous beliefs there is little difference between the simulations when credit conditions contract earlier in the boom (thin and thick dashed purple lines).



(a) House prices, 1997=1 (b) House price forecast errors, percentage points

Figure 9: House prices and forecast errors with counterfactually tighter credit conditions midway through the boom. The model's forecast error is $\log p_t - \mathbb{E}_{t-1}[a^0_{\mathcal{Z},t-1} + a^1_{\mathcal{Z},t-1} \log p_{t-1}]$.

While Figures 8 and 9 show that higher interest rates and tighter credit conditions, respectively, have large effects on house prices under endogenous beliefs, Figure 10 compares the house price response to interventions of various sizes or timing in the cases of interest rate increases or tight credit conditions, respectively. These figures show that the response of house prices does not scale linearly with the size or timing of an intervention and the response of the homeownership rate is highly non-monotonic. Appendix F.7 shows the full panel of simulations.

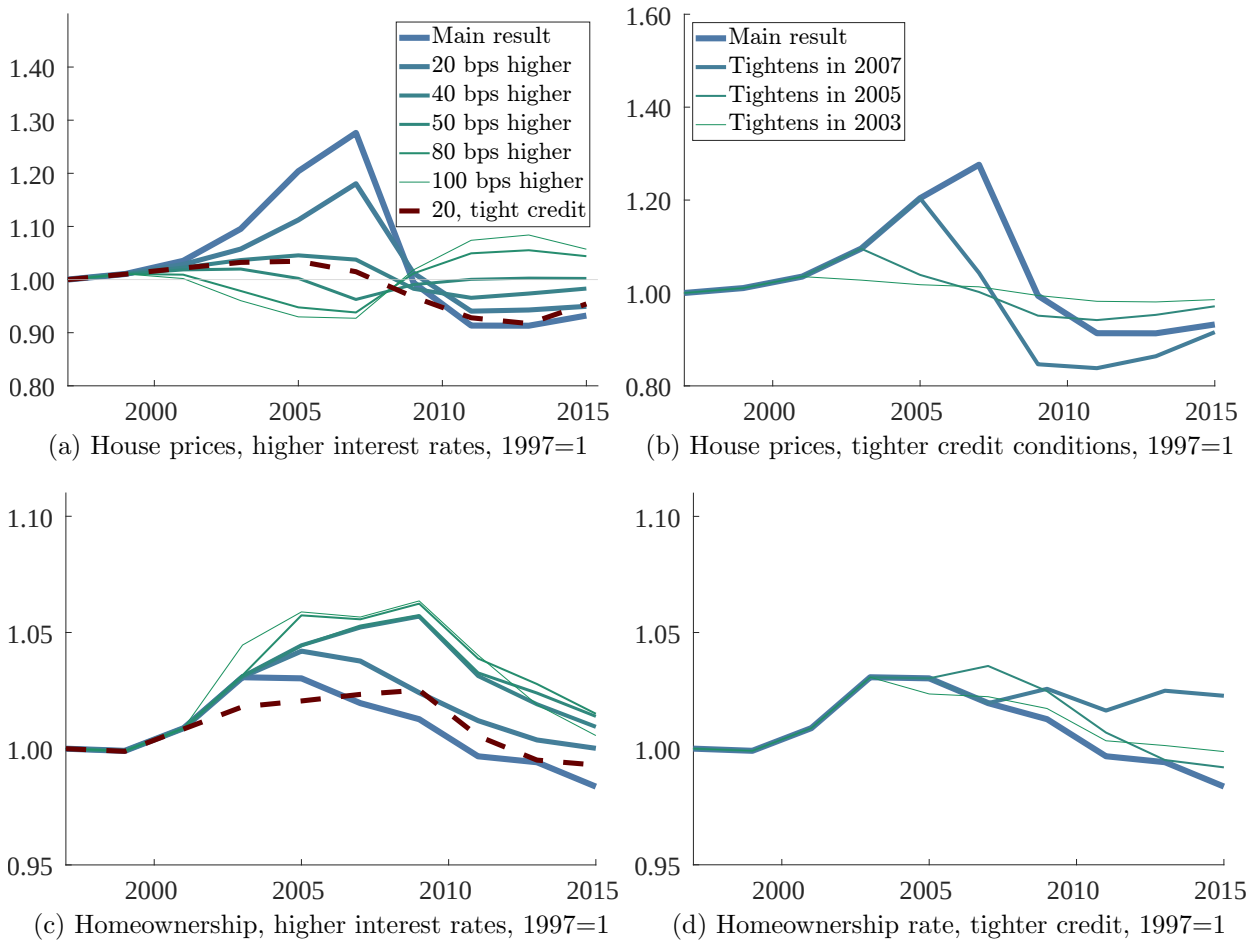
In Panel 10a, a 20 basis point increase in interest rates results in house prices that peak 18 p.p. above their pre-boom values, which is 10 p.p. below their baseline peak of 28 p.p. Doubling the size of the rate hike to 40 basis points results in a peak of 5 p.p., 23 p.p. below the peak in the main simulation. If the house price response scaled linearly, a 40 basis point increase would result in an increase of 8 p.p. instead of 5. An interest rate hike of 50 basis points causes a slight contraction in house prices of about 4 p.p.—a 32 p.p. differential between the main simulation. Doubling this rate hike to 100 basis points only results in a contraction of 7 p.p.—a 35 p.p. differential. Taken together, once house prices have contracted, as is the case with a 50 basis point hike, the risk of deepening the

contraction is quite limited. In fact, the difference between an 80 and 100 basis point hike is a few percentage points, while the difference between a 20 and 40 basis point hike is about 15 p.p.

The limited downside risk of a deepened contraction likely stems from offsetting effects of looser credit conditions and higher interest rates on demand for housing services coupled with general equilibrium feedback from house prices. As credit conditions loosen at the start of the boom, a decrease in house prices stemming from a rate hike leads to housing being more affordable for constrained agents and the homeownership rate increasing. This demand for housing then helps stabilize house prices, which leads to a less dramatic contraction. In fact, Figure 40 in Appendix F.7 confirms this by finding that most younger households prefer whichever simulation has lower house prices. That is, they prefer the prudential policy simulation with no boom until 2007 but then prefer the boom-bust simulation once house prices drop in the bust. While it may seem counterintuitive that higher interest rates lead to a higher path of homeownership, this is not the case if credit conditions did not loosen. The dashed red lines in Panels 10a and 10c show that a 20 basis point increase in interest rates without an increase in credit conditions results in a homeownership rate below that of the main simulation.

Changing the timing of the tighter credit conditions also uncovers nonlinearities in the response of house prices and the homeownership rate, as shown in Panels 10b and 10d, respectively. Because house price growth in the boom is amplified under endogenous beliefs, intervening sooner is more effective. House prices only rise by 4 p.p. relative to their pre-boom values if credit conditions are tightened immediately in 2003. If tightened in 2005, they rise by 10 p.p. instead of 8 if the response were linear. Finally, tightening in 2007 results in an increase of about 20 p.p., which is only about 8 p.p. below the main simulation where credit conditions tighten in 2009. In the bust, the house price response is non-monotonic: tightening loan-to-value constraints later in the boom leads to a larger bust in house prices. The path of the homeownership rate is actually quite similar in all the simulations shown, which suggests that short periods of looser credit conditions may be effective at increasing the homeownership rate.

Figure 10 also presents a useful comparison across types of macroprudential policies by showing that raising interest rates to lean against housing booms may be the more potent tool. While a 50 basis point increase in interest rates can eliminate a boom in house prices and lead to a contraction, only when credit conditions are tightened very late in the boom do prices fall below pre-boom levels. In fact, if credit conditions are tightened early, house prices are almost flat or a little higher than in 1997. By contrast, tightening interest rates can lead to a contraction, but one that is only mildly increasing in the size of the tightening.



(a) House prices, higher interest rates, 1997=1

(b) House prices, tighter credit conditions, 1997=1

(c) Homeownership, higher interest rates, 1997=1

(d) Homeownership rate, tighter credit, 1997=1

Figure 10: House prices and homeownership rates with counterfactually higher interest rates (Panels a and c) or tighter credit conditions (Panels b and d).

The two counterfactual scenarios studied have several caveats. First, the model is assumed to be a small open economy where interest rates are taken as given and fixed rather than endogenously responding to macroeconomic conditions as they would in other settings. Second, the rich household heterogeneity along with incomplete markets and aggregate risk necessitates simplifying some of the general equilibrium feedback from other sectors of the economy. Nonetheless, these counterfactual scenarios show that belief formation affects the sensitivity of house prices to macroprudential policy, which is important for the management of housing booms.

6 Conclusion

This paper presents a one-step solution method to endogenize beliefs about future prices in general equilibrium models with incomplete markets and aggregate risk. In an application to the U.S. housing boom of the 2000s, this method can address 1) *why* beliefs about future house prices shifted in the early 2000s to push up house prices and 2) how belief formation

affects prudential policy.

First, this paper shows that limited knowledge of the evolution of house prices in an economic expansion with loose credit conditions can account for optimistic beliefs. When agents underpredict future house prices and update their beliefs via adaptive learning, the model generates persistently positive forecast errors that are externally validated with empirical evidence obtained via a novel proxy developed from the University of Michigan Survey of Consumers. The resulting model-generated house prices match the time path, autocorrelation, and standard deviation of empirical aggregate house prices in addition to the level increase. As noted by Piazzesi and Schneider (2016), existing models typically struggle to generate any volatility in house prices, let alone match the high volatility of house price data.

Next, this paper shows that the effectiveness of prudential policies can depend on belief formation. Prudential policies like higher interest rates or tighter credit conditions can dampen endogenously formed optimistic beliefs such that the boom in house prices is shorter or non-existent. By contrast, these policies have little material effect on house prices under exogenously optimistic beliefs. Because raising interest rates may lower output, and tightening credit conditions may decrease homeownership, studying the effectiveness of these policies is important for understanding when benefits outweigh costs.

Finally, this paper's contribution of a one-step solution method to endogenize beliefs in a model with incomplete markets and aggregate risk can generalize to settings beyond adaptive learning and the 2000s housing boom. For example, Figure 41 in Appendix F.8 shows that an unexpected shift to an economic state without precedent could describe events such as the onset of the COVID-19 pandemic in 2020 where house prices also boomed.[36]

[36]Loewenstein and Willen (2024) find that beliefs explain the 2000s boom, but a combination of beliefs and preferences for larger house account for that of the 2020s.

A Appendix: Recursive Problems

A.1 Recursive Households' Problem

Households are renters r or homeowners h with distribution $\mu^r + \mu^h = \mu = 1$. In the final period, $(j = J)$, all solve the bequest problem. Let $\Upsilon_{j+1|j}(y)$ denote the distribution of $y'|y$ which embeds the Markov transition for the individual shocks $\epsilon_j(z)$ and deterministic χ_j.

Renters' Problem ($\mathbf{j < J}$)

Renters have two choices: stay renters or purchase a house and become homeowners.

$$V_j^r(b, y; \mu, Z) = \max\{V_j^{rent}(b, y; \mu, Z), V_j^{own}(b, y; \mu, Z)\} \tag{6}$$

Rent to rent: if renters choose to stay renters, they solve

$$V_j^{rent}(b, y; \mu, Z) = \max_{\{b', \tilde{h}', c\}} \left\{u_j(c, s) + \beta \mathbb{E}_{Z', \epsilon'|Z, \epsilon}[V_{j+1}^r(b', y'; \mu', Z')]\right\} \tag{6.a}$$

$$\begin{aligned} s.to. \quad c + q_b b' + \rho(\mu, Z)\tilde{h}' =& y - \mathcal{T}(y) + b \\ 0 \leq& b' \\ s =& \tilde{h}' \in \tilde{\mathcal{H}} \\ \mu' =& \Gamma_\mu(\mu; Z, Z') \\ Z' \sim& \Gamma_Z(Z) \\ y' \sim& \Upsilon_{j+1|j}(y) \end{aligned}$$

Liquid financial instruments b, income y, and age j are individual state variables. Rental housing $\tilde{h}'$ costs $\rho(\mu, Z)$ and enters the budget constraint as a cost of foregone consumption.
Rent to own: if renters choose to purchase a house and become homeowners, they solve

$$V_j^{own}(b, y; \mu, Z) = \max_{\{b', h', c\}} \left\{u_j(c, s) + \beta \mathbb{E}_{Z', \epsilon'|Z, \epsilon}[V_{j+1}^h(b', h', m', y'; \mu', Z')]\right\} \tag{6.b}$$

$$\begin{aligned} s.to. \quad c + p(\mu, Z)h' + q_b b' =& y - \mathcal{T}(y) + b + q_j(b', h', m', y; \mu, Z)m' - \kappa^m(Z) \\ 0 \leq& b' \\ 0 \leq m' \leq& \theta^{LTV}(Z) p_h(\mu, Z) h' \\ \pi_j^{min}(m') \leq& \theta^{PTI}(Z) y \\ s =& \omega h', \quad h' \in \mathcal{H} \\ \mu' =& \Gamma_\mu(\mu; Z, Z') \\ Z' \sim& \Gamma_Z(Z) \\ y' \sim& \Upsilon_{j+1|j}(y) \end{aligned}$$

Where the minimum mortgage payment is defined in equation (4) as:

$$\pi_j^{min}(m) = \left[\frac{r_m(1+r_m)^{J-j+1}}{(1+r_m)^{(J-j+1)}-1}\right] m \tag{4}$$

Renters who purchase a house have the homeowners' continuation value $V_{j+1}^h(b', h', m', y'; \mu, Z)$ which has a state space that also includes the newly originated mortgage m' and the purchased house h'.

Homeowners' Problem $(j < J)$

Homeowners have five options: sell their house and purchase a new house, sell their house and become a renter, default and become a renter, stay in their house and pay their existing mortgage, or stay in their house and refinance a new mortgage. Let $\mathbf{x} \equiv \{b, h, m\}$.

$$V_j^h(\mathbf{x}, y; \mu, Z) = \max \begin{cases} V_j^{sell,buy}(b^n, y; \mu, Z) \\ V_j^{sell,rent}(b^n, y; \mu, Z) \\ V_j^{default}(b, y; \mu, Z) \\ V_j^{stay,pay}(\mathbf{x}, y; \mu, Z) \\ V_j^{stay,refi}(\mathbf{x}, y; \mu, Z) \end{cases} \tag{9}$$

Sell to rent or buy: homeowners who sell their house solve the renters' problem (6) with financial assets b^n equal to one-period liquid financial instruments b plus the equity from the sale of the house—the sale price less housing costs and the outstanding mortgage balance.

$$b^n = b + (1 - \delta_h - \tau_h - \kappa_h)p(\mu, Z)h - (1 + r_m)m$$

Default: homeowners who default become renters, incur a utility penalty ξ, and live in the smallest rental housing unit $\tilde{h}^0$. Their property is foreclosed and the lender receives the proceeds from the sale of the housing unit less property taxes and depreciation. In default, depreciation is higher $\delta_h^\delta > \delta_h$.[37] Defaulters are not subject to recourse meaning that the lender cannot lay claim to other assets should the housing collateral not cover all of the defaulted mortgage balance.

[37] As studied by Gabriel et al. (2021), not all defaults lead to foreclosure and there is sometimes a time lag between the two events. A default is failure to meet the terms of mortgage contract and a defaulted mortgage is foreclosed when the homeowner's rights are to the property are eliminated. See Fannie Mae's glossary: https://www.knowyouroptions.com/find-resources/information-and-tools/glossary.

$$V_j^{default}(b,y;\mu,Z) = \max_{\{b',c\}} \left\{ u_j(c,\tilde{h}^0) - \xi + \beta \mathbb{E}_{Z',\epsilon'|Z,\epsilon}[V_{j+1}^r(b',y';\mu',Z')] \right\} \tag{9.a}$$

$$\begin{aligned} s.to. \quad c + q_b b' + \rho(\mu,Z)\tilde{h}^0 =& y - \mathcal{T}(y) + b \\ 0 \leq& b' \\ \tilde{h}^0 =& \min \tilde{\mathcal{H}} \\ \mu' =& \Gamma_\mu(\mu;Z,Z') \\ Z' \sim& \Gamma_Z(Z) \\ y' \sim& \Upsilon_{j+1|j}(y) \end{aligned}$$

Stay and pay: homeowners who stay in their house and pay their existing mortgage solve

$$V_j^{stay,pay}(\mathbf{x},y;\mu,Z) = \max_{\{b',m',c\}} \left\{ u_j(c,s) + \beta \mathbb{E}_{Z',\epsilon'|Z,\epsilon}[V_{j+1}^h(\mathbf{x}',y';\mu',Z')] \right\} \tag{9.b}$$

$$\begin{aligned} s.to. \quad c + q_b b' + (\delta_h + \tau_h)p(\mu,Z)h + m(1+r_m) =& y - \mathcal{T}(y,m) + b + m' \\ \pi_j^{min}(m) \leq& (1+r_m)m - m' \\ 0 \leq& m' \\ -b' \leq& \theta^{HELOC} p(\mu,Z)h \\ s =& \omega h, \quad h' = h \\ \mu' =& \Gamma_\mu(\mu;Z,Z') \\ Z' \sim& \Gamma_Z(Z) \\ y' \sim& \Upsilon_{j+1|j}(y) \end{aligned}$$

Because these homeowners stay in the same house $h' = h$, there is no housing choice. Instead, these homeowners pay the housing depreciation costs and tax $(\delta_h + \tau_h)p(\mu,Z)h$ and a mortgage payment greater than the minimum payment $\pi_j^{min}(m)$ described in equation (4). These homeowners can also borrow against the value of their house in the form of HELOCs.

Stay and refi: homeowners who stay in their house and refinance a new mortgage solve:

$$V_j^{stay,refi}(\mathbf{x}, y; \mu, Z) = \max_{\{b', m', c\}} \left\{ u_j(c, s) + \beta \mathbb{E}_{Z', \epsilon' | Z, \epsilon} [V_{j+1}^h(\mathbf{x}', y'; \mu', Z')] \right\} \tag{9.c}$$

$$\begin{aligned} s.to. \quad c + q_b b' + (\delta_h + \tau_h) p(\mu, Z) h + m(1 + r_m) =& y - \mathcal{T}(y, m) + q_j(\mathbf{x}', y; \mu, Z) m' - \kappa^m(Z) \\ 0 \leq m' \leq& \theta^{LTV}(Z) p_h(\mu, Z) h' \\ \pi_j^{min}(m') \leq& \theta^{PTI}(Z) y \\ -b' \leq& \theta^{HELOC} p(\mu, Z) h \\ s =& \omega h, \quad h' = h \\ \mu' =& \Gamma_\mu(\mu; Z, Z') \\ Z' \sim& \Gamma_Z(Z) \\ y' \sim& \Upsilon_{j+1|j}(y) \end{aligned}$$

These homeowners stay in their house h and pay housing maintenance costs $(\delta_h + \tau_h) p(\mu, Z)$. They pay off their existing mortgage m with the proceeds of a new mortgage m' that is subject to points $q_j(\mathbf{x}', y; \mu, Z) \leq 1$, the fixed mortgage origination cost $\kappa^m(Z)$, and loan-to-value and payment-to-income constraints described in equations (7)-(8).

Homeowners can thus extract equity from HELOCS or mortgage refinancing which is essentially a cash-out equity extraction because all mortgages amortize at the same rate r_m.[38] Because mortgage refinancing requires the payment of fixed origination cost $\kappa^m(Z)$ and mortgage points $q_j(\mathbf{x}', y; \mu, Z)$, it is preferred to HELOCS when homeowners are extracting relatively large amounts of equity or have a higher income such that $q_j(\mathbf{x}', y; \mu, Z)$ is closer to 1. Otherwise, HELOCs may be preferred.

Bequest Problems ($\mathbf{j = J}$)

When households exit economic life, they solve the following problems with the bequest motive $V_{J+1} = v(\flat + \underline{\flat})$ described in equation (1).

Renters' Bequest Problem

Renters must stay renters and cannot become homeowners in the final period of economic life. Their value function is thus $V_J^r(b, y; \mu, Z) = V_J^{rent}(b, y; \mu, Z)$.

[38] Wong (2019) relaxes the fixed interest rate assumption when modeling mortgage refinancing and finds that younger households or those with larger mortgages are more likely to refinance.

$$V_J^{rent}(b, y; \mu, Z) = \max_{\{b', \tilde{h}', c\}} \{u_j(c, s) + \beta v(\flat + \underline{b})\} \tag{16}$$
$$\begin{aligned} s.to. \quad c + q_b b' + \rho(\mu, Z)\tilde{h}' =& y - \mathcal{T}(y) + b \\ \flat =& b' \\ 0 \leq& b' \\ s =& \tilde{h}' \in \tilde{\mathcal{H}} \end{aligned}$$

Homeowners' Bequest Problem

Homeowners have three choices in the final period of economic life. They can sell their houses and become renters, default and become renters, or stay in their house and leave it as a bequest after paying off the residual mortgage.[39] Homeowners can neither purchase a new house nor refinance a new mortgage in the final period of economic life.

$$V_J^h(\mathbf{x}, y; \mu, Z) = \max \begin{cases} V_J^{sell,rent}(b^n, y; \mu, Z) \\ V_J^{default}(b, y; \mu, Z) \\ V_J^{stay,pay}(\mathbf{x}, y; \mu, Z) \end{cases} \tag{17}$$

Sell to rent: when homeowners sell their house and choose to become renters, they solve the renters bequest problem (16) with liquid financial instruments b^n.

$$b^n = b + (1 - \delta_h - \tau_h - \kappa_h)p(\mu, Z)h - (1 + r_m)m$$

Default: homeowners who default in the last period of economic life solve:

$$V_J^{rent}(b, y; \mu, Z) = \max_{\{b', c\}} \left\{ u_j(c, \tilde{h}^0) - \xi + \beta v(\flat + \underline{b}) \right\} \tag{17.a}$$
$$\begin{aligned} s.to. \quad c + q_b b' + \rho(\mu, Z)\tilde{h}^0 =& y - \mathcal{T}(y) + b \\ 0 \leq& b' \\ \flat =& b' \end{aligned}$$

[39] Strategic bequests and dynasties are beyond the scope of this paper, see Brandsaas (2024) for a study.

Stay and pay: homeowners who pay off their mortgage and leave their house in bequest solve:

$$V_J^{pay,stay}(\mathbf{x}, y; \mu, Z) = \max_{\{b', c\}} \left\{ u_j(c, s) + \beta \mathbb{E}_{Z', \epsilon' | Z, \epsilon} [v(\flat + \underline{b})] \right\} \tag{17.b}$$

$$\begin{aligned} s.to. \quad c + q_b b' + (\delta_h + \tau_h) p(\mu, Z) h + (1 + r_m) m' =& y - \mathcal{T}(y, m) + b \\ 0 \leq& b' \\ \flat =& b' + (1 - \kappa_h) p'(\mu'; Z, Z') h \\ s =& h\omega, \quad h \in \mathcal{H} \\ \mu' =& \Gamma_\mu(\mu; Z, Z') \\ Z' \sim& \Gamma_Z(Z) \end{aligned}$$

A.2 Final Goods, Construction, and Rental Firms' Problems

Final goods firms: V_c denotes the value function of final goods producing firms.

$$\begin{aligned} V_c(N_c; \mu, Z) =& \max_{N_c} \{Y - w(\mu, Z) N_c\} \\ s.to. \qquad Y =& \Theta(Z) N_c \\ 0 \leq& N_c \\ \mu' =& \Gamma_\mu(\mu; Z, Z') \end{aligned}$$

Because final goods consumption C is the numeraire, the price for final goods has been normalized to one. Taking the first order condition with respect to labor N_c pins down the aggregate equilibrium wage:

$$w(\mu, Z) = \Theta(Z) \tag{11}$$

Housing construction firms: V_h denotes the value function of housing construction firms.

$$\begin{aligned} V_h(N_h; \mu, Z) =& \max_{N_h} \{p(\mu, Z) H_h - w(\mu, Z) N_h\} \\ s.to. \qquad H_h =& [\Theta(Z) N_h]^\alpha \bar{L}^{1-\alpha} \\ 0 \leq& N_h \\ \mu' =& \Gamma_\mu(\mu; Z, Z') \end{aligned}$$

Taking the first order condition with respect to labor N_h and using the above equilibrium expression for aggregate wage $w(\mu, Z)$ yields:

$$\alpha \Theta(Z) p(\mu, Z) [\Theta(Z) N_h]^{\alpha - 1} \bar{L}^{1-\alpha} = \Theta(Z)$$

Re-arranging delivers the expression for housing supply:

$$\underbrace{[\Theta(Z)N_h]^{\alpha}\bar{L}^{1-\alpha}}_{\equiv H_h} = [\alpha p(\mu, Z)]^{\frac{\alpha}{1-\alpha}}\bar{L} \tag{12}$$

Rental housing sector: V_r denotes the value function of the rental housing sector with the stock of housing units owned by the rental company given as $\tilde{H}'$ and the rental ready housing units given as X'.

$$\begin{aligned} V_r(\tilde{H}, X; \mu, Z) = &\max_{\tilde{H}', X'} \big\{ [\rho(\mu, Z) - \Xi]X' - p(\mu, Z)[\tilde{H}' - (1 - \delta_h - \tau_h)\tilde{H}] \dots \\ &\dots + q_b \mathbb{E}_{Z', \epsilon'|Z, \epsilon}[V_r(\tilde{H}', X'; \mu', Z')] \big\} \\ s.to. \qquad X' \leq &\tilde{H}' \\ 0 \leq &\tilde{H}', X' \\ \mu' = &\Gamma_{\mu}(\mu; Z, Z') \end{aligned}$$

First order and envelope conditions yield the rental housing pricing equation:

$$\rho(\mu, Z) = \Xi + p_h(\mu, Z) - (1 - \delta_h - \tau_h) q_b \mathbb{E}_{Z', \epsilon'|Z, \epsilon}[p'(\mu', Z')] \tag{13}$$

See Appendix D of Kaplan et al. (2020) for a more general version that allows for financial and convertibility frictions.

Greenwald and Guren (2025) show that looser credit conditions can quantitatively account for the drop in the rent-price ratio observed throughout the housing boom when landlords are heterogeneous with disperse ownership costs. Although Kaplan et al. (2020, Appendix D) and Arslan et al. (2026) introduce these important frictions in their frameworks, doing so in this paper is complicated by aggregate risk. With costs on the conversion of newly purchased housing units into rentals, the aggregate stock of rental housing $\tilde{H}'$ becomes an additional state variable with an unknown evolution that must be approximated via a forecasting rule like that for house prices $p(\mu, Z)$.

A.3 Recursive Competitive Equilibrium

A *recursive competitive equilibrium* consists of:

- A sequence of income endowments y to households.
- Prices for owner-occupied housing $p(\mu, Z)$, rental housing $\rho(\mu, Z)$, wages $w(\mu, Z)$, and mortgages $q_j(\mathbf{x}', y; \mu, Z)$ where $\mathbf{x} = \{b, h, m\}$.
- Government parameters for constraints on loan-to-value $\theta^{LTV}(Z)$ and payment-to-income $\theta^{PTI}(Z)$, land permits $\bar{L}$, taxes $\mathcal{T}(y, m)$, and social security payments ρ_{SS}.
- Perceived laws of motion for the state space $\mu = \Gamma_\mu(\mu; Z, Z')$ where $\mu = \mu^r + \mu^h$ is the measure over the set of individual states for renters r and homeowners h. Let $\mu = 1$ and the respective state spaces be defined as $\mathcal{X}^h = (\mathcal{B} \times \mathcal{H} \times \mathcal{M} \times \mathcal{E} \times \mathcal{J})$ and $\mathcal{X}^r = (\mathcal{B} \times \mathcal{E} \times \mathcal{J})$.
- Value functions for renters $V_j^r(b, y; \mu, Z)$ with policy functions for consumption, liquid financial instruments, owner-occupied housing, and rental housing $\{c, b', h', \tilde{h}'\}$ solve the renters' problem. Value functions for homeowners $V_j^h(\mathbf{x}, y; \mu, Z)$ with policy functions for consumption, liquid financial instruments, owner-occupied housing, rental housing, and mortgages $\{c, b', h', \tilde{h}', m'\}$ solve the homeowners' problem.

Markets clear:

- The lending sector maximizes profits and the mortgage market clears loan-by-loan with homeowner specific pricing functions $q_j(\mathbf{x}', y; \mu, Z)$.

$$\int_{\mathcal{X}^h} m' d\mu^h = M'$$

- The rental sector maximizes profits with policy function $\tilde{H}'$ at price $\rho(\mu, Z)$ to supply rental housing.

$$\underbrace{\int_{\mathcal{X}^r} \tilde{h}' d\mu^r}_{\text{renters}} + \underbrace{\int_{\mathcal{X}^h} \tilde{h}' d\mu^h}_{\substack{\text{sellers} \\ \& \\ \text{defaulters}}} = \tilde{H}'$$

- Housing construction firms maximize profits with policy functions for labor demand N_h and the supply of new housing units H_h. There is a single housing price $p(\mu, Z)$ that clears the housing market so that housing outflows (LHS) equal inflows (RHS).

$$\underbrace{\tilde{H}' - (1 - \delta_h)\tilde{H}}_{\substack{\text{rental sector} \\ \text{housing purchases}}} + \underbrace{\int_{\mathcal{X}^r} h' d\mu_r}_{\substack{\text{purchases} \\ \text{by renters}}} + \underbrace{\int_{\mathcal{X}^h} h' d\mu_h}_{\substack{\text{purchases} \\ \text{by owners}}}$$

$$= \underbrace{H_h - \delta_h \int_{\mathcal{X}^h} h d\mu^h}_{\substack{\text{new construction} \\ \text{less depreciation}}} + \underbrace{\int_{\mathcal{X}^h} h[\mathbb{1}_{sell} + \mathbb{1}_{default}(1 - \delta_h^d + \delta_h)] d\mu^h}_{\text{sales by owners \& sales of foreclosures}} + \underbrace{\int_{\mathcal{X}^h} \mathbb{1}_{bequest} h' d\mu^h}_{\text{estate sales}}$$

- Final goods firms maximize profits so that the labor market clears at $\Theta(Z) = w(\mu, Z)$ with the total labor supply from housing construction and final goods firms $N_h + N_c$ normalized to 1.

$$\int_{\mathcal{X}} \exp(\chi_j + \epsilon_j(z)) d\mu_{\mathcal{J}_{work}} = \underbrace{N_h + N_c}_{1}$$

- The government collects revenue from income taxes $\mathcal{T}(y, m)$, property taxes τ_h, and the sale of land permits $[p(\mu, Z)H_h - w(\mu, Z)N_h]$ to finance expenditures on social security income for retirees y_{ret} and non-valued government spending G:

$$\mathcal{T}(y, m) + \tau_h p(\mu, Z) \int_{\mathcal{X}^h} h d\mu^h + [p(\mu, Z)H_h - w(\mu, Z)N_h] = \rho_{ss} \int_{\mathcal{X}} y_{ret} d\mu_{\mathcal{J}_{ret}} + G$$

- Profits or losses from the financial and rental sectors are expressed as net exports:

$$\begin{aligned} NX = & \underbrace{\int_{\mathcal{X}^r} [b - q_b b'] d\mu^r + \int_{\mathcal{X}^h} [b - q_b b' \mathbb{1}_{[b'>0]} - \left(r_b(1+\iota)\right)^{-1} b' \mathbb{1}_{[b'<0]}] d\mu^h}_{\text{lenders' net expenses from liquid financial instruments}} \\ & \ldots + \underbrace{\int_{\mathcal{X}^h} [(1 + r_m)m + q_j(\mathbf{x}', y; \mu, Z)m'] d\mu^h}_{\text{lenders' net revenue from mortgages}} \\ & \ldots + \underbrace{(\rho(\mu, Z) - \Xi)\tilde{H}'}_{\substack{\text{rental sector revenue} \\ \text{less operating costs}}} - \underbrace{p(\mu, Z)[\tilde{H}' - (1 - \delta_h - \tau_h)\tilde{H}]}_{\substack{\text{rental sector housing} \\ \text{purchases}}} \end{aligned}$$

- The aggregate resource constraint is satisfied where household consumption C, government spending G, net exports NX equal output Y less housing and mortgage adjustment costs.

$$\begin{aligned} & \int_{\mathcal{X}^h} c d\mu^h + \int_{\mathcal{X}^r} c d\mu^r + G + NX + \Xi \tilde{H}' \\ = & Y - \kappa p(\mu, Z) \int_{\mathcal{X}^h} h(\mathbb{1}_{sell} + \mathbb{1}_{default}) d\mu^h - \iota r_b \int_{\mathcal{X}^h} (m + b\mathbb{1}_{\{b<0\}}) d\mu^h \\ & - (\zeta + \kappa^m) \int_{\mathcal{X}^h} m'(\mathbb{1}_{buy} + \mathbb{1}_{refi}) d\mu^h \end{aligned}$$

- Consistency is satisfied and perceived laws of motion of the state space $\mu' = \Gamma_\mu(\mu, Z, Z')$ is consistent with individual behavior

B Appendix: One-Step Solution Method

B.1 An Example of the Number of Grids in the One-Step Method

The example below illustrates how using as many additional dimensions as there are future aggregate states has the advantage of state-dependent beliefs. If a single additional grid were used instead, it would imply that beliefs depart from their near-rational counterparts in the same manner in all possible future states of the economy. This would imply no variation in optimism or pessimism in booms, busts, and normal times. This is especially important in light of the findings of Roschitsch and Twieling (2025), who find that households are relatively more pessimistic in the bust than they are optimistic in the boom.

Case 1: Standard Krusell and Smith (1998), no beliefs

A simplified recursive value function for households aged j with individual state vector $\boldsymbol{x}$, and current and future aggregate states $\mathcal{Z} = \{Z, Z'\}$ solves a problem of the form:

$$V_j(\boldsymbol{x}; p(Z), Z) = \max_{\{c', b', h'\}} \left\{ u_j(c, h') + \beta \mathbb{E}_{Z', \epsilon' | Z, \epsilon} \left[V_{j+1}\Big(\boldsymbol{x}'; p'(p(Z); \mathcal{Z}), Z'\Big) \right] \right\}$$

$$\text{where} \quad \log p'(p(Z); \mathcal{Z}) = a^0_{\mathcal{Z}} + a^1_{\mathcal{Z}} \log p(Z)$$

Notice that $\log p'(p(Z); \mathcal{Z})$ is pinned down by $\{\boldsymbol{a}_{\mathcal{Z}}, p(Z), Z, Z'\}$ so that no additional grid is needed.

Case 2: One-step method, single belief grid

for the one-step method, suppose that beliefs can vary from their near-rational fixed point values $\boldsymbol{a}_{\mathcal{Z}}$ by φ so that there can be optimism or pessimism, $\varphi > 0$ or $\varphi < 0$. This variation need not be linear or the form chosen here for illustrative purposes. Then, future prices are pinned down by $\{p(Z), Z, Z', \varphi\}$ and one additional grid for φ must be added as a state variable.

$$\log p'(p(Z); \mathcal{Z}, \varphi) = a^0_{\mathcal{Z}} + a^1_{\mathcal{Z}} \log p(Z) + \varphi$$

Assuming that agents do not internalize that they update beliefs each period, $\varphi' = \varphi$, then the value function is:

$$V_j\Big(\boldsymbol{x}; p(Z), Z, \varphi\Big) = \max_{\{c', b', h'\}} \left\{ u_j(c, h') + \beta \mathbb{E}_{Z', \epsilon' | Z, \epsilon} \left[V_{j+1}\Big(\boldsymbol{x}'; p'(p(Z); \mathcal{Z}, \varphi), Z', \varphi\Big) \right] \right\}$$

With only one additional grid for beliefs, expected future prices are not state-dependent. That is, agents are either optimistic or pessimistic in all future economic states. The next case relaxes this assumption and allows for state-dependent beliefs, $\varphi(Z')$.

Case 3: One-step method, state-dependent belief grids

Allowing beliefs in the one-step method to vary based on current and future aggregate states expands the additional grids from φ to $\#Z'$, which is $\{\varphi(Z'_{low}), \varphi(Z'_{high})\}$ in this setting. When $\varphi(Z'_{low}) \neq \varphi(Z'_{high})$, there could be different degrees of optimism or pessimism for booms or non-booms. Future prices then vary by $\{p(Z), Z, Z', \varphi(Z')\}$.

$$\log p'(p(Z); \mathcal{Z}, \varphi(Z')) = a^0_{\mathcal{Z}} + a^1_{\mathcal{Z}} \log p(Z) + \varphi(Z')$$

Assuming that agents do not internalize that they update their beliefs, $\varphi(Z') = \varphi'(Z'')$. Because agents need to take into account that $\varphi(Z'_{low}) \neq \varphi(Z'_{high})$, there must be two additional belief grids (or $\#Z'$ in the general case).

$$V_j\Big(\boldsymbol{x}; p(Z), Z, \varphi(Z'_{low}), \varphi(Z'_{high})\Big) = \max_{\{c', b', h'\}} \Bigg\{ u_j(c, h') + ...$$

$$\beta \mathbb{E}_{Z', \epsilon' | Z, \epsilon} \Big[V_{j+1}\Big(\boldsymbol{x}'; p'(p(Z); \mathcal{Z}, \varphi(Z')), Z', \varphi(Z'_{low}), \varphi(Z'_{high})\Big)\Big] \Bigg\}$$

Evaluating the continuation value $V_{j+1}(\cdot)$ at Z'_{high} and Z'_{low} could yield very different outcomes in terms of renting, selling, defaulting because the households problem is highly non-linear with many discrete choices:

$$V_j\Big(\boldsymbol{x}; p(Z), Z, \varphi(Z'_{low}), \varphi(Z'_{high})\Big) = \max_{\{c', b', h'\}} \Bigg\{ u_j(c, h') + ...$$

$$\beta \pi_{Z, Z'_{high}} \sum_{\epsilon'} \pi_{\epsilon, \epsilon'} \Big[V_{j+1}\Big(\boldsymbol{x}'; p'(p(Z); Z, Z'_{high}, \varphi(Z'_{high})), Z'_{high}, \varphi(Z'_{low}), \varphi(Z'_{high})\Big)\Big] ...+$$

$$\beta \pi_{Z, Z'_{low}} \sum_{\epsilon'} \pi_{\epsilon, \epsilon'} \Big[V_{j+1}\Big(\boldsymbol{x}'; p'(p(Z); Z, Z'_{low}, \varphi(Z'_{low})), Z'_{low}, \varphi(Z'_{low}), \varphi(Z'_{high})\Big)\Big] \Bigg\}$$

Simply taking the expectation of the belief component, $\mathbb{E}_{Z'|Z}[\varphi(Z')] = \pi_{Z, Z'_{low}} \varphi(Z'_{low}) + \pi_{Z, Z'_{high}} \varphi(Z'_{high})$, so that the problem collapses onto a single grid as in case 2, could result in entirely different continuation values and thus housing choices. In fact, suppose $\mathbb{E}_{[Z'|Z]}[\varphi(Z')] = 0$, such that optimism in the high-income state is equal to pessimism in the low-income state, then expected future house prices would simply be at their near-rational values and there would be no role for beliefs, even though there may be large degrees of

optimism or pessimism.

$$\beta\pi_{Z,Z'_{low}}\left[V_{j+1}\Big(\boldsymbol{x}';\mathbb{E}_{Z'|Z}\left[\log p'(p(Z);\mathcal{Z},\varphi(Z'))\right],Z'_{low},\mathbb{E}_{Z'|Z}\left[\log p'(p(Z);\mathcal{Z},\varphi(Z'))\right]\Big)\right]\Bigg\}$$

As the average expected future price could lead to different continuation values then the actual state-dependent price.

B.2 Computational Algorithm

Both the multi- and one-step algorithms with either fixed or time-varying forecasting coefficients, respectively, rely on the assumption that agents keep track of house prices via a log-linear forecasting rule for each combination of current and future aggregate states $\mathcal{Z} \in \{Z, Z'\}$ as an approximation of the entire distribution over individual states μ and its law of motion $\Gamma_\mu(\mu;\mathcal{Z})$. While the aggregate capital stock in Krusell and Smith (1998) is predetermined, aggregate housing is not and requires explicit market clearing, which makes the computational algorithm in this paper closer to that of Krusell and Smith (2006).

$$\log p'(p(Z);\mathcal{Z}) = a^0_{\mathcal{Z}} + a^1_{\mathcal{Z}}\log p(Z) \quad \Longleftrightarrow \quad \mu' = \Gamma_\mu(\mu;\mathcal{Z})$$

A log-linear forecasting rule for house prices is standard in macro housing applications and assumes that the conditional sample log means of house prices $\mu^P_{\mathcal{Z}} = a^0_{\mathcal{Z}}/(1-a^1_{\mathcal{Z}})$ derived from forecasting coefficients $\boldsymbol{a}_{\mathcal{Z}}$, coupled with the current price $p(\mu, Z)$ and aggregate states $\{Z, Z'\}$ is sufficient information for agents to predict future prices such that they do not need to keep track of the entire wealth distribution of households.

The model is first solved with the multi-step fixed-point method to obtain fixed coefficients. Next, the model is solved with the one-step method that endogenizes beliefs by tracking time-varying coefficients throughout the simulation. Because the one-step method with endogenous beliefs requires as many additional grids as there are aggregate states in the one-step method, the households' problem must be solved $3^2 = 9$ times for two belief grids with 3 points each.[40] Fortunately, the increase in dimensionality is not too much of a computational burden because the belief grids can be parallelized.[41] The model has been solved on the high-throughput computing clusters at Indiana University (Karst and Carbonate), the University of Texas at Dallas (BigTex), and the Federal Reserve Board.

[40] Experiments with 5 grid points instead of 3 show that although individual aggregate housing demand may vary, the demand schedule for housing is roughly similar.

[41] Allowing independent aggregate shocks to credit conditions and income would require defining four belief grids instead of two. With 3 points for each grid, the households' problem would then need to be solved $3^4 = 81$ times instead of $3^2 = 9$ times. Computation times increase drastically as a result of increasing the dimensions of the arrays and interpolation in two additional dimensions.

1. Define grids over ages $j = 1, \ldots, J$, liquid financial instruments b, liquid financial instrument choices b', rental housing choices $\tilde{h}'$, owner-occupied housing h, owner-occupied housing choices h', mortgage balances m, mortgage choices m', aggregate income $\Theta(Z) \in \{\Theta(Z_{high}), \Theta(Z_{low})\}$, aggregate credit conditions $\mathcal{C}(Z) \in \{\mathcal{C}(Z^{tight}), \mathcal{C}(Z^{loose})\}$, and house prices p.

2. Define coefficients:
   - Fixed: to pin down next period house prices, guess coefficients $\boldsymbol{a}_{\mathcal{Z}}$ for each $\mathcal{Z}$ for a total of $\#Z' = 2$ vectors of coefficients. If credit conditions are Markov this is 4.[42]

$$\log p'(p(Z); \mathcal{Z}) = a^0_{\mathcal{Z}} + a^1_{\mathcal{Z}} \log p(Z)$$

   - Beliefs: define additional grids over future house prices in each future state $\{p'(Z'_{low}), p'(Z'_{high})\}$ for a total of $\#Z' = 2$ additional grids. Whether these are grids over expected future prices $\{p'(Z'_{low}), p'(Z'_{high})\}$ or beliefs $\{\boldsymbol{a}_{Z,Z'_{low}}, \boldsymbol{a}_{Z,Z'_{hight}}\}$ is equivalent.

   With a risk-neutral lending sector, the mortgage price $q_j(\mathbf{x}', y; p(Z), Z)$ is pinned down by the lenders' mortgage pricing function in equation (10). The price for financial assets q_b is taken as given. The price of rental housing $\rho(\mu, Z)$ is pinned down by the rental sector's maximization problem (13) and aggregate wages $w(\mu, Z)$ are pinned down by the final goods firms' maximization problem (11).

3. Solve the households' problem at each point on the house price grid p (and at each point on the belief grids if endogenizing beliefs). I use value function iteration with grid search for housing and mortgages and a golden section solver for liquid financial instruments.

4. Simulate a long time series of aggregate states Z_t for $t = 1, \ldots, T$ where T =5000 with a burn in period of 100.

5. Fix an initial distribution of liquid financial instruments, idiosyncratic incomes, and ages assuming that all households are initially renters $\mu_t \in \mathcal{B} \times \mathcal{E} \times \mathcal{J}$ for $t = 1$. Beginning realizations of $\{b, \epsilon, j\}$ denote liquid financial instruments, income realizations, and ages, respectively. There are N =125000 households.

   Some notes on initialization:
   - endowments of liquid financial instruments equal zero for all households at $t = 1$.

[42] The forecasting equation is $\log p_{\mathcal{Z},t+1} = \mathbb{1}_{\mathcal{Z}} a^0_{\mathcal{Z}} + a^1_{\mathcal{Z}} \log p_{Z_t}$ for $\mathcal{Z} \in \{Z, Z'\}$ which is slightly different, but practically equivalent to equation (14), $\log p_{\mathcal{Z},t+1} = a^0_{\mathcal{Z}} + a^1_{\mathcal{Z}} \log p_{\mathcal{Z},t}$.

- because all agents are renters at $t = 1$, there is no initial housing endowment
- initial household ages are drawn from a uniform distribution[43] $j \sim [1, \ldots, J]$
- when household i reaches the final period of economic life, $j_{i,t} = J$, a new household replaces them with $j_{i,t+1} = 1$ as renters with no mortgage debt $m_{i,t} = 0$. These new households inherit a random draw of liquid financial instruments $b_{i,t}$ that are correlated with individual income so that household with a higher income are more likely to inherit larger quantities.
- if solving with beliefs, set calibrated initial lagged coefficients $\boldsymbol{a}_0$.

6. Simulate
   - Fixed: coefficients are already set at $\mathbf{a}_{\mathcal{Z}}$ at the beginning of the problem.
   - Beliefs: compute period t coefficients to pin down $p'(p(Z); \mathcal{Z})$. Coefficients for each future aggregate state $\boldsymbol{a}_{Z_t, Z'_{low}}$ and $\boldsymbol{a}_{Z_t, Z'_{high}}$ are necessary because the value of Z' is not yet known. For this reason, endogenous beliefs require as many additional grids as there are aggregate states.

7. Compute the aggregate demand schedule for housing at each point on the house price grid p from the housing policy functions for renters $\tilde{h}'$ and homeowners h'. Interpolate over $b_{i,t}$ for renters and $\{b_{i,t}, h_{i,t}, m_{i,t}\}$ for homeowners. Average by housing type:

$$\begin{aligned} \text{Renters:} \quad \tilde{H}_{t+1}(p, \boldsymbol{X}_t) &= \frac{1}{N} \sum\nolimits_{i=1}^{N^r} \tilde{h}'(b_{i,t}, y_{i,t}, j_{i,t}; p, \boldsymbol{X}_t) \\ \text{Homeowners:} \quad H_{t+1}(p, \boldsymbol{X}_t) &= \frac{1}{N} \sum\nolimits_{i=1}^{N^h} h'(b_{i,t}, h_{i,t}, m_{i,t}, y_{i,t}, j_{i,t}; p, \boldsymbol{X}_t) \end{aligned}$$

   The number of aggregate states in the vector $\boldsymbol{X}_t$ depends on the algorithm:
   - Fixed: $\boldsymbol{X}_t = \{Z_t\}$
   - Beliefs: $\boldsymbol{X}_t = \{Z_t, p'(Z'_{low}), p'(Z'_{high})\}$ which requires additional interpolation over $p'(Z'_{low})$ and $p'(Z'_{high})$.

8. Compute excess demand for aggregate housing at each point on the house price grid p and recover the equilibrium house price $p^*_t(Z_t)$ from the market clearing condition:

$$H_{t+1}(p^*_t(Z_t), \boldsymbol{X}_t) + \tilde{H}_{t+1}(p^*_t(Z_t), \boldsymbol{X}_t) = H_h + (1 - \delta_h)[H_t + \tilde{H}_t]$$

9. Interpolate the individual policy functions b', h', $\tilde{h}'$, and m' at the equilibrium house price $p^*_t(Z_t)$ and then average to calculate the aggregate equilibrium quantities.

[43] This assumption follows Kaplan and Violante (2014). Matching the distribution of ages would introduce generational demographic changes into equilibrium dynamics which is beyond the scope of this paper.

$$
\begin{aligned}
B^*_{t+1}(p^*_t(Z_t), \boldsymbol{X}_t) =& \frac{1}{N} \sum\nolimits_{i=1}^{N} b'(b_{i,t}, h_{i,t}, m_{i,t}, y_{i,t}, j_{i,t}; p^*_t(Z_t), \boldsymbol{X}_t) \\
\tilde{H}^*_{t+1}(p^*_t(Z_t), \boldsymbol{X}_t) =& \frac{1}{N} \sum\nolimits_{i=1}^{N^r} \tilde{h}'(b_{i,t}, y_{i,t}, j_{i,t}; p^*_t(Z_t), \boldsymbol{X}_t) \\
H^*_{t+1}(p^*_t(Z_t), \boldsymbol{X}_t) =& \frac{1}{N} \sum\nolimits_{i=1}^{N^h} h'(b_{i,t}, h_{i,t}, m_{i,t}, y_{i,t}, j_{i,t}; p^*_t(Z_t), \boldsymbol{X}_t) \\
M^*_{t+1}(p^*_t(Z_t), \boldsymbol{X}_t) =& \frac{1}{N} \sum\nolimits_{i=1}^{N} m'(b_{i,t}, h_{i,t}, m_{i,t}, y_{i,t}, j_{i,t}; p^*_t(Z_t), \boldsymbol{X}_t)
\end{aligned}
$$

Again, the number of aggregate states $\boldsymbol{X}_t$ depends on the algorithm:

- Fixed: $\boldsymbol{X}_t = \{Z_t\}$
- Beliefs: $\boldsymbol{X}_t = \{Z_t, p'(Z'_{low}), p'(Z'_{high})\}$

$$
\boldsymbol{p'}(p^*_t(Z_t)) = \begin{pmatrix} \exp\{a^0_{Z_t,Z'_{low}} + a^1_{Z_t,Z'_{low}} \log(p^*_t(Z_t))\} \\ \exp\{a^0_{Z_t,Z'_{high}} + a^1_{Z_t,Z'_{high}} \log(p^*_t(Z_t))\} \end{pmatrix}
$$

10. Simulate for $t = 1, \ldots, T$ periods by repeating steps (6)-(9).

11. Compare coefficients:
    - Fixed: partition the time series of market clearing house prices $\{p^*_t(Z_t)\}^T_{t=burn}$ by $\mathcal{Z} \in \{Z, Z'\}$ to generate $\#Z^2 = 4$ sub-samples. Estimate new forecasting coefficients for each sub-sample $\mathcal{Z}$ via ordinary least squares regression:

$$
\boldsymbol{a}^{new}_{\mathcal{Z}} = \left(\sum_{t=burn}^{T} \boldsymbol{x}_{\mathcal{Z},t} \boldsymbol{x}'_{\mathcal{Z},t} \right)^{-1} \sum_{t=burn}^{T} \boldsymbol{x}_{\mathcal{Z},t} \log p_{\mathcal{Z},t+1}
$$

      Repeat steps (2) - (11) until coefficients converge, $\boldsymbol{a}^{new}_{\mathcal{Z}} \approx \boldsymbol{a}_{\mathcal{Z}}$
    - Beliefs: verify that the time-varying coefficients $\{\boldsymbol{a}_{\mathcal{Z},t}\}^T_{t=burn}$ from step (9) converge to their near-rational counterparts $\boldsymbol{a}_{\mathcal{Z}}$.

Chipeniuk et al. (2022) propose a solution check called auctioneer iteration to avoid relying on potentially misleading R^2 statistics as measures of accuracy. Their method adds the following steps after coefficients are obtained via a fixed point:

12. Use converged $\boldsymbol{a}_{\mathcal{Z}}$ and a price grid that is a subset of simulated prices, $p =\subset \{p^*_t(Z_t)\}^T_{t=1}$.
13. Repeat steps (1)-(11) to obtain new converged coefficients and new prices $\{p^{new}_t(Z_t)\}^T_{t=1}$.
14. Stop if $\{p^{new}_t(Z_t)\}^T_{t=1} \approx \{p^*_t(Z_t)\}^T_{t=1}$, otherwise go to step (1).

B.3 Computational Details

The known forecasting coefficients $\boldsymbol{a}_{\mathcal{Z}}$ shown in Table 5 are solved as a fixed point in the model's stochastic steady state characterized by tight aggregate credit conditions and fluctuations in aggregate income. The model is then solved under loose aggregate credit conditions and fluctuations in income. The coefficients have similar values in all combinations of current and future income states which is consistent with Kaplan et al. (2020) and suggests that shocks to aggregate income do not have much effect on house price forecasts. Approximating the sample mean of house prices $\bar{p}_{\mathcal{Z}}$ via the coefficients points to slightly more variation in the highest and lowest values attained in the transitions between income states.

Income states	$a^0_{\mathcal{Z}}$	$a^1_{\mathcal{Z}}$	$\bar{p}_{\mathcal{Z}} \approx \frac{a^0_{\mathcal{Z}}}{1-a^1_{\mathcal{Z}}}$	R^2	Den Haan	Chipeniuk et. al
High, High	-0.102	0.838	0.53	0.9999	0.01	0.01
High, Low	-0.111	0.838	0.50			
Low, High	-0.091	0.847	0.55			
Low, Low	-0.100	0.847	0.52			

Table 5: Converged forecasting coefficients under tight aggregate credit conditions.

Like Kaplan et al. (2020) and Favilukis et al. (2017), the model's R^2 statistic is close to 1 which suggests adequate goodness of fit standards. Because R^2 statistics can be misleading gauges of accuracy, Den Haan (2010) and Chipeniuk et al. (2022) propose alternative tests which both come in sufficiently low at 0.01.

C Appendix: Calibration

C.1 Parameter Values

The model's parameters are set to resemble the U.S. economy in the late 1990s with the cross-sectional moments from the 1998 Survey of Consumer Finances to nest Kaplan et al. (2020).

Demographics: Each model period is equal to two years. Households begin economic life at age 21 ($j = 1$), retire at age 65 ($J^{ret} = 23$), and exit economic life at age 79 ($J = 30$).[44]

Preferences: The elasticity of substitution between goods consumption and housing expenditures $1/\gamma$ is set to 1.25 based on the estimates of Piazzesi et al. (2007). The elasticity of intertemporal substitution equals 0.5 by setting $\sigma = 2$. A McClements (1977) scale sets the consumption equivalence scale $\{e_j\}$ to match the OECD average number of children across different age groups. The discount factor β is set to replicate the 1998 ratio of aggregate net worth to annual labor income of 5.5 for which the model comes in slightly below at 4.9.

Two parameters, the strength of the bequest motive ψ and the extent to which bequests are luxuries $\underline{b}$, pin down the warm-glow bequest motive given in equation (1). The strength of bequests ψ is chosen to replicate the ratio of net worth at age 75 to age 50 of 1.55 which indicates the importance of bequests as a saving motive. The model comes in slightly below at 1.48. The luxuriousness of bequests $\underline{b}$ is chosen so that households in the bottom half of the model's wealth distribution do not leave bequests, as observed in the data.

Homeowners' additional utility ω sets the average homeownership rate which was 66 percent in the late 1990s and is slightly higher at 68 percent in the model. Defaulters' disutility ξ is set to match the average foreclosure rate in the late 1990s of 0.5 percent and is slightly lower at 0.02 percent in the model.

Income endowments: The deterministic life-cycle component of earning $\{\chi_j\}$ is from Kaplan and Violante (2014). Stochastic individual earnings $\epsilon_j(z)$ in equation (5) follow an AR(1) process in logs with an annual persistence of 0.97, annual standard deviation of 0.2, and an initial standard deviation of 0.42. The variance of log earnings rises by 2.5 between ages 21 and 64 which follows Heathcote et al. (2010). Bequested inheritances are correlated with individual income so that higher income households are more likely to inherit more.

[44] Krueger and Kübler (2004) find that computational accuracy decays rapidly if the length of economic life extends beyond 30 periods when using aggregate risk in a life-cycle economy.

Housing: The following three parameters discipline the size of owner-occupied housing units $\mathcal{H}$: the size of the smallest unit h^0, the number of house sizes available $\#\mathcal{H}$, and the gap between housing sizes. The values of these parameters are obtained from targeting the 10th, 50th, and 90th percentiles of the distribution of the ratio of housing net worth to total net worth. The model matches the 10th percentile but comes in a bit below the 50th and 90th percentiles, as is common this class of model. The size of rental housing units $\tilde{\mathcal{H}}$ is chosen to target a ratio of median owner-occupied to rental housing of 1.5 square feet per person as detailed in Chatterjee and Eyigungor (2015) and a ratio of the average earnings of homeowners to renters equal to 2.1. The model generates moments that are mostly in line with these empirical targets.

The maintenance cost of housing that offsets depreciation δ_h equals 0.015 and replicates the empirical depreciation rate of the housing stock of 1.5 percent per year.[45] Defaulted housing has a higher depreciation rate δ_h^d equal to 0.22 to account for the loss of value from foreclosure. The linear transaction cost of housing adjustments κ_h equals 7 percent which falls within the 6 to 12 percent range estimated by Quigley (2002).[46] In line with the data, 9 percent of homeowners in the model buy or sell houses each year. The relative cost of renting versus owning is determined in part by the operating cost of the rental company Ξ and is particularly salient for young households. Ξ is chosen to match the 39 percent homeownership rate of households younger than 35 and comes in slightly lower at 34 percent.

Housing construction technology α is set so that the price elasticity of housing supply $\alpha/(1-\alpha)$ equals 1.5, the median among MSAs in Saiz (2010). Land permits $\bar{L}$ pin down employment in the construction sector at 5% of total employment which is consistent with the 1998 employment share of construction measured by the Bureau of Labor Statistics.

Financial instruments: The risk-free rate r is 2.5 percent per year and the lending wedge ι is 0.33 so that the interest rate on loans r_b is equal to 3 percent per year to replicate the gap between the average rate on 30-year fixed-term mortgage and the 10-year Treasury rate in the late 1990s. The maximum HELOC value θ^{HELOC} is 0.2.

Government: The property tax τ_h is set to 1% per year which is the median tax rate across U.S. states according to the Tax Policy Center. The income tax function in equation (3) follows the functional form of Heathcote et al. (2017). The parameter τ_y^0 indicates the average level of taxation and is set so that aggregate income tax revenues are 20% of income.

[45] Kaplan et al. (2020, p. 18) use the Bureau of Economic Analysis' Table 7.4.5 which details the consumption of fixed capital of the housing sector divided by the stock of residential housing at market value.

[46] Ghent (2012) finds a value of 13 percent and Ngai and Sheedy (2020) settle on 10 percent suggesting that 7 percent may be on the lower end of the established range.

The parameter τ_y^1 measures the degree of progressivity of the tax and transfer system and is set at 0.15 based on the estimates of Heathcote et al. (2017). The share of mortgage interest deducted ϱ is set at 0.75 to so that only the first \$1,000,000 of mortgage debt is deductible. Social security payments in equation (2) maintain income heterogeneity by scaling the last realization of earnings y^w_{Jret-1} by the replacement rate ρ_{ss}. Kaplan et al. (2020) compute the ratio of average benefits to average lifetime earnings and find a replacement rate of 0.4.

Interpretation	Parameter	Value	Annualized
Demographics			
Maximum age	J	30	N
Retirement age	J^{ret}	23	N
Preferences			
Inverse elasticity of substitution	γ	0.8	N
Risk aversion	σ	2	N
Discount factor	β	0.97	Y
Strength of bequest motive	ψ	100	N
Extent of bequests as a luxuries	$\underline{b}$	7.7	N
Taste for housing	ϕ	0.13	N
Additional utility from owning	ω	1.015	N
Utility cost of foreclosure	ξ	0.8	N
Individual income			
Deterministic income	$\{\chi_j\}$	Kaplan & Violante (2014)	N
Annual persistence, ind. income	ρ_ϵ	0.97	Y
Annual st. dev., ind. income	σ_ϵ	0.20	Y
Initial st. dev., ind. income	σ_{ϵ_0}	0.42	Y
Distribution of bequest to new hhs	$b_{j=1} = b'_{j=J}$	Kaplan & Violante (2014)	N
Housing			
Owner-occupied housing unit sizes	$\mathcal{H}$	$\{1.5, 1.92, 2.46, \ldots$ $\ldots 3.15, 4.03, 5.15\}$	N
Rental housing unit sizes	$\tilde{\mathcal{H}}$	$\{1.125, 1.5, 1.92\}$	N
Depreciation rate of housing	δ_h	0.015	Y
Housing loss in foreclosure	δ_h^d	0.22	Y
Housing transaction cost	κ_h	0.07	N
Operating cost of rental company	Ξ	0.002	N
Housing supply elasticity	$\alpha/(1-\alpha)$	1.5	N
New land permits	$\bar{L}$	0.311	N
Financial instruments			
Risk-free interest rate	r	0.025	Y
Interest rate wedge on borrowing	ι	0.33	N
Maximum HELOC	θ^{HELOC}	0.2	N
Government			
Property tax on housing	τ_h	0.01	Y
Income tax function	τ_y^0, τ_y^1	0.75,0.151	N
Mortgage interest deduction fraction	ϱ	0.75	N
Mortgage interest deduction limit	$\bar{m}$	\$1 mil.	N
Social Security replacement rate	ρ_{SS}	0.42	N

Table 6: Parameter values. A period is two years and annualized values are noted in the final column with a Y. 1=\$52,000 which is the average value of income in the 1998 SCF.

To improve the fit of the 90th percentile of housing to net worth, the share of housing

expenditure to total expenditures can be increased from 0.16 to 0.26 by increasing the preference over housing ϕ from 0.13 to 0.2. As shown in Table 7, this adjustment increases the P90 housing to total net worth of owners from 0.74 to 0.85, which is closer to the 0.95 observed in the data. However, this makes housing less affordable for younger and worse-off households, which are more likely to be sensitive to credit conditions and thus important to discipline in this episode. For example, the average size owned house to its rented counterpart jumps up from 2 to 2.4, which is much larger than the 1.5 in the data. Relatedly, the homeownership rate of those under 35 drops to 0.3, nearly 10 percentage points below the 0.39 in the data. Housing boom simulations with this calibration are broadly similar to the main results.

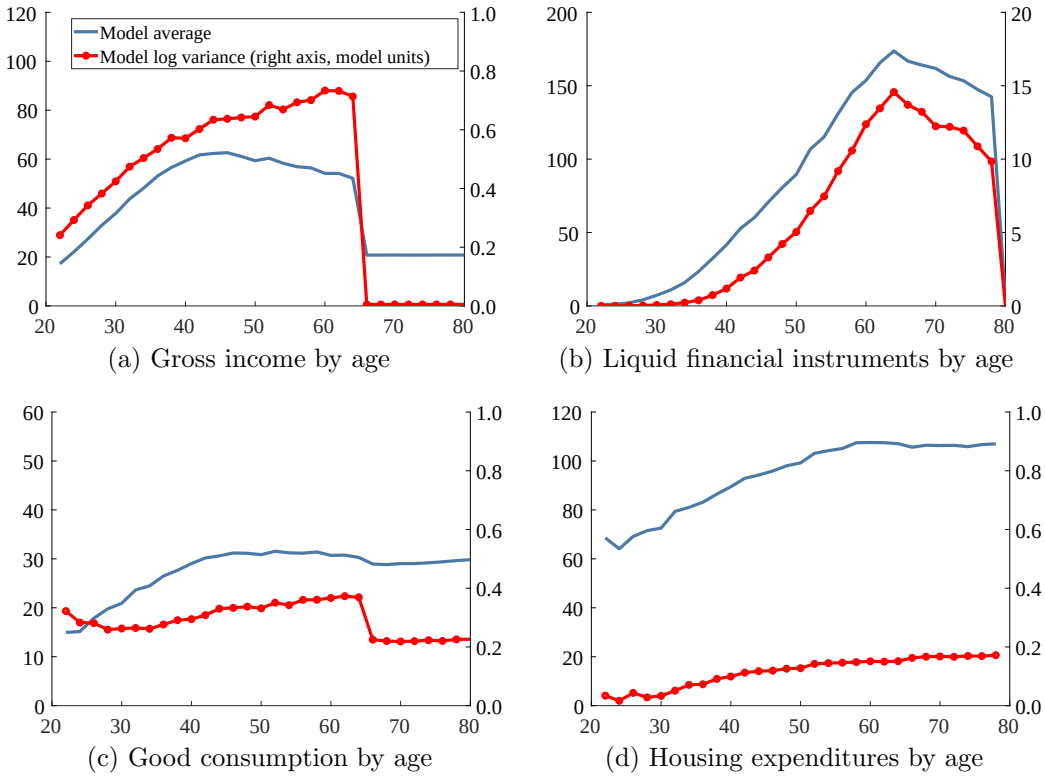
Moment	Param.	Empirical	Model Value	
			Main	Alt
Agg. net worth/annual agg. labor income	β	5.5	4.9	5.3
Median ratio of net worth to labor income	β	1.2	1.2	1.4
Median net worth: age 75/age 50	ψ	1.55	1.48	1.57
% of bequests in bottom 1/2 of wealth dist.	$\underline{b}$	0	0	0
Housing/total cons. expenditures	ϕ	0.16	0.16	0.26
Aggregate home-ownership rate	ω	0.66	0.68	0.67
Foreclosure rate	ξ	0.005	0.0002	0.002
P10 housing/total net worth of owners	$\min \mathcal{H}$	0.11	0.13	0.2
P50 housing/total net worth of owners	$\#\mathcal{H}$	0.5	0.33	0.41
P90 housing/total net worth of owners	gap $\mathcal{H}$	0.95	0.74	0.85
Average sized owned house/rented house	$\min \tilde{\mathcal{H}}$	1.5	2	2.4
Average earnings of owners to renters	$\#\tilde{\mathcal{H}}$	2.1	2.8	2.7
Annual fraction of houses sold	κ_h	0.1	0.09	0.1
Homeownership rate of < 35 y.o.	Ξ	0.39	0.34	0.3
Employment in construction sector	$\bar{L}$	0.05	0.04	0.06

Table 7: Targeted moments in calibration corresponding to model parameters.

C.2 Cross-sectional Distribution across the Life Cycle

Before simulating the U.S. housing boom under this paper's contribution of endogenously optimistic beliefs, it is important to first discuss the cross-sectional distribution of households in the model's stochastic steady state which assumes known forecasting coefficients, tight aggregate credit conditions, and fluctuations in aggregate income.[47] Figure 11 shows that the life cycle profiles of the means and variances of model quantities are largely consistent with a typical incomplete markets model where households succeed at some smoothing of goods consumption and housing expenditures by accumulating a buffer stock of liquid financial instruments.

[47] Appendix C.4 shows that the calibration is similar assuming the credit conditions can fluctuate between their loose and tight values via a Markov chain. Appendices E and F.1 confirm for impulse response functions are similar under credit conditions that shift one time or are Markov.



(a) Gross income by age

(b) Liquid financial instruments by age

(c) Good consumption by age

(d) Housing expenditures by age

Figure 11: Life cycle statistics from the model in thousands of 1998 dollars.

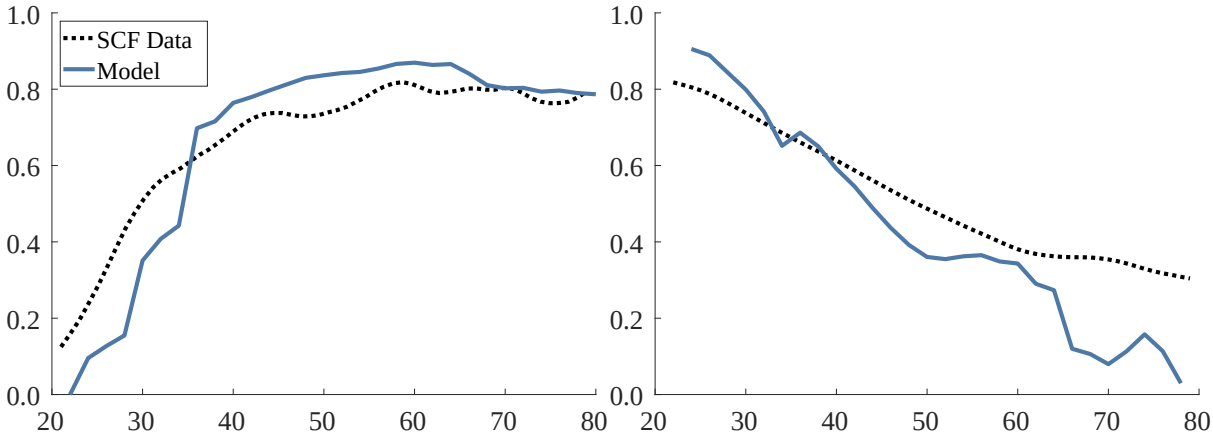
Panel 11a shows that the average and log variance of income rise over the life cycle consistent with the data from the 1998 Survey of Consumer Finances. The non-stationary age-dependent variance for idiosyncratic income helps match the more volatile earnings in the pre-retirement phase as is observed in the data.

Panel 11b shows that households accumulate liquid financial instruments, on average, as a buffer against both idiosyncratic shocks and the decrease in earnings at retirement. The pronounced hump-shape profile in both the average and variance (in levels, not logs) arises from a desire to smooth both consumption and housing expenditures in retirement. Average liquid financial instruments drop at the end of economic life because of bequest heterogeneity—only households in the top half of the wealth distribution leave inheritances.

The relatively flat profiles of both goods consumption and housing expenditures over the life cycle shown in Panels 11c-11d suggest that households partially self-insure against idiosyncratic shocks and retirement. Housing illiquidity and the availability of liquid financial assets are also both key for generating the flat profile of average housing expenditures, which is, in turn, key for matching the data and solving for a Krusell and Smith (1998) approximate equilibrium. Iacoviello and Pavan (2007) explain that illiquid housing results in households accumulating a buffer stock of wealth in liquid financial instruments instead of housing since households cannot adjust housing without paying additional transaction costs. If liquid

financial instruments were not available, the average housing expenditure profile would look like that of liquid financial instruments with the hump-shape indicating more variation in marginal propensities to consume within and across households. The simple AR(1) forecast given by equation (14) would then be insufficient to approximate the evolution of house prices and would generate worse goodness of fit measures than those in Appendix B.2.

An empirically consistent distribution of housing is key for pinning down aggregate house prices throughout the housing boom and Figure 12 shows the model also matches the homeownership rate and mortgage leverage ratio over the life cycle. Young households are particularly important because they are less likely to own homes and more likely to have maximum leverage which makes them the most sensitive to looser credit conditions.



(a) Homeownership rate by age, % (b) Mortgage leverage by age, % loan to value

Figure 12: Housing statistics over the life cycle, 1998 dollars.

Panel 12a shows that households remain renters, on average, in the first part of economic life in order to accumulate enough wealth and earnings to overcome the loan-to-value and payment-to-income constraints. The size of inheritances and initial income endowments determine the age in which households can afford to purchase housing.[48] Consistent with the data, the worse off households can never afford housing and remain life-long renters.

Panel 12b shows that mortgage leverage—homeowners' ratio of mortgage debt to housing value—is highest at age 22 and declines thereafter as households amortize mortgage balances. Model simulated leverage is slightly below but mostly in line with its empirical counterpart until age 65. Thereafter, it takes on counterfactual oscillations and low values due to assumptions on mortgages. In the model, homeowners can take out a mortgage that lasts for only a few years while in practice mortgages may only be available longer tenures.

[48] See Brandsaas (2024) for an overview of how parental transfers help young households overcome mortgage constraints. Including intervivos transfers is beyond the scope of this paper.

C.3 Segmentation of Housing Unit Sizes

This Appendix evaluates the assumptions on the partial segmentation of housing unit sizes and shows that the model's distributions match those of the data minus a few discrepancies. Because housing size enters the households' utility function, empirically consistent partial segmentation can be represented by certain sizes of housing units only being available to a subset of households.[49] To best fit the pre-boom homeownership rate of 66 percent, I follow Kaplan et al. (2020) and assume that the smallest housing unit cannot be owned and the larger housing units cannot be rented. Table 8 shows that in the data only 9 percent of homeowners occupy the smallest sized unit and only 10 percent of renters occupy the largest sized unit which suggests that the partial segmentation assumption only restricts the model from matching a fraction of the overall size distribution. The model's distributions of owner-occupied and rental units overstate the demand for the smallest available units but match the rest of the distributions relatively well. Alternative segmentation assumptions shown in Table 8 provide small fixes to these shortcomings at the expense of distorting other targeted moments such as the homeownership rate or the housing expenditure share as shown in Table 9.

House Size	Data Owners	Benchmark Model	No seg.	Full seg.	Partial seg.	Smaller size 1	Larger size 1
1	9		19		19		
2	24	55	23		22	46	49
3	25	9	23		24	17	14
4	18	13	13	89	12	13	13
5	10	17	16	2	16	18	18
6	9	4	5	9	5	5	4
7	6	2	1	0	1	1	1
House Size	Data Renters	Benchmark Model	No seg.	Full seg.	Partial seg.	Smaller size 1	Larger size 1
1	51	79	76	75	76	73	77
2	28	14	14	12	14	18	14
3	11	7	6	13	10	9	9
4	5		2				
5	2		2				
6	1		1				
7	2		0				

Table 8: Distribution of housing unit sizes in percentage points by segmentation assumptions. Source: American Housing Survey, Kaplan et al. (2020).

[49] Landvoigt et al.'s (2015) empirical evidence points to partial segmentation of housing markets by quality of units rather than size which is beyond the scope of most macro housing models. Absent a standard segmentation convention, Arslan et al. (2026) rely on rental sector convertibility frictions and Greenwald and Guren (2025) assume separate housing stocks similar to the full segmentation shown in Table 8.

Moment	Empirical Value	Benchmark Model	No seg.	Full seg.	Smaller size 1	Larger size 1
Housing/total cons. expenditures	0.16	0.16	0.15	0.18	0.15	0.16
Aggregate home-ownership rate	0.66	0.68	0.7	0.54	0.71	0.68
Av. sized owned/rented house	1.5	2	1.6	2	1.6	1.7
Av. earnings of owners/renters	2.1	2.8	2.6	2.6	2.6	2.7
Homeownership rate of < 35 y.o.	0.39	0.34	0.38	0.26	0.38	0.36

Table 9: Moments by segmentation assumptions. Source: American Housing Survey, Kaplan et al. (2020).

The segmentation assumptions are:

- **Benchmark**: renters can only choose the smallest three housing unit sizes and owners cannot choose the smallest size.
- **No segmentation**: renters and owners can choose all available housing unit sizes.
- **Full segmentation**: homeowners can only choose the largest four housing unit sizes so that there is no overlap between the units available to renters and homeowners.
- **Smaller unit 1**: size 1 is decreased to 1.07 (1.125 in the benchmark).
- **Larger unit 1**: size 1 is increased to 1.17 (1.125 in the benchmark).

Table 10 shows that the model matches the data in terms of the change in housing size as households change dwellings.

Data	Rent-to-own	Rent-to-rent	Own-to-rent	Own-to-own
Data	0.25 (0.01)	0.04 (0.01)	-0.29 (0.01)	0.04 (0.01)
Model	0.29	0.15	-0.34	0.03

Table 10: Average log-changes in house size by type of transition. House size is the number of rooms of a house in the data which is the PSID form 1961996 via Kaplan et al. (2020). Standard errors are in parentheses.

C.4 Cross-sectional Calibration under Markov Credit Conditions

Moment	Parameter	Empirical Value	Model Value
Agg. net worth/annual agg. labor income	β	5.5	4.9
Median ratio of net worth to labor income	β	1.2	1.2
Median net worth: age 75/age 50	ψ	1.55	1.47
% of bequests in bottom 1/2 of wealth dist.	$\underline{b}$	0	0
Housing/total cons. expenditures	ϕ	0.16	0.16
Aggregate home-ownership rate	ω	0.66	0.7
Foreclosure rate	ξ	0.005	0.0014
P10 housing/total net worth of owners	$\min \mathcal{H}$	0.11	0.08
P50 housing/total net worth of owners	$\#\mathcal{H}$	0.5	0.29
P90 housing/total net worth of owners	gap $\mathcal{H}$	0.95	0.76
Average sized owned house/rented house	$\min \tilde{\mathcal{H}}$	1.5	2
Average earnings of owners to renters	$\#\tilde{\mathcal{H}}$	2.1	2.7
Annual fraction of houses sold	κ_h	0.1	0.1
Homeownership rate of < 35 y.o.	Ξ	0.39	0.42
Employment in construction sector	$\bar{L}$	0.05	0.04

Table 11: Targeted moments in calibration corresponding to model parameters. The table is similar to Table 1 in the main text but with credit conditions that evolve according to a Markov process rather than a one-time shift.

Parameter	Empirical Value	Model Value
Fraction of homeowners w/mortgage	0.66	0.72
Fraction of Homeowners w/HELOC	0.06	0.01
Aggr. mortgage debt/housing value	0.42	0.51
P10 LTV ratio for mortgages	0.15	0.02
P50 LTV ratio for mortgages	0.57	0.52
P90 LTV ratio for mortgages	0.92	0.92
Share of NW held by bottom quintile	0	0
Share of NW held by middle quintile	0.05	0.09
Share of NW held by top quintile	0.81	0.66
Share of NW held by top 10 percent	0.7	0.43
Share of NW held by top 1 percent	0.46	0.06
P10 house value/earnings	0.9	0.94
P50 house value/earnings	2.1	1.8
P90 house value/earnings	5.5	4.1

Table 12: Untargeted moments in calibration. The table is similar to Table 3 in the main text but with credit conditions that evolve according to a Markov process rather than a one-time shift.

D Appendix: Proxy Construction and Learning Calibration

D.1 Proxy Construction

When constant, the learning gain g is set to minimize the mean squared errors of house price forecasts from the model relative to an empirical proxy constructed from the University of Michigan Surveys of Consumers. Because the Michigan Survey's questions about future house prices are only available starting in 2007, there are no direct measures of house price expectations from 1999 to 2007 as shown by Kuchler et al. (2023, Table 2). To construct a house price expectations proxy, I exploit the high correlation between expected house price growth in the next 12 months (2007 start) and whether or not it is a relatively good time to sell a house (1992 start) as shown in Table 13 and Figure 13.[50]

	75th Percentile	Mean	Median
Correlation of Survey with Selling Conditions	0.92	0.92	0.71
R^2 of Survey vs. Selling Conditions	0.85	0.85	0.50

Table 13: Comparing Proxy Constructions, 2007:Q1 to 2022:Q1.

Besides the high correlation, why can one intuitively think of selling conditions as a proxy for expectations? The link between the selling conditions and expectations series comes from a housing asset valuation equation like that in Famiglietti et al. (2023) shown in equation (18) where the price of housing $p(\mu, Z)$ is equal to the cashflow of rents $\rho(\mu, Z)$ and the discounted expected capital gain from selling next period $p'(\mu'; \mathcal{Z})$.[51] Suppose $p'(\mu; \mathcal{Z}) = \exp(a^0_{\mathcal{Z},t} + a^1_{\mathcal{Z},t} \log p_{\mathcal{Z},t})$ as in equation (15), then as beliefs $(a^0_{\mathcal{Z},t}, a^1_{\mathcal{Z},t})$ become more optimistic, expected future capital gains from selling housing necessarily rise as well.

$$p(\mu, Z) = \rho(\mu, Z) + \mathbb{E}_{Z',\epsilon'|Z,\epsilon}\left[\frac{p'(\mu'; \mathcal{Z})}{1 + r_b}\right] \tag{18}$$

Given the tight correlation between the selling conditions and expectations series, I create backcasts of the expected change in house prices going back to 1992 by projecting expectations y^j_t for $j = \{\text{mean, median, 75th percentile}\}$ onto selling conditions x_t for $t = \{\text{Jan. } 2007, \ldots, \text{Jun. } 2022\}$.

$$y^j_t = \beta_0 + \beta_1 x_t + e^j_t$$

[50] The `Relative` column of Table 43 is the `Good time to sell` column minus the `Bad time to sell` column plus 100. The survey asks, "Generally speaking, do you think now is a good time or a bad time to sell a house?" The `Median`, `Mean`, and `75th Percentile` columns of Table 46 are the responses to the question, "By about what percent do you expect prices of homes like yours in your community to go (up/down), on average, over the next 12 months?"

[51] This is valuation equation is a simpler version of the housing demand equation for an unconstrained household who buys a house in this period and sells it next period as in equation (6).b in Appendix A.1.

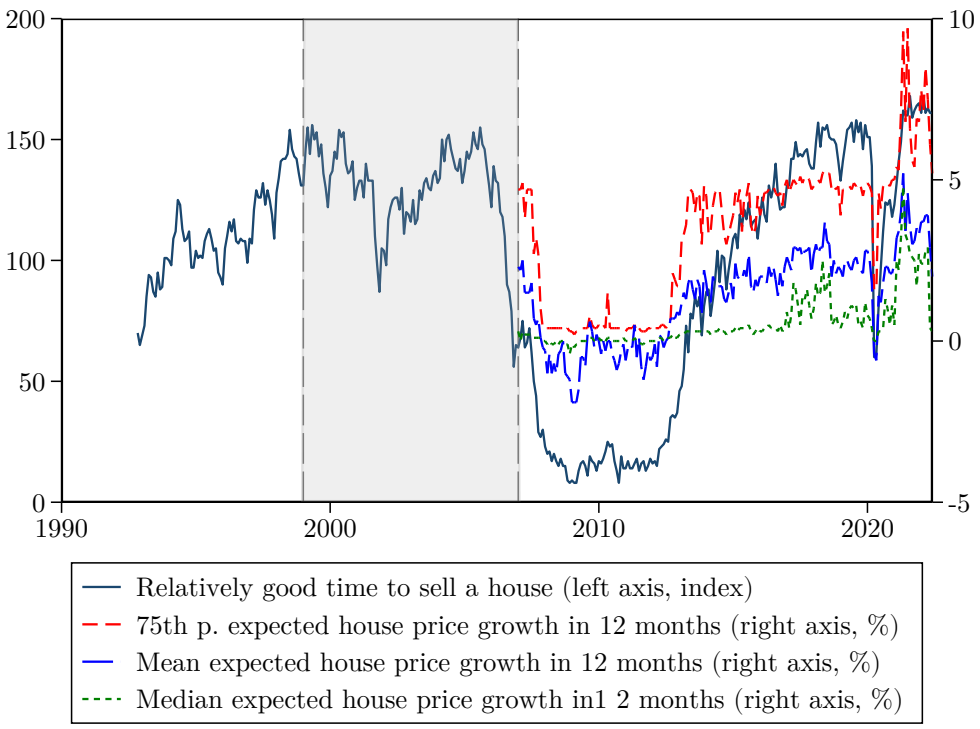


Figure 13: Responses from the University of Michigan Surveys of Consumers, percent of respondents who say it is a good time to sell a house less those who say it is a bad time plus 100 and the 75th percentile, mean, and median expected house price growth in the next 12 months expressed as 12-month percentage change. The shaded bar between the dashed vertical lines denotes the U.S. housing boom (1999 to 2007).

Using the estimated coefficients $\hat{\boldsymbol{\beta}}$, I then backcast the expectations series $\hat{y}_t$ from the selling conditions series x_t when the former is not yet available, $t = \{\text{Jan. } 1992, \ldots, \text{Dec. } 2006\}$.

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_t$$

Table 13 and Figure 14 show that empirical proxies have good in-sample fits with an R^2 of 0.85 for the mean and 75th percentile, and a lower one of 0.50 for the median. Although the higher frequency movements of the selling conditions and expectations series differ, the backcast captures the path of expectations which is key for the study of the U.S. housing boom. The implications of the different levels of the series are discussed in Appendix D.2.

Figure 15 shows that results are similar if the backcasting exercise stops in 2016 instead of extending through 2022. In fact, the R^2 is higher in all three proxies. By contrast, in the case of the median—the proxy with the lowest correlation with the selling series—removing the COVID period has the largest change. However, in the case of the mean and the 75th percentile, the proxies are quite similar regardless of the end date. This ensures that the

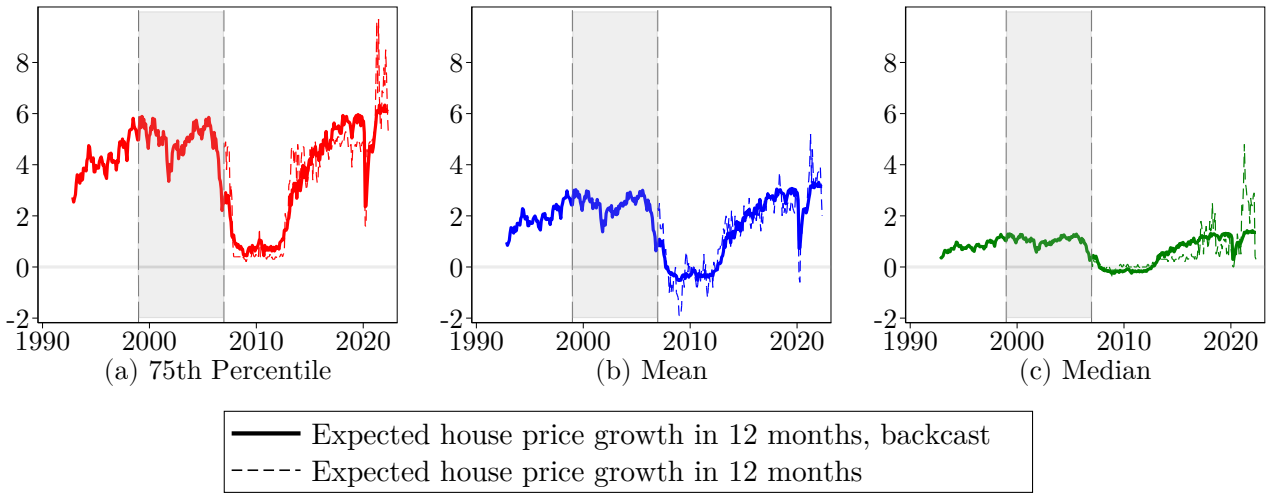


Figure 14: Expected house price growth in the next 12 months from the University of Michigan Survey of Consumers, 12-month percentage change. Shaded bar between the dashed vertical lines marks the U.S. housing boom (1999 to 2007).

COVID period is not driving the goodness of fit—at least for the mean and 75th percentile series.

House price expectations from the model and the empirical proxy have four key differences that affect their comparability. Those from the model are for two-year real detrended house prices in levels while those from the data are for nominal expected 12-month growth rates. Calculations below discuss which differences affect the mean squared forecast error used to calibrate the constant learning gain. These are shown for the 75th percentile which is the main calibration. Why this is the main calibration is discussed in Appendix D.2.

Growth rates vs. levels: Forecast errors from growth rates and levels are roughly equivalent as shown by equations (19) and (20). Assuming the current month's house prices p_τ are known to survey participants, the h period ahead forecast error e_τ^{survey} for month τ can be defined as:

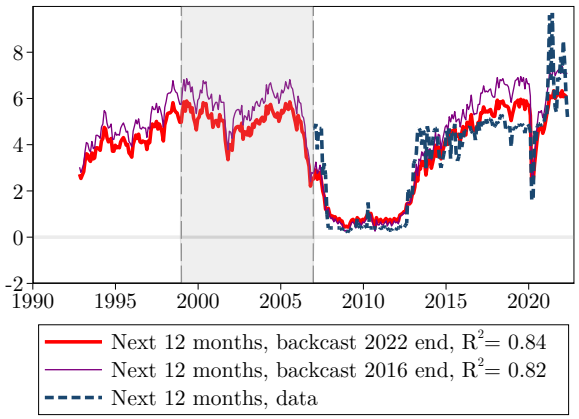
$$e_\tau^{survey} \equiv \log p_{\tau+h} - \log F_\tau p_{\tau+h}$$

Noting that the Michigan survey provides an expected 12-month growth rate $F_\tau g_{\tau+h}$ for $h = 12$, the above definition becomes:

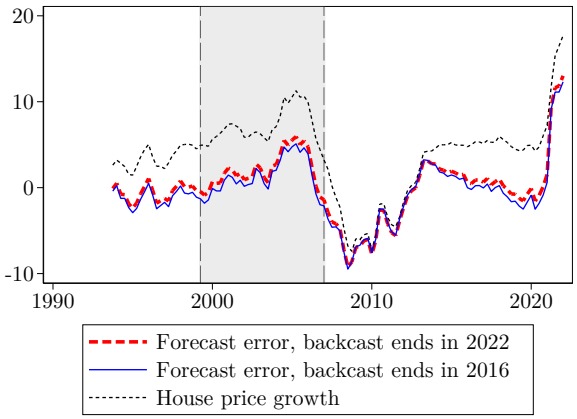
$$e_\tau^{survey} \equiv \log p_{\tau+h} - \log \left(p_\tau (1 + F_\tau g_{\tau+h})\right)$$

Re-arranging and letting $\log(1 + F_\tau g_{\tau+h}) \approx F_\tau g_{\tau+h}$, the above expression becomes:

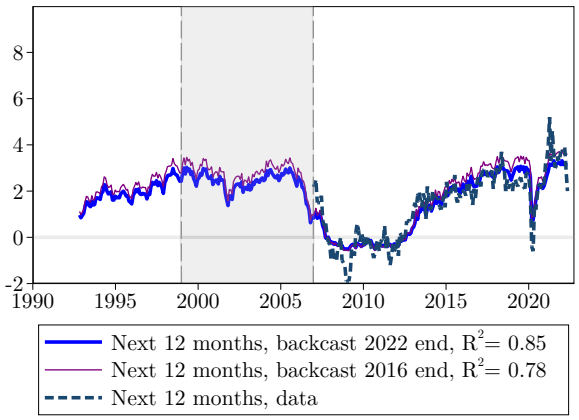
$$e_\tau^{survey} \equiv \log \left(p_{\tau+h}/p_t\right) - F_\tau g_{\tau+h} \tag{19}$$



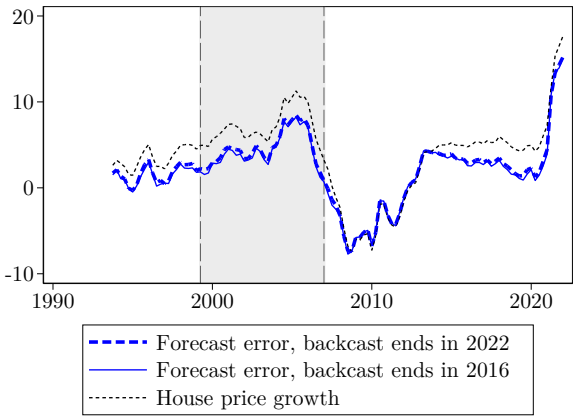
(a) 75th. expectations of house price growth, %



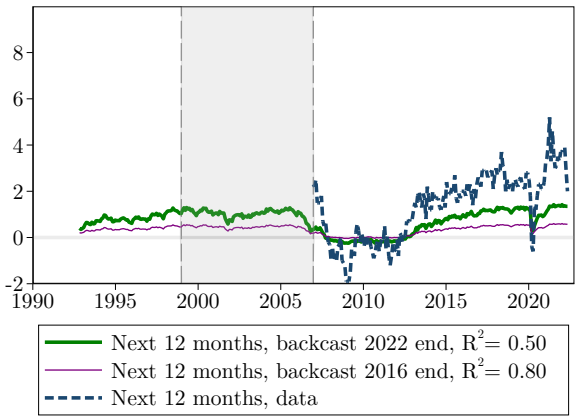
(b) House price growth and 75th p. forecast errors



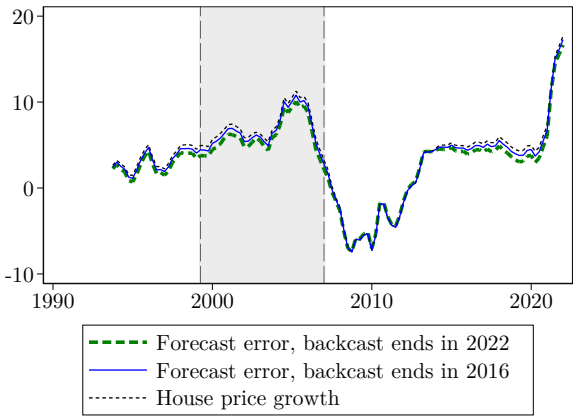
(c) Mean expectations of house price growth, %



(d) House price growth and mean forecast errors



(e) Median expectations of house price growth, %



(f) House price growth and median forecast errors

Figure 15: 12-month percent change in expected house price growth and 4-quarter percent change in house price growth with forecast errors calculated as house price growth less the proxy. Shaded bar between the dashed vertical lines marks the U.S. housing boom (1999 to 2007).

In the model, the f period ahead forecast for period t in aggregate state Z_t with $\mathcal{Z}, t \in \{Z, Z'\}$ is given as:

$$e_t^{model} \equiv \log p_{t+f} - \mathbb{E}_{Z_{t+f}}[a^0_{\mathcal{Z},t} + a^1_{\mathcal{Z},t} \log p_t] \tag{20}$$

Where the period t is two years and $f = 1$ so that the forecast is one period ahead.

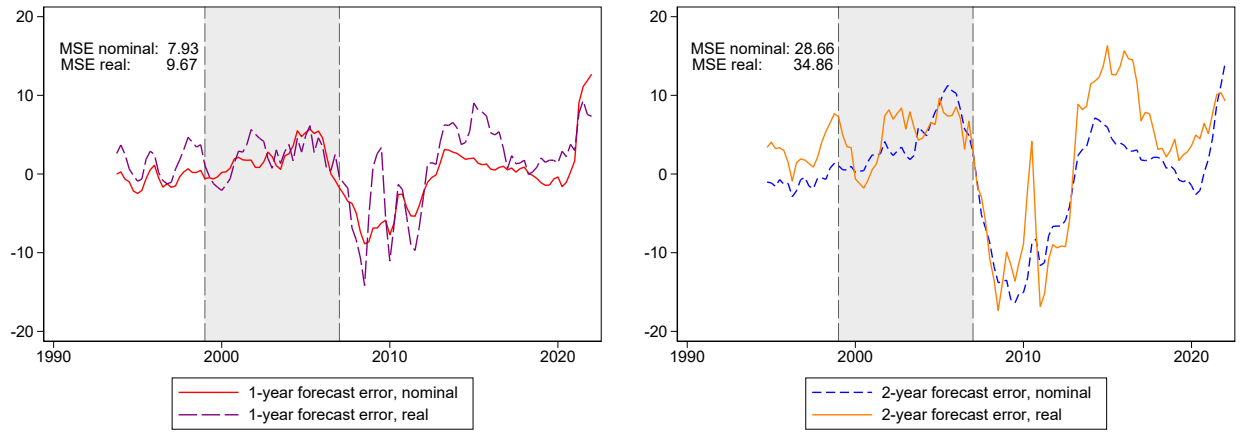
Forecast horizon: To compare the empirical proxy's 12-month forecast horizon to the two-year forecast horizon from the model, I assume that survey respondents expect house price growth for the next 12 months to remain for the following 12 months. The imputed forecast for the next $2 \times h = 24$ can be written as:

$$\begin{aligned} e_\tau^{survey} \equiv & \log p_{\tau+2\times h} - \log F_\tau p_{\tau+2\times h} \\ & \log p_{\tau+2\times h} - \log \left(p_\tau (1 + F_\tau g_{\tau+h})^2\right) \\ & \log(p_{\tau+2\times h}/p_\tau) - 2 \times F_\tau g_{\tau+h} \end{aligned} \tag{21}$$

Imputing the forecast for the next two years via compounding may overstate expected house price growth in the next two years, $2 \times F_\tau g_{\tau+h} > F_\tau g_{\tau+2\times h}$. Although Figure 16 shows that the forecast errors of house price growth at the 12-month and two-year horizons (equations 19 and 21) are quite similar throughout the housing boom, they differ in the later part when the two-year series is higher resulting in a larger mean squared error. However, because expected house prices for the next five years are at their all-time high at the end of the housing boom (not shown), house price expectations for the next two years may have indeed been higher than those for the next 12 months.

Given the concerns about imputing the house price expectations for the next two years, one could instead calibrate the learning gain to the one-year forecast error. The resulting housing boom simulations would be like those shown in Figure 4 with a gain lower than g=0.508. With the lower value of the learning gain, the mean squared forecast error is closer to the 7.93 value of the one-year horizon rather than the 28.66 value at the two-year horizon. House prices still boom, but to a peak that is lower than the 75 percent match to the data observed in the main specification.

Nominal vs. real: The real house price forecast error can be calculated by dividing both realized and expected house prices by actual and expected inflation. Let $\pi_{\tau+h}$ be one-year



(a) 1-year nominal and real forecast errors

(b) 2-year nominal and real forecast errors

Figure 16: Panel (a): Real house price growth is the FHFA house price index scaled by the price index for non-durable consumption. Expected house price growth for the next 12 months is the empirical proxy constructed from the University of Michigan Survey of Consumers and is scaled by responses for expected inflation from the survey. Panel (b) is same as Panel (a), but with house price and inflation expectations both imputed to a two-year forecast horizon. The panels are given in the quarterly average of the 12-month percentage change. Shaded bar between the dashed vertical lines marks the U.S. housing boom (1999 to 2007).

realized inflation and $F_\tau \pi_{\tau+h}$ be its expected value for month τ. The real forecast error is:

$$e_\tau^{survey,real} \equiv \log\left(\frac{p_{\tau+h}}{1+\pi_{\tau+h}}\right) - \log\left(\frac{p_\tau(1+F_\tau g_{\tau+h})}{(1+F_\tau \pi_{\tau+h})}\right)$$
$$\underbrace{\log(p_{\tau+h}/p_\tau) - F_\tau g_{\tau+h}}_{e_\tau^{survey}} - \left(\pi_{\tau+h} - F_\tau \pi_{\tau+h}\right) \tag{22}$$

The real house price forecast error is the forecast error of nominal house price growth less the forecast for the inflation rate.[52] Figure 16 shows that the real and nominal house price forecasts are roughly similar for both the one-year expectations and the two-year imputed expectations throughout the housing boom.[53] Within each forecast horizon, there is little difference between the real or nominal mean squared errors at the one-year horizon. At the two-year horizon, the real error is larger than the nominal error and corresponds most closely to the value of 37.88 obtained under a larger constant shown in Figure 4.

De-trending: The empirical real house price forecast errors are not affected by subtracting out a time trend like the one used to construct house prices in the model. The predicted

[52]The realized inflation series is the PCE price index for durables less consumption which is what is used elsewhere in the paper. Results are similar if the headline PCE price index is used instead.

[53]The 2-year real forecast error is: $\log(p_{\tau+2\times h}/p_\tau) - 2 \times F_\tau g_{\tau+2\times h} - \left(\pi_{\tau+2\times h} - 2 \times F_\tau \pi_{\tau+2\times h}\right)$.

linear time trend $\hat{\Gamma} p_{\tau+h}$ cancels from both sides of equation (22):

$$\log\left(p_{\tau+h}(1-\Gamma)/(1+\pi_{\tau})\right) - \log\left(p_{\tau}(1-\Gamma)(1+F_{\tau}g_{\tau+h})/(1+F_{\tau}\pi_{\tau+h})\right)$$

D.2 Why the 75th Percentile is the Main Calibration

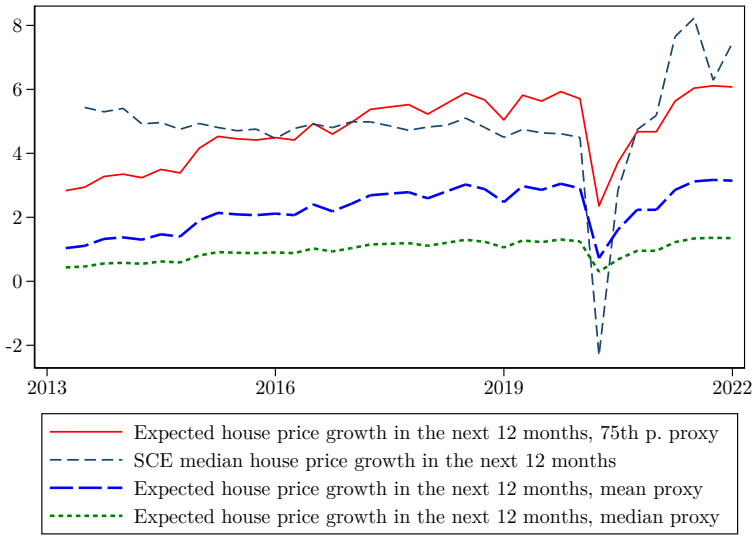


Figure 17: Expected house price growth for the next 12 months is the empirical proxy constructed from the University of Michigan Survey of Consumers and the median point prediction of expected house price growth for the next 12 months from the Federal Reserve Bank of New York Survey of Consumer Expectations, 12-month percentage change.

Evidence from other surveys suggests that responses from the University of Michigan Survey of Consumers may actually underestimate expected house price growth which could lead to higher forecast errors than those presented above.[54] For this reason, I use the 75th percentile of house price expectations for the next 12 months instead of the mean or the median as the 75th percentile tends to run higher than the latter two thus resulting in lower forecast errors. Moreover, Figure 17 shows that the median point prediction for house price growth in the next 12 months from the NY Fed Survey of Consumer Expectations is not always lower than the empirical proxy constructed from the 75th percentile used in this paper in which they are both available. This suggests that the 75th percentile is not too optimistic. By contrast, the proxies constructed from the median and mean are persistently below that of the other survey that is less prone to underestimating house price growth.

[54]Kuchler et al. (2023) note in footnote 5 that house price expectations from the NY Fed Survey of Consumer Expectations (2013 start) show patterns similar to those of the University of Michigan Survey of Consumers with average one-year expectations usually about 2 percentage points higher. De Stefani (2021) notes that house prices expectations from the University of Michigan Survey of Consumers are only available for homeowners which may be problematic because Kindermann et al. (2024) find that renters looking to buy houses are the most informed about house price expectations.

In terms of the implications for calibrating the model, Table 14 shows that the MSEs and initial forecast errors—two key moments used in calibration—are higher under the proxies constructed from the mean and median than that of the 75th percentile. This is because the forecast errors for the mean and median are systematically higher than those of the 75th percentile, as shown in Figure 18. Reassuringly, Figure 19 shows that targeting the statistics in Table 14 can produce a similar boom in house prices under any of the proxies, albeit with a counterfactually large drop in the rent-price ratio under the mean and median calibration.

		MSE		Initial lagged	Initial Forecast Error	
Proxy	Gain, g	Data	Model	beliefs, $\boldsymbol{a}_0$	Data	Model
75th P. (Main)	0.30	28.66	28.26	$(a^0_{t-1}, a^1_{t-1})'$	[2.35, 4.1]%	2.36%
Mean	0.31	94.27	91.28	$(a^0_{t-1} + 0.037, a^1_{t-1} - 0.03)'$	[7.7, 9.0]%	9.00%
Median	0.32	152.24	149.53	$(a^0_{t-1} + 0.065, a^1_{t-1} - 0.011)'$	[10.7, 11.7]%	11.8%

Table 14: Housing boom mean squared forecast errors (MSE), corresponding constant learning gains along with initial house price forecast errors and adjustments to lagged beliefs.

Note: The constant gain g is annualized. Initial lagged beliefs $\boldsymbol{a}_0$ can be ad-hoc adjusted by a small factor to set the initial forecast error when learning starts. Without this adjustment, $\boldsymbol{a}_0 = \boldsymbol{a}_{t-1} = \boldsymbol{a}^{tight}_{Z_{high},Z'}$, which are beliefs from the period before learning starts, i.e. the coefficients associated with an economic expansion and tight credit conditions. The MSE from the model is calculated from beliefs formed in 1997 through 2005 and prices from 1999 (first period of learning) through 2007. In the data, the mean-squared forecast error is also formed from beliefs starting in 1997 through those formed through 2005 and prices from 1999 through 2007. The initial forecast error in the model is formed from 2001 prices and 1999 beliefs. 2001 is the first period of adaptive learning corresponding to the loosening of credit conditions and thus the initial forecast error is formed from 2001 prices and 1999 beliefs. Table 15 shows how the ranges for initial forecast errors are taken from the data. These match the model—time 2001 prices with 1999 beliefs of various aggregation methods.

	75th	Mean	Median
2001:Q1	2.43	7.64	10.71
2001:Q2	2.35	7.70	10.98
2001:Q3	2.61	7.91	11.09
2001:Q4	4.12	8.99	11.68
mean(2001:Q1 -2002:Q4)	2.90	7.92	10.94
Range	[2.35,4.12]	[7.64,8.99]	[10.71,11.68]

Table 15: House price forecast errors from the 2-year ahead proxies constructed from 1992 to 2022 from the University of Michigan Survey of Consumers. The dates corresponds to the dates of the price formed by one-quarter earlier beliefs.

But which calibration should be used for the main result? Several pieces of evidence, including previous work, point to the 75th percentile.

First, the model calibrated to the 75th percentile proxy better matches the path of forecast errors. Table 14 shows that the model calibrated to the 75th percentile starts with

an initial forecast error based on the realized lagged beliefs, $\boldsymbol{a}^{tight}_{Z_{high},Z'}$ which implies that initial beliefs in learning $\boldsymbol{a}_0$ are simply adjusted by the weighted forecast error. By contrast, to target the larger forecast errors observed in the mean or median, this involves setting initial lagged beliefs different from actual lagged beliefs, $\boldsymbol{a}_0 \neq \boldsymbol{a}^{tight}_{Z_{high},Z'}$. This implies a jump in belief formation—and thus a jump in forecast errors from about 1 percent to values of about 9 to 12 percent, as seen in Figure 19g which seems contrary to what is observed in Figure 18—at no point is there a jump in forecast errors around 2001, instead it is rather gradual.

Second, the replication of the findings of Adam et al. (2025) shown in Appendix D.3 presents untargeted exercises that show that the model calibrated to the 75th percentile can match the data in slightly better ways.

Finally, using the beliefs of relatively more optimistic households is consistent with existing work that studies the expectations of the marginal buyer. As mentioned in section 2, existing work shows that relative optimists tend to push up house prices, which suggests that the beliefs of more optimistic marginal buyers may matter more than those of the average buyer. Although expectations are the same for all buyers in this framework, by using the more optimistic 75th percentile for the main simulation, I am better connecting with documented dynamics on price-relevant beliefs.

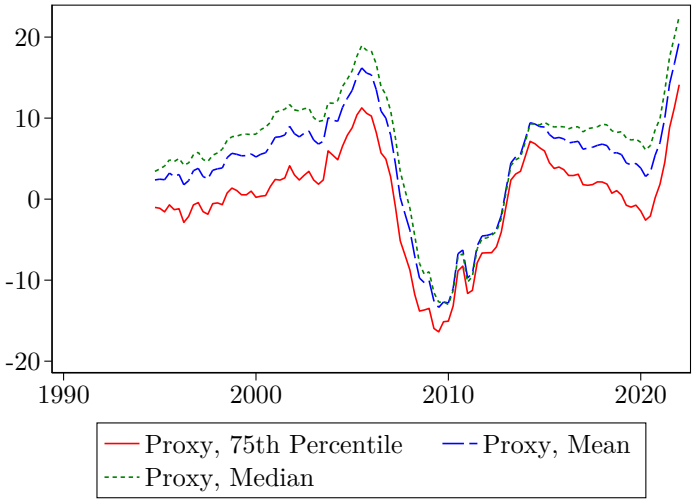
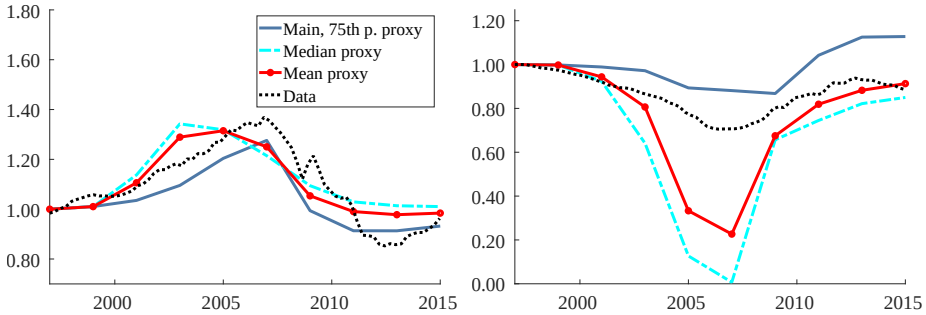
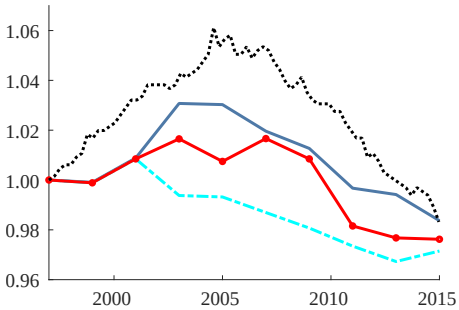


Figure 18: House price forecast errors from various empirical proxies constructed from the University of Michigan Survey of Consumers, 12-month percentage change.

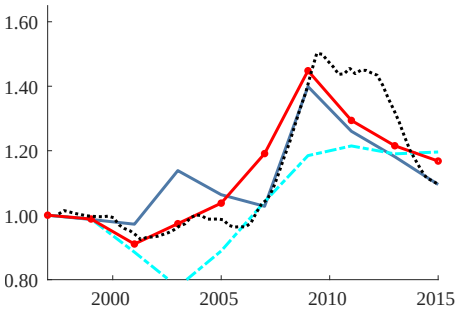


(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1



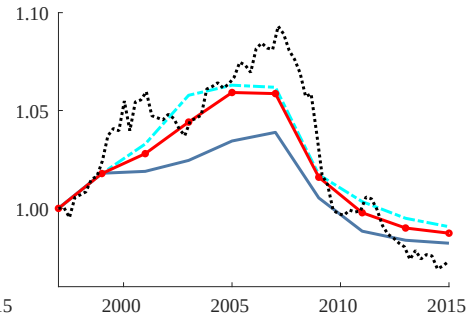
(c) Homeownership rate, 1997=1



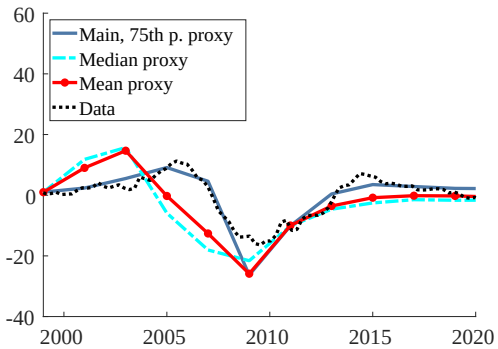
(d) Mortgage leverage, 1997=1



(e) Foreclosure rate, percent



(f) Goods consumption, 1997=1



(g) House price forecast error, percent

Figure 19: Housing boom simulations from the model that target the initial forecast error and mean-squared forecast errors for various proxies of house price expectations in the University of Michigan Survey of Consumers. The model's forecast error is $\log p_t - \mathbb{E}_{t-1}[a^0_{\mathcal{Z},t-1} + a^1_{\mathcal{Z},t-1} \log p_{t-1}]$. See Appendix G for sources and definitions of the data series and Appendix D for details on the construction of the proxy for empirical forecast errors from University of Michigan Survey of Consumers.

D.3 Reproducing the Results of Adam et al. (2025)

Adam et al. (2025) provide recent empirical facts about subjective house price beliefs for the 2007 to 2021 period. This Appendix compares their findings to those of this paper's proxy and model and finds that they match. In fact, among the model calibrated to the 75th percentile, mean, and median, the 75th percentile matches slightly better as it can match all three exercises rather than just two.

Because the model period is two years, model-generated house price expectations are necessarily at a two-year frequency. By contrast, the expectations data from the Michigan survey and the proxies I construct are one-year expectations sampled at a quarterly frequency. Consequently, some of the magnitudes of the model-generated regressions may differ. However, there is no issue in scaling for reproducing the results of Adam et al. (2025) with the proxy. Because their results are similar if real or nominal values are used, all data is nominal for ease of interpretation.

Table 1: Coibion and Gorodnichenko (2015) Regressions

Table 16 reproduces Table 1 in Adam et al. (2025) with the proxy and with model-generated data. For the survey data and the proxy, the table shows estimates of a Coibion and Gorodnichenko (2015) regression of the form:

$$q_{t+4} - \mathbb{E}_t^{\mathcal{P}}[q_{t+4}] = a^{CG} + b^{CG} \cdot (\mathbb{E}_t^{\mathcal{P}}[q_{t+4}] - \mathbb{E}_{t-1}^{\mathcal{P}}[q_{t+3}]) + \epsilon_t$$

Where q_{t+4} is the S&P/Case-Shiller house price index 4-quarters-ahead, $\mathbb{E}_t^{\mathcal{P}}[q_{t+4}] = \mathbb{E}_t^{\mathcal{P}}[q_{t+4}/q_t]q_t$ is the expected house price index 4-quarters-ahead from the Michigan Survey, and $\mathbb{E}_t^{\mathcal{P}}[q_{t+4}] - \mathbb{E}_{t-1}^{\mathcal{P}}[q_{t+3}]$ is the change in expectations. If $\hat{b}^{CG} \neq 0$ then past forecasts can predict future forecast errors and full-information rational expectations can be rejected.

For the model, the regression is instead at a two-year frequency, τ.

$$q_{\tau+1} - \mathbb{E}_\tau[q_{\tau+1}] = \mathbb{E}_\tau[q_{\tau+1}] - \mathbb{E}_{\tau-1}[q_\tau]$$

And can also be estimated on expectations that are transformed to a 1-year frequency via, $\mathbb{E}_\tau[q_{\tau+0.5}] = (\mathbb{E}_\tau[q_{\tau+1}]/q_\tau)^{1/2} q_\tau$ and $q_{\tau+0.5} = (q_{\tau+1}/q_\tau)^{1/2} q_\tau$

$$q_{\tau+0.5} - \mathbb{E}_\tau[p_{\tau+0.5}] = \mathbb{E}_\tau[p_{\tau+0.5}] - \mathbb{E}_{\tau-1}[p_{\tau-0.5}]$$

For all data series in Table 16, $\hat{b}^{CG} \neq 0$ using either expectations or the proxy in all time periods and for the mean, median, and 75th percentile.

In the model-generated data, the coefficients are also positive, but only those for the proxy constructed from 75th percentile are statistically significant once an outlier is removed.

Because the model sample covers the boom and bust when house prices and forecast errors are particularly volatile, there is an exceptionally large and empirically consistent forecast error (per Panel 19g) at the onset of the housing bust in 2009. When this outlier is removed from the regression, the R^2 jumps up from below 0.1 to above 0.8. However, none of the coefficients are statistically significant in the model calibrated to either the mean or median.

Comparing the magnitudes of the coefficients involves a few calculations. According to Coibion and Gorodnichenko (2015), the coefficient $\hat{b}^{CG} = (1 - G)/G$ is how much weight is placed on new information. From the data and proxy, G ranges from 0.25 to 0.33 for one-year expectations at quarterly frequency.[55] Transforming the model-generated expectations calibrated to the 75th percentile to 1-year results in $\hat{b}^{CG} = 0.21$, as shown in column 10, and $G = 0.35$ at a quarterly frequency which is close to the [0.25, 0.33] range of estimates from columns 1 to 6 using the proxy and data.[56]

Taken together, this table shows that the proxy and model-generated data can reproduce the results of Table 1 in Adam et al. (2025). That is, forecast error revisions can predict future forecast errors.

Table 2: Seemingly Unrelated Regressions

Table 17 reproduces Table 2 in Adam et al. (2025) by estimating the following seemingly unrelated regression where PR_{t-1} is the lagged price-rent ratio.

$$\begin{aligned}\mathbb{E}_t^{\mathcal{P}}[q_{t+4}/q_t] =& a + c \cdot PR_{t-1} + u_t \\ q_{t+4}/q_t =& \boldsymbol{a} + \boldsymbol{c} \cdot PR_{t-1} + \boldsymbol{u}_t\end{aligned}$$

While there are some differences between my inputs due to data series and scaling, the inference is unchanged from Adam et al. (2025) as shown in rows 1 and 2 of Table 17.

Row 3 reports the estimates using the proxy instead of the actual Michigan Survey series over the same sample (2007-2021). These results are similar if the mean or median expectations are used instead of 75th percentile to construct the proxy. The coefficients are broadly similar and the Wald test is rejected. Taken together, this evidence points to the estimates holding over the 2007-2021 period. Rows 4 and 5 use samples from 1992-2006 and 1992-1999, respectively, and shows that the coefficients on the beliefs regression $\hat{c}$ are positive and significant as in Adam et al. (2025). However, the coefficient on actual

[55]The extremum of $\hat{b}^{CG}$ in Table 16 columns 1 to 6 are 1.99 and 2.93. Since $G = 1/(1 + \hat{b}^{CG})$, then $G = [0.25, 0.33]$.

[56]Transforming $G^{1Y} = 1/(1 + \hat{b}^{CG}) = 0.826$ to quarterly frequency, $G^{Q} = 1 - (1 - G^{1Y})^{1/4} = 0.35$. The two-year expectation transformed to a quarterly frequency is $G^{2Y} = 1/(1 + \hat{b}^{CG}) = 0.704$ to $G^{Q} = 1 - (1 - G^{2Y})^{1/8} = 0.14$, which is much lower and suggests more sluggish updating over the larger interval.

	(1) APR 2007-21	(2) Data 2007-21	(3) Proxy 1992-2006	(4) Proxy 2007-21	(5) Proxy 1992-2021	(6) Proxy+data 1992-2021	(7) Model 1997-2021	(8) Model 1997-2021 ex 2009	(9) Model 1997-2021	(10) Model 1997-2021 ex 2009
75th p.										
$\hat{b}^{CG}$		1.99***	2.09***	2.08***	2.13***	2.05***	0.18	0.42***	0.08	0.21***
		(0.410)	(0.598)	(0.472)	(0.409)	(0.359)	(0.247)	(0.059)	(0.131)	(0.023)
R^2		0.53	0.46	0.48	0.50	0.51	0.03	0.81	0.03	0.83
Mean										
$\hat{b}^{CG}$	2.22***	2.30***	2.57***	2.50***	2.52***	2.37***	0.34	0.23	0.23	0.15
	(0.507)	(0.515)	(0.638)	(0.478)	(0.414)	(0.429)	(0.20)	(0.186)	(0.12)	(0.093)
R^2	0.59	0.59	0.56	0.57	0.59	0.57	0.10	0.10	0.15	0.14
Median										
$\hat{b}^{CG}$	2.85***	2.93***	2.94***	2.91***	2.90***	2.89***	0.33	0.18	0.24	0.14
	(0.513)	(0.528)	(0.695)	(0.508)	(0.438)	(0.439)	(0.30)	(0.27)	(0.19)	(0.17)
R^2	0.65	0.65	0.60	0.62	0.64	0.62	0.09	0.04	0.15	0.07
N	52	52	56	54	109	109	12	11	12	11
Freq.										
Expect.	1-year	1-year	1-year	1-year	1-year	1-year	2-year	2-year	1-year	1-year
Sampling	quarter	quarter	quarter	quarter	quarter	quarter	2-year	2-year	2-year	2-year

Table 16: Table 1 from Adam et. al (2025). Standard errors in parentheses are robust with respect to heteroskedasticity and autocorrelation (Newey-West with four lags). Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Note: The table shows estimates of $\hat{b}^{CG}$ from $q_{t+4} - \mathbb{E}_t^{\mathcal{P}}[q_{t+4}] = a^{CG} + b^{CG} \cdot (\mathbb{E}_t^{\mathcal{P}}[q_{t+4}] - \mathbb{E}_{t-1}^{\mathcal{P}}[q_{t+3}]) + \epsilon_t$ where q_t is the house price index, and $\mathbb{E}_t^{\mathcal{P}}[q_{t+4}]$ is the either the mean, 75th percentile, or median one-year house price growth expectations from the Michigan survey. Data and proxy results are thus nominal. In the years prior to 2007, $\mathbb{E}_t^{\mathcal{P}}[q_{t+4}]$ is the proxy constructed as described in Appendix D. See Appendix G for data sources. Model-generated regressions are estimated at a 2-year frequency τ using either 2- or 1-year ahead expectations. These specifications are, $q_{\tau+1} - \mathbb{E}_\tau[q_{\tau+1}] = \mathbb{E}_\tau[q_{\tau+1}] - \mathbb{E}_{\tau-1}[q_\tau]$ and $q_{\tau+1} - \mathbb{E}_\tau[q_{\tau+1}] = \mathbb{E}_\tau[q_{\tau+1}] - \mathbb{E}_{\tau-1}[q_\tau]$, respectively. Where $\mathbb{E}_\tau[q_{\tau+0.5}] = (\mathbb{E}_\tau[q_{\tau+1}]/q_\tau)^{1/2} q_\tau$ and $q_{\tau+0.5} = (q_{\tau+1}/q_\tau)^{1/2} q_\tau$.

house price growth, $\hat{\boldsymbol{c}}$, is not statistically significant in either specification and the Wald-test on the differences in the coefficients cannot be rejected. Because full information rational expectations implies that $\hat{c} = \hat{\boldsymbol{c}}$, failing to reject rational expectations in the pre-boom sample implies that subjective beliefs are not detected in the pre-boom era. This lends to the question, if the 2007-2021 sample is extended earlier, at what point does $\hat{c} = \hat{\boldsymbol{c}}$? Rows 6 and 7 show that the sample starting in 2004:Q4 detects subjective beliefs (for all but the median), but the sample starting one quarter earlier fails to reject $\hat{c} = \hat{\boldsymbol{c}}$ in all versions. This evidence suggests that subjective beliefs were more prominent in the later phase of the boom than in the earlier phase.

The specification for the model at a two-year frequency τ is:

$$\mathbb{E}_\tau[q_{\tau+1}/q_\tau] = a + c \cdot PR_{\tau-1} + u_\tau$$

$$q_{\tau+1}/q_\tau = \boldsymbol{a} + \boldsymbol{c} \cdot PR_{\tau-1} + \boldsymbol{u}_\tau$$

And for expectations and growth imputed to a 1-year frequency, but still sampled at a

	75th percentile				mean				median			
	$\hat{c}$	Bias	pval	$\hat{\mathbf{c}}$	$\hat{c}$	Bias	pval	$\hat{\mathbf{c}}$	$\hat{c}$	Bias	pval	$\hat{\mathbf{c}}$
Adam et al. (2025)												
(1) In paper,	-	-	-		0.033	0.001	0.00	−0.102	0.014	0.009	0.00	−0.102
2007-21, N=50	-	-	-	-	(0.008)			(0.007)	(0.001)			(0.007)
(2) Updated Inputs,	0.039	0.015	0.00	−0.525	0.018	0.020	0.00	−0.525	0.009	0.027	0.00	−0.525
2007-21, N=50	(0.062)			(0.105)	(0.038)			(0.105)	(0.013)			(0.105)
Proxy												
(3) 2007-2021,	0.036	0.016	0.00	−0.525	0.023	0.021	0.00	−0.525	0.010	0.026	0.00	−0.525
N=50	(0.066)			(0.105)	(0.042)			(0.105)	(0.017)			(0.105)
(4) 1992-2006,	0.026	−0.000	0.49	0.11	0.017	−0.000	0.45	0.11	0.007	−0.000	0.42	0.11
N=54	(0.010)			(0.130)	(0.007)			(0.130)	(0.003)			(0.130)
(5) 1992-1999,	0.301	0.092	0.46	0.925	0.194	0.108	0.44	0.925	0.081	0.124	0.42	0.925
N=26	(0.280)			(1.118)	(0.180)			(1.118)	(0.076)			(1.118)
(6) 2004:Q4-2021,	0.098	0.114	0.02	−0.128	0.063	0.134	0.07	−0.128	0.026	0.154	0.18	−0.128
N=58	(0.037)			(0.122)	(0.024)			(0.122)	(0.010)			(0.122)
(7) 2004:Q3-2021,	0.112	0.150	0.11	−0.055	0.072	0.173	0.26	−0.055	0.030	0.197	0.48	−0.055
N=59	(0.037)			(0.128)	(0.024)			(0.128)	(0.010)			(0.128)
Model												
(8) 2-y exp., N=12,	0.128	-0.10	0.01	−0.48	0.020	-0.015	0.00	-0.056	0.016	-0.003	0.00	-0.022
1997-2021	(0.013)			(0.231)	(0.006)			(0.016)	(0.002)			(0.006)
(9) 1-y exp., N=12	0.064	-0.049	0.01	−0.241	0.010	-0.008	0.00	-0.028	0.01	-0.002	0.00	-0.011
1997-2021	(0.006)			(0.115)	(0.003)			(0.008)	(0.001)			(0.003)

Table 17: Table 2 from Adam et. al (2025). Standard errors in parentheses are robust with respect to heteroskedasticity and autocorrelation (Newey-West with four lags). The Stambaugh (1999) small-sample bias correction is reported in columns 3, 7, and 11 and the associated p-value for the null hypothesis $c = \boldsymbol{c}$ in columns 4, 8, and 12 using the small sample bias correction. Expectations data are nominal.

Note: The table shows SUR estimates of $\hat{c}$ from $\mathbb{E}_t^{\mathcal{P}}[q_{t+4}/q_t] = a + c \cdot PR_{t-1} + u_t$ and $\hat{\boldsymbol{c}}$ from $q_{t+4}/q_t = \boldsymbol{a} + \boldsymbol{c} \cdot PR_{t-1} + \boldsymbol{u}_t$. Where q_t is aggregate house prices, and $\mathbb{E}_t^{\mathcal{P}}[q_{t+4}]$ is the either the mean, 75th percentile, or median one-year house price growth expectations from the Michigan survey or the proxy constructed as described in Appendix D. PR_{t-1} is the lagged price-rent ratio. See Appendix G for data sources. Model generated data are at a two-year frequency τ and one-year expectations are imputed by $\mathbb{E}_\tau[q_{\tau+0.5}/q_\tau] = (\mathbb{E}_\tau[q_{\tau+1}]/q_\tau)^{1/2}$ and $q_{\tau+0.5} = (q_{\tau+1}/q_\tau)^{1/2}$. The median calibration of the model excludes an observation that is an outlier in the price-rent ratio.

two-year frequency:

$$\mathbb{E}_\tau[q_{\tau+0.5}/q_\tau] = a + c \cdot PR_{\tau-1} + u_\tau$$

$$q_{\tau+0.5}/q_\tau = \boldsymbol{a} + \boldsymbol{c} \cdot PR_{\tau-1} + \boldsymbol{u}_\tau$$

Where, $\mathbb{E}_\tau[q_{\tau+0.5}/q_\tau] = (\mathbb{E}_\tau[q_{\tau+1}]/q_\tau)^{1/2}$ and $q_{\tau+0.5} = (q_{\tau+1}/q_\tau)^{1/2}$

The coefficients in the model for 2- and 1-year expectations in rows 8 and 9, respectively, are largely in line with those in the data, especially the upper panel of the table. That is, the coefficient on the lagged price-rent ratio and forecast errors, $\hat{c}$, is positive, and that of lagged price-rent ratio on house price growth is negative, $\hat{\boldsymbol{c}}$. While these results are estimated using 4 Newey-West lags as in the quarterly data, inference is similar with only one Newey-West lag, except that the pvalue is 0.02 instead of 0.01. However, both of these values are still statistically significant at the 5% level. These results hold for all the

model calibrated to all three different proxies.

In summary, Table 17—like Table 16— shows that the model can reproduce results that are similar to those observed in the data.

Figure 1: Local Projections

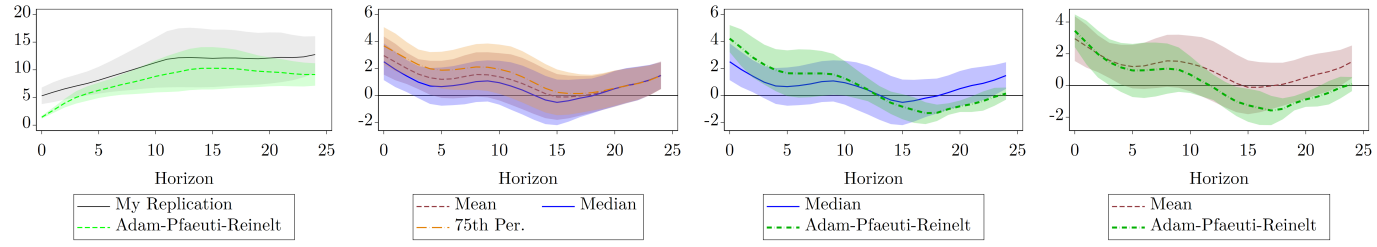
To document that household's expectations initially underreact to observed housing price growth, but overreact later on, Adam et al. (2025) estimate the following local projections:

$$\text{Cumulative House Price Growth:} \quad \frac{q_{t+4+h}}{q_t} = a^h + b^h \left(\frac{q_{t-1}}{q_{t-2}} \right) + u_t^h$$

$$\text{House Price Forecast Errors:} \quad \frac{q_{t+4+h}}{q_{t+h}} - \mathbb{E}_{t+h}^{\mathcal{P}} \left[\frac{q_{t+h+4}}{q_{t+h}} \right] = a^h + b^h \left(\frac{q_{t-1}}{q_{t-2}} \right) + u_t^h$$

Standard errors are Newey-West with $h+1$ lags.

Figure 20 compares the local projections estimates to those of Figure 1 of Adam et al. (2025) using survey data over the 2007 to 2021 sample. There are slight differences in the house price data, which results in the point estimates being similar but not exactly the same. None of these differences affect inference. The coefficient on cumulative house price growth (Panel 20a) increases and those on forecast errors of house price growth diminish over the sample. Panels 20b-20c show that these results hold whether the median, mean, or 75th percentile expectations are used.

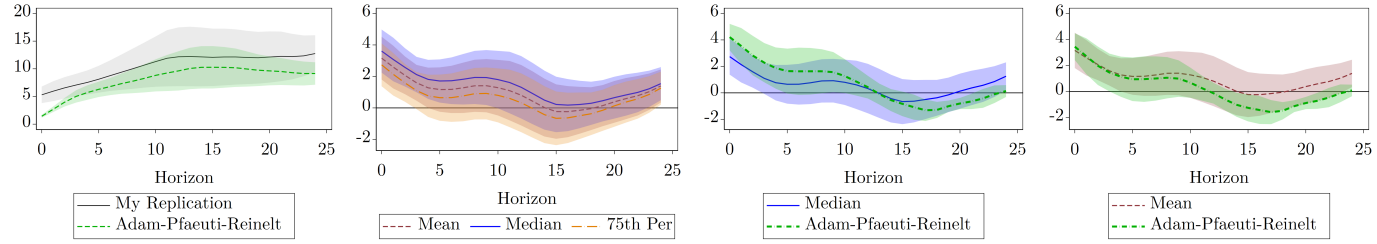


(a) Cumulative house price growth (b) All forecast errors (c) Median forecast errors (d) Mean forecast errors

Figure 20: Survey Data, 2007-2021

Figure 21 compares estimates of Figure 1b in Adam et al. (2025) to those using the proxy over the same sample, 2007-2021. The results using the proxy are similar to those using the actual survey (Figure 20). This suggests that the proxy adequately tracks key properties of the survey data.

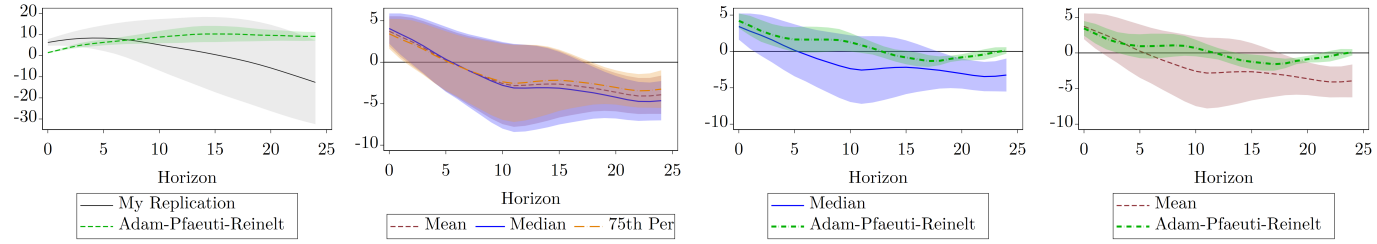
Figure 22 shows that results hold using the proxy from the pre-boom period of 1992 to 2006 as they are similar to those in Figures 20 and 21. Because the 1992 to 2006 period excludes the house boom and cumulative house price growth is quite low, the coefficients



(a) Cumulative house price growth (b) All forecast errors (c) Median forecast errors (d) Mean forecast errors

Figure 21: Proxy, 2007-2021

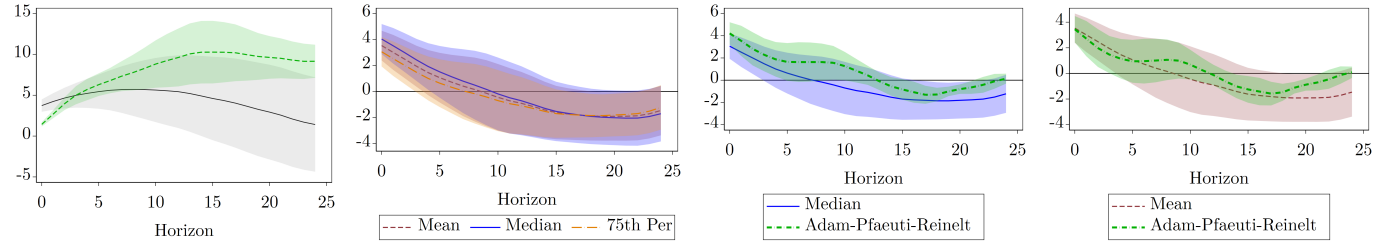
in Panel 22a. shows that cumulative house price growth is imprecisely estimated with large standard errors.



(a) Cumulative house price growth (b) All forecast errors (c) Median forecast errors (d) Mean forecast errors

Figure 22: Proxy, 1992-2006

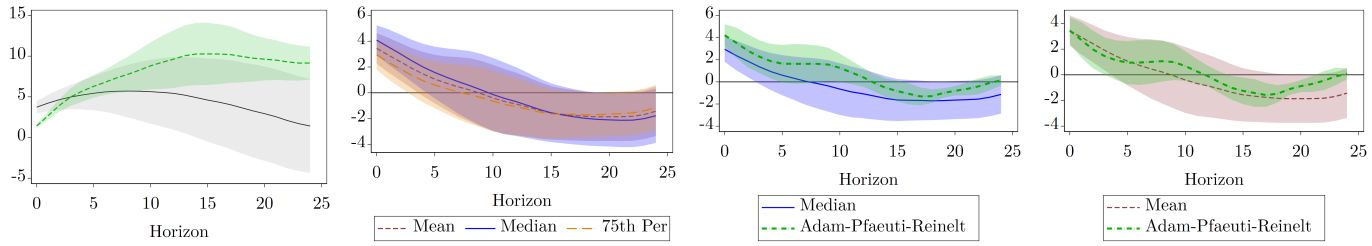
Figures 23 and 24 show that the results are similar for the full sample (1992 to 2021) whether the proxy is used as in Figure 23 or the proxy with actual data once it is available as in Figure 24. Again, the coefficients on cumulative house price growth are different from Adam et al. (2025), once the pre-boom period is included where there is low house price growth.



(a) Cumulative house price growth (b) All forecast errors (c) Median forecast errors (d) Mean forecast errors

Figure 23: Proxy, 1992-2021

Figure 25 compares model-generated house-price forecast errors from the model of Adam et al. (2025) to the main results of this paper. These specifications are the quarterly t



(a) Cumulative housing price growth (b) All forecast errors (c) Median forecast errors (d) Mean forecast errors

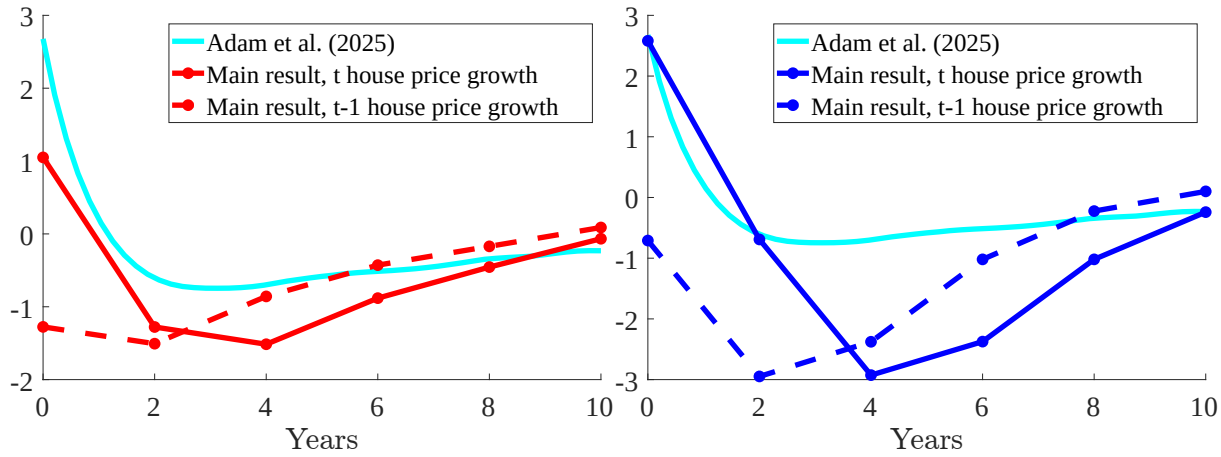
Figure 24: Proxy and Survey Data, 1992-2021

lagged house price growth q_{t-1}/q_{t-2} on forecast errors for one-year ahead expectations. The specification for this paper transforms two-year house price growth to a quarterly frequency and house price growth forecast errors to a yearly frequency.

$$\text{Theirs:} \quad \frac{q_{t+h+4}}{q_{t+h}} - \left(\mathbb{E}_t^{\mathcal{P}}\left[\frac{q_{t+1+h}}{q_{t+h}}\right]\right)^4 = a^h + b^h\left(\frac{q_{t-1}}{q_{t-2}}\right) + u_t^h$$

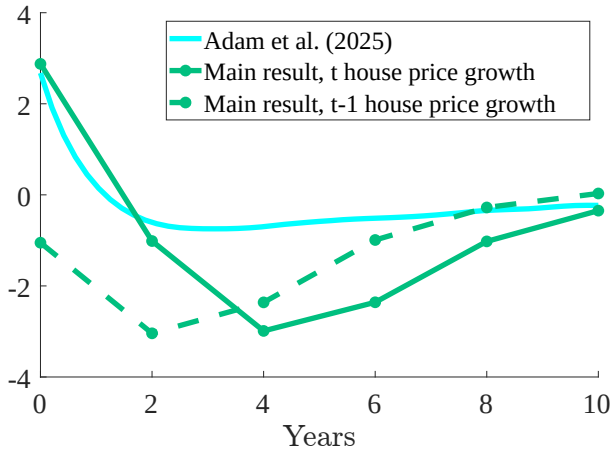
$$\text{Mine:} \quad \left(\frac{q_{\tau+h+1}}{q_{\tau+h}}\right)^{1/2} - \left(\mathbb{E}_t^{\mathcal{P}}\left[\frac{q_{t+1+h}}{q_{\tau+h}}\right]\right)^{1/2} = a^h + b^h\left(\frac{q_{\tau-1}}{q_{\tau-2}}\right)^{1/8} + u_\tau^h$$

Figure 25 shows that the model can reproduce similar patterns in local projection point estimates as Adam et al. (2025) for horizons of 3 years and beyond (dashed red lines) in all three of the calibrations. However, using a specification with contemporaneous house price growth $(q_t/q_{t-1})^{1/8}$ instead of lagged house price growth results in point estimates that better match those of Adam et al. (2025) in the earlier projection horizons (solid red line with circles). These differences can likely be attributed to the differences in sampling frequencies. Because my model is solved at a 2-year frequency, it will not capture the richer earlier arriving dynamics as their model solved at a quarterly frequency.



(a) 75th Percentile Proxy

(b) Mean Proxy



(c) Median Proxy

Figure 25: Local Projection of 1-year-ahead house price growth forecast errors on realized house price growth from Adam et al. (2025) and this paper's model.

Note: The figure shows point estimates from the following local projections. For quarterly frequency t, theirs is: $q_{t+h+4}/q_{t+h} - \left(\mathbb{E}_t^{\mathcal{P}}\left[q_{t+1+h}/q_{t+h}\right]\right)^4 = a^h + b^h\left(q_{t-1}/q_{t-2}\right) + u_t^h$ where q_t is house prices. For a two-year frequency τ, mine is, $\left(q_{\tau+h+1}/q_{\tau+h}\right)^{1/2} - \left(\mathbb{E}_t^{\mathcal{P}}\left[q_{t+1+h}/q_{\tau+h}\right]\right)^{1/2} = a^h + b^h\left(q_{\tau-1}/q_{\tau-2}\right)^{1/8} + u_\tau^h$. Standard errors are not shown because they are imprecisely estimated for my housing boom simulation from 1997 to 2021. They are very precisely estimated for the larger quarterly simulation sample of Adam et al. (2025).

E Appendix: Comparison to Kaplan et al. (2020)

As noted in section 3.1, there are some small differences between the implementation in this paper relative to that of Kaplan et al. (2020). These differences require a slightly lower operating cost of the rental company Ξ to calibrate to the correct homeownership rate for households under 35 years of age and are listed below:

- The McClement's scale on utility e_j is set so that $\{e_j\}_{j=21}^{30} = 1$ while in Kaplan et al. (2020) it is instead set at $\{e_j\}_{j=21}^{30} = \{e_j\}_{j=1}^{8} > 1$.
- At age 30, agents who choose to stay in their house do not borrow HELOCs.
- Retired agents have $-\zeta(Z)$ in the mortgage pricing function.[57]

Figure 26 shows that these differences have almost no implications on the main findings of Kaplan et al. (2020) and only affect magnitudes and time paths of impulse responses. All purple lines shown in Figure 26 are results from Kaplan et al. and all blue lines are from the code in this paper solved under the latter's forecasting coefficients for house prices.

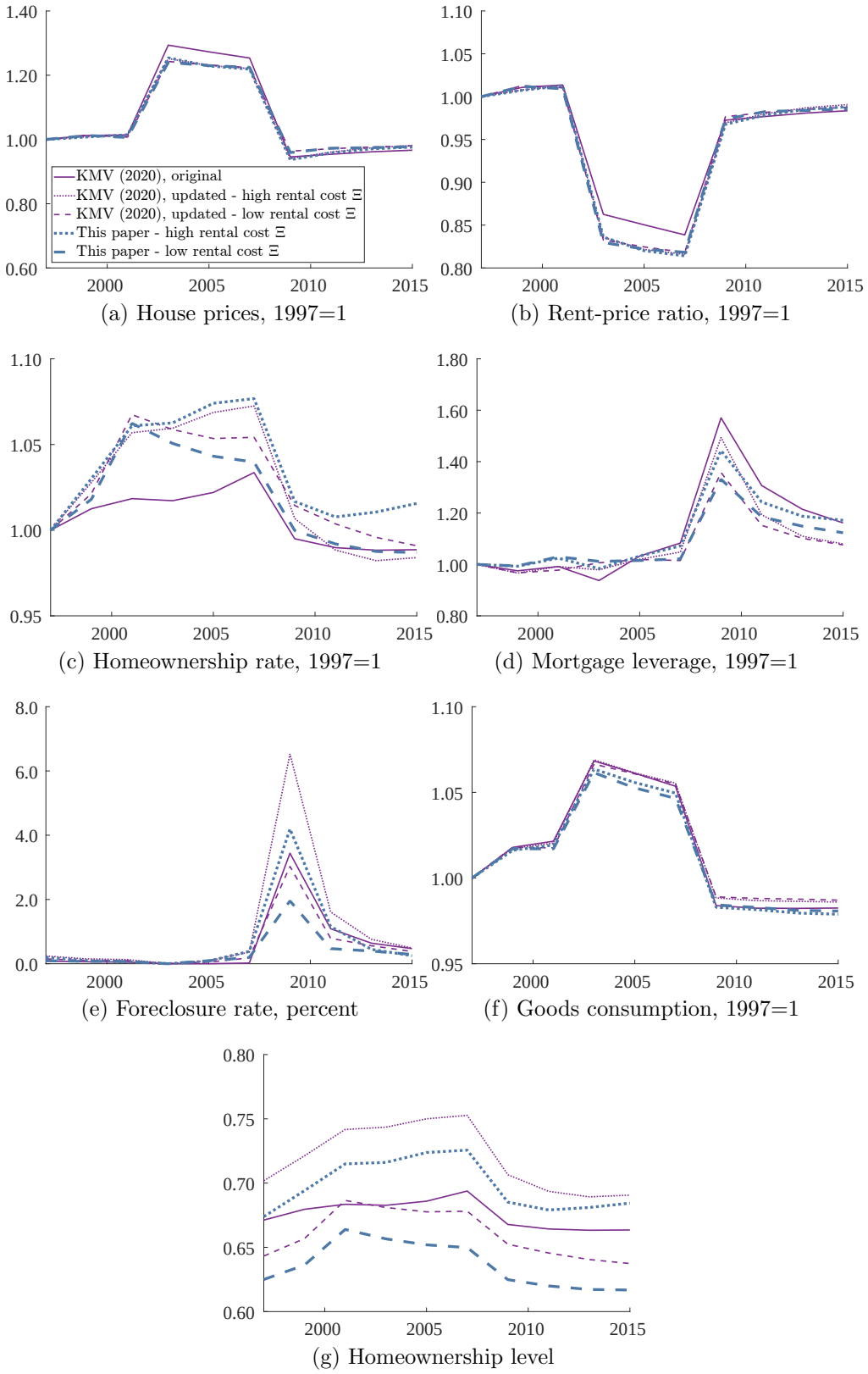
The differences noted above result in a slightly smaller boom in house prices, slightly larger drop in the rent-to-price ratio, and smaller rise in consumption, as shown in Panels 26a, 26b, and 26f, respectively.

The effects are more substantial for the impulse responses for the normalized and un-normalized homeownership rate, leverage, and foreclosure rate as shown in Panels 26c, 26g, 26d, and 26e, respectively. First, I note that leverage and the foreclosure rate (Panels 26d and 26e, respectively) both increase sharply in the bust with the degree of the increases depending on the operating cost of the rental company Ξ. The increases in the bust are relatively higher under the original high value of Ξ as shown by the dotted purple and blue lines than under the re-calibrated low value of Ξ as shown by the dashed purple and blue lines. Although the increase in both leverage and the foreclosure rate are lowest under this paper's implementation with the re-calibrated low value of Ξ, Panels 26d and 26e show that this can be accounted for, in part, due to the re-calibration.

Furthermore, the homeownership rate can also account for differences in the increase in leverage and foreclosures in the bust. The un-normalized homeownership rate shown in Panel 26g is relatively higher under the original high value of Ξ and lower under the re-calibrated lower value, as seen by the differences between the dotted and dashed lines. The purple lines from Kaplan et al. (2020) generally start from a higher homeownership rate than the blue lines of this paper. Finally, I note that for both the dashed lines and the dotted lines, the shape of the homeownership rate is quite similar.

[57] The mortgage pricing function was updating before it finished the age loop. This was fixed by closing off several loops before going into the value function iteration step.

Figure 27 compares the exogenous beliefs version of Kaplan et al. (2020) to a version with credit conditions that loosen one time instead of credit conditions that evolve according to a Markov process. The figure shows that the impulse responses of housing boom simulations are nearly identical for all prices and quantities. This is the case because the probability of Markov credit conditions shifting is incredibly low at 1% so shutting it to 0% and having the shift be a surprise is quantitatively little different. This figure suggests that the model solved under endogenous beliefs and a one-time shift in credit conditions is readily comparable to its endogenous beliefs counterpart with Markov credit conditions.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

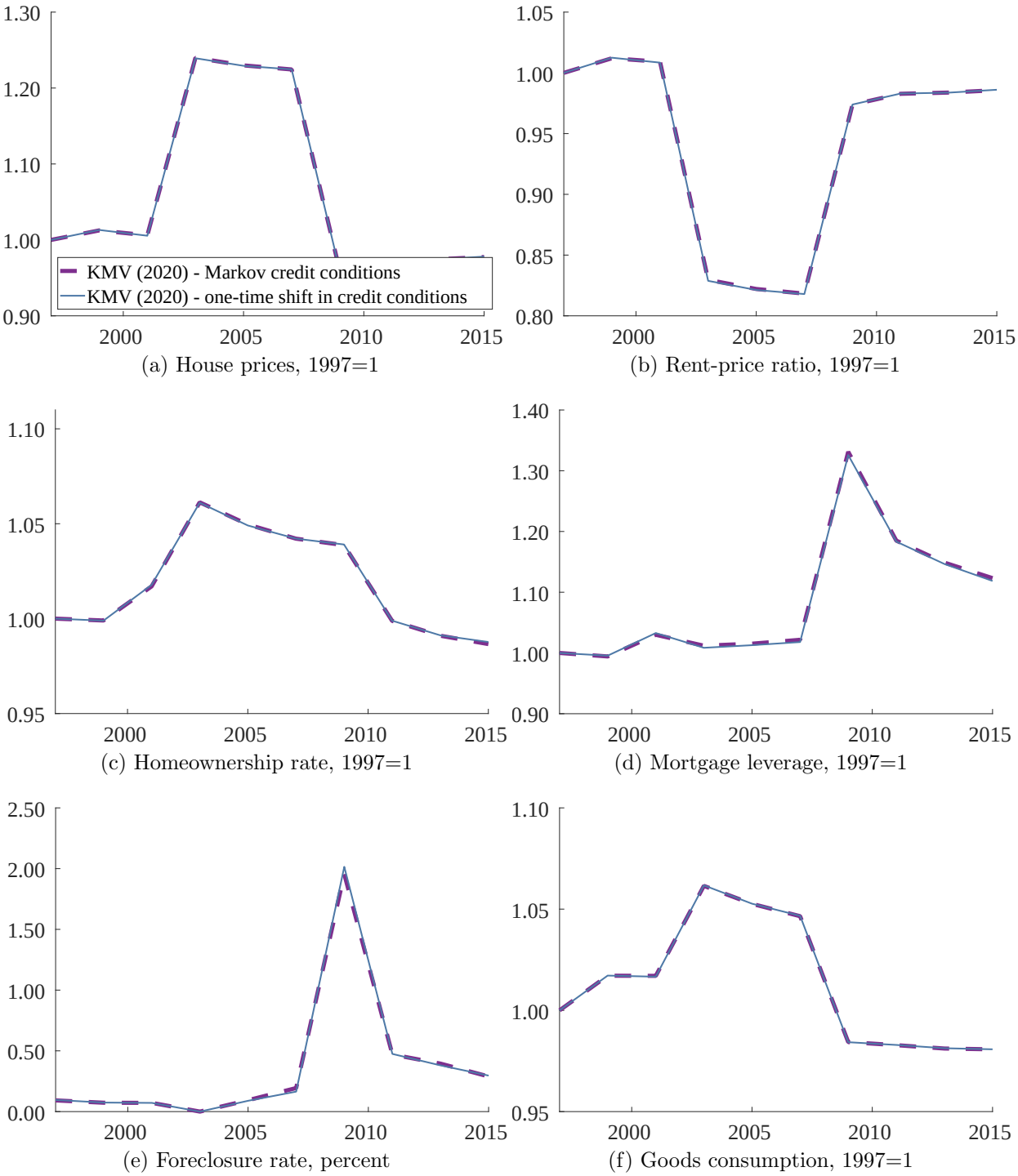
(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

(g) Homeownership level

Figure 26: Housing boom impulse responses under exogenous beliefs, 1997=1.

Note: purple lines are results from Kaplan et al.'s (2020) code repository and blue lines are from the code of this paper, set to the same values of forecasting rule coefficients. The solid purple line is the original impulse responses. The dashed and dotted lines correct from some small discrepancies with the former solved under the original value of the operating cost of the rental company Ξ which is higher than the re-calibrated lower value.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

Figure 27: Housing boom impulse responses under exogenous beliefs, 1997=1.

Note: simulations are from this paper's code solved under Markov credit conditions as in (2020) and under a one-time shift in credit conditions as in this paper.

	St. dev., σ	Autocorrelation
House price growth, p_t/p_{t-1}		
Model, KMV	3.89	-0.05
Model, Jacobson	2.60	0.43
Data	7.85	0.46
Price-rent ratio		
Model, KMV	12.23	0.40
Model, KMV (ex 230 outliers)	8.60	0.85
Model, Jacobson	60	0.49
Model, Jacobson (ex 3 outliers)	8.7	0.62
Data	11.38	0.67
Homeownership rate		
Model, KMV	2.0	0.80
Model, Jacobson	2.06	0.80
Data	1.70	0.86
Leverage		
Model, KMV	3.87	0.89
Model, Jacobson	4.73	0.96
Data	6.01	0.89

Table 18: Unconditional Moments from 6,600 periods including 173 housing booms (2.6 %), learning starts in period 200

Note: All statistics are calculated using the series described in Appendix G. Samples are from 1992 to 2022, but are largely robust to different end points such as starting in 1975 (when data is available) or stopping in 2007. Because the model frequency is two years, data series are aggregated to a two-year frequency via: end of quarter (house prices), end of quarter (price-rent-ratio), two-year yearly average (homeownership rate), and end of quarter at a two-year frequency (leverage).

F Appendix: Alternative Housing Boom Simulations

F.1 Alternative fundamentals

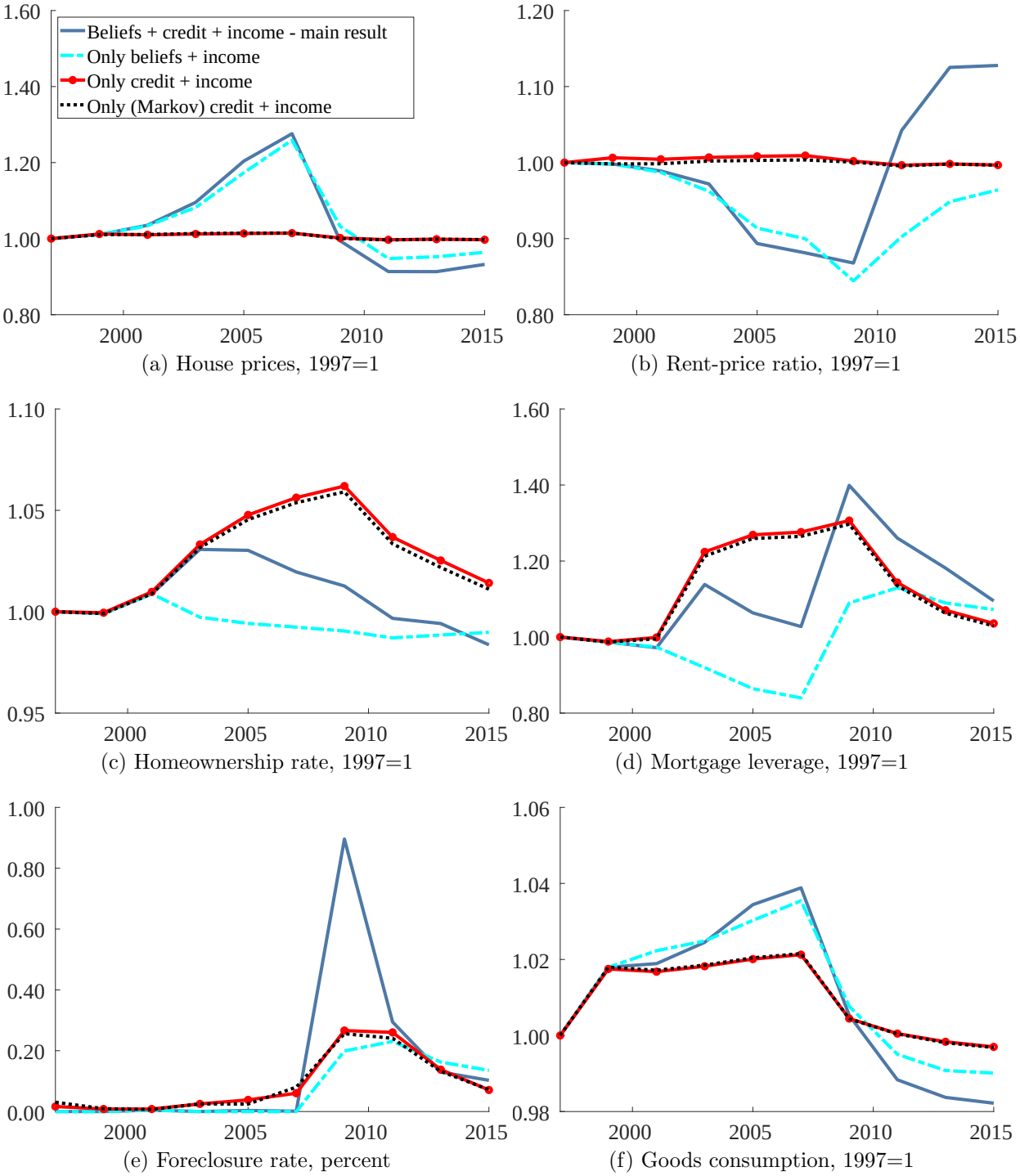
Figure 28 shows counterfactual housing boom simulations that quantify the contributions of beliefs and credit conditions. Panel 28a shows that house prices are highest and thus closest to matching the data in the main specification when there is a shift in income and credit conditions accompanied by endogenously optimistic beliefs. However, counterfactual simulations show that beliefs are quantitatively most important for determining booming house prices but credit conditions must loosen to account for other housing boom dynamics.

Panel 28a shows via the dashed light blue lines that shut down the shift in credit conditions, but still allowing for endogenous beliefs and the shock to income results in house prices that boom, albeit less than in the main specification. This suggests that beliefs play an important role in generating an empirically consistent boom in house prices. Panels 28c-28e show that without the loosening of credit conditions there are counterfactually low responses in the homeownership rate, mortgage leverage, and the foreclosure rate, respectively.

Shutting down beliefs, but allowing for shocks to income and credit conditions (red lines with circles), results in no boom in house prices as shown in Panel 28a.[58] Although there is an increase in homeownership as shown in Panel 28c, new homeowners mostly purchase the same sized house as they previously rented. Homeowners must upsize to push up house prices and this does not occur without optimistic beliefs about future house prices. The rent-price ratio (Panel 28b) remains flat, mortgage leverage counterfactually rises (Panel 28d) and consumption and foreclosures remain counterfactually low (Panels 28f-28e).

Finally, Figure 28 shows that simulations are similar whether credit conditions loosen via a one-time shock (red lines with circles) or are Markov (dotted black lines).

[58] Forecasting coefficients $\boldsymbol{a}_{\mathcal{Z}}^{loose}$ are obtained via the multi-step fixed point method with fluctuations in aggregate income only and credit conditions set at their loose values. As in the main specification, the housing boom simulation then starts with the economy in the state with low aggregate income and tight aggregate credit conditions $\{\Theta(Z_{low}), \mathcal{C}(Z^{tight})\}$ along with the corresponding coefficients $\boldsymbol{a}^{tight}_{Z_{low},Z'}$. As the economy transitions to the state with high aggregate income and loose credit conditions $\{\Theta(Z_{high}), \mathcal{C}(Z^{loose})\}$ coefficients take on the values $\boldsymbol{a}^{loose}_{Z_{high},Z'}$.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

Figure 28: Counterfactual housing boom simulations from the model decomposition the effects of beliefs and credit conditions.

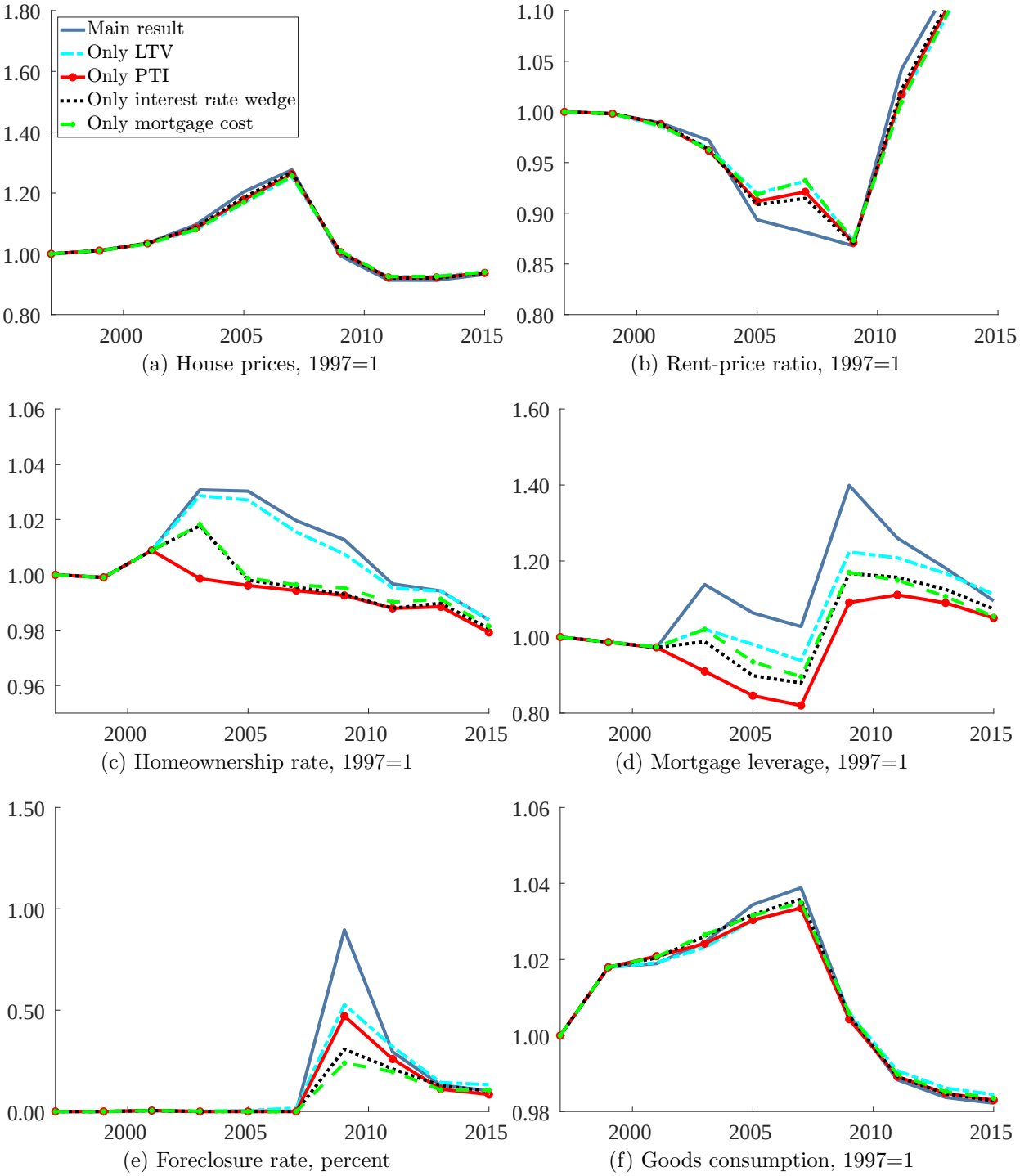
Panel 29a shows that shifting only one of the four credit conditions—LTV constraint $\theta^{LTV}(Z)$, PTI constraint $\theta^{PTI}(Z)$, mortgage origination cost $\kappa^{M}(Z)$, and mortgage wedge $\zeta^{M}(Z)$—can result in a similar boom in house prices as in the baseline when all four shift. This is because any increase in housing demand can only be partially internalized when adaptive learning begins and results in a positive forecast error. The positive forecast error then induces optimism in house price expectations and the boom dynamics in house prices.

Figure 29 highlights the importance of non-linearities preserved via the one-step solution method. The contribution of each credit condition in isolation will not sum to those of the main result with all four shifting. For example, one household may be constrained by both the PTI and the LTV constraints, but if only one constraint loosens then the household may still be unable to purchase a house because they are constrained by the other. However, if both borrowing constraints loosen simultaneously, as in the main simulation, then they will no longer be constrained.

Because consumption and the rent-price ratio largely track house prices, the similarity of the house price simulations results in similar consumption and price-rent ratios, as shown in Panels 29f and 29b, respectively.

Panel 29c shows that the loan-to-value ratio $\theta^{LTV}(Z)$ is the most responsible for the increase in the homeownership rate, as the simulation with just this credit condition shifting is closest to the main result where they all shift. Relatedly, leverage is lower in all of the simulations with just a single credit condition shifting than in the main result. This is because the homeownership rate is relatively lower, which results in less borrowers, and indicates that all credit conditions contribute to increased borrowing.

Panel 29e shows that all credit conditions contribute to the increase in foreclosure in the bust, as none of the credit conditions in isolation achieve the same increase in the foreclosure rate as in the main result.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

Figure 29: Counterfactual housing boom simulations from the model decomposition the effects of each of the credit conditions.

F.2 Alternative Forecast Error Formations

In equation (15) the lagged forecast error is defined as:

$$\boldsymbol{a}_{\mathcal{Z},t} = \boldsymbol{a}_{\mathcal{Z},t-1} + g_t \boldsymbol{x}_{t-2} \underbrace{(\log p_{t-1} - \boldsymbol{x}'_{t-2}\boldsymbol{a}_{\mathcal{Z},t-1})}_{e_{t-1}} \tag{15}$$

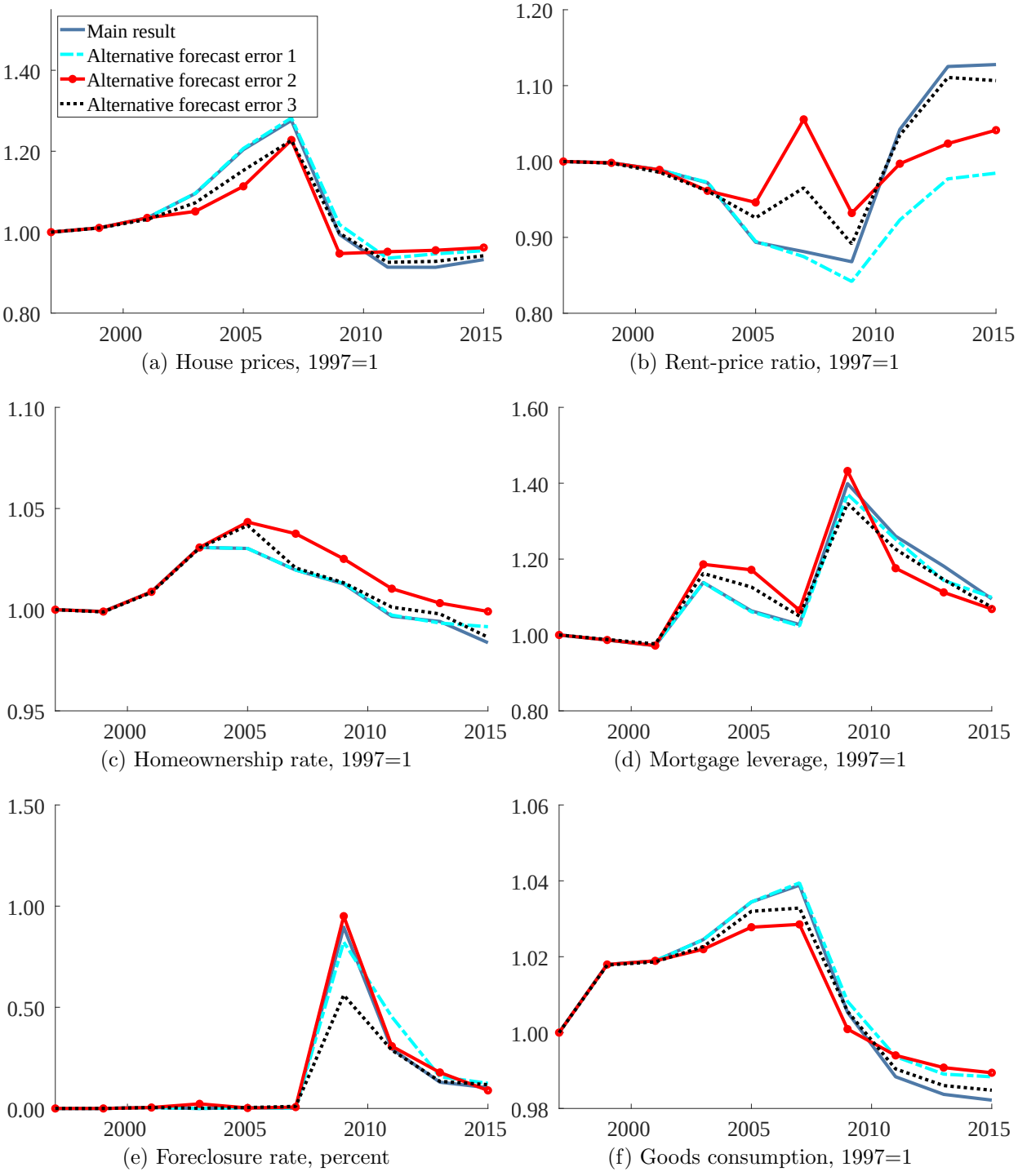
Because there is no definitive convention, this forecast error could take on other forms such as the lagged forecast error from a particular aggregate state or the realized forecast error shown below in equation (23) and (24), respectively.

$$\text{Alternative forecast error 1:} \quad e_{t-1} = \log p_{t-1} - \boldsymbol{x}'_{t-2}\boldsymbol{a}_{\mathcal{Z},t-2} \tag{23}$$

$$\text{Alternative forecast error 2:} \quad e_{\mathcal{Z},t-1} = \log p_{\mathcal{Z},t-1} - \boldsymbol{x}'_{\mathcal{Z},t-2}\boldsymbol{a}_{\mathcal{Z},t-1} \tag{24}$$

$$\text{Alternative forecast error 3:} \quad e_{t-1} = \log p_{t-1} - \boldsymbol{x}'_{t-2}\boldsymbol{a}_{t-1} \tag{25}$$

Figure 30 shows that these forecast error formations, "Alternative forecast error 1", "Alternative forecast error 2", and "Alternative forecast error 3" are shown by the dashed light blue, red solid lines with a circle, and dotted black lines respectively, have little material effect on housing boom simulations in most cases. They are quite close to the main results shown by the solid blue line.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

Figure 30: Housing boom simulations from the model under alternative forecast error formations.

F.3 Initial Forecast Error Calibrations

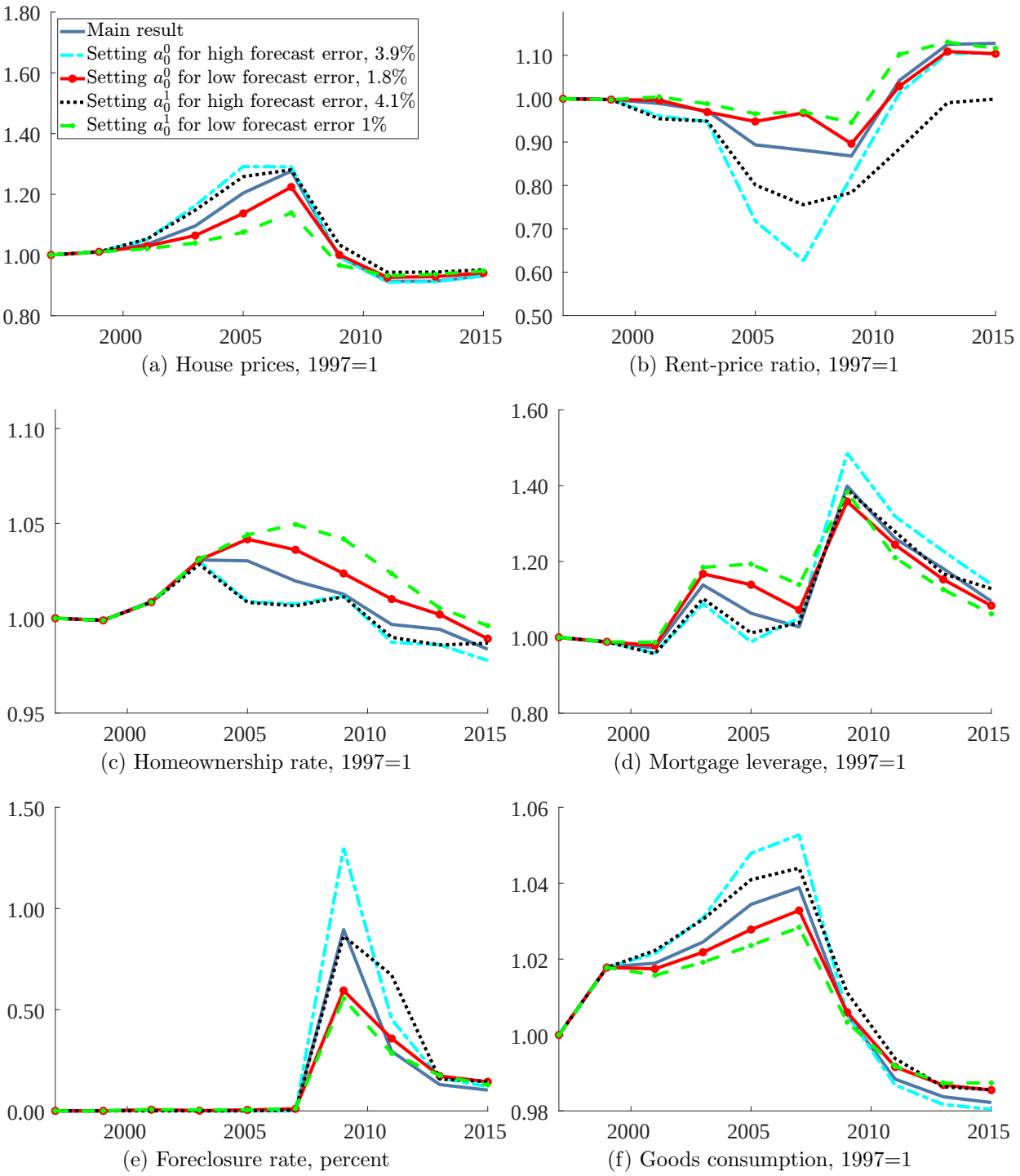
With the initial lagged learning coefficients $\boldsymbol{a}_0$ set at values for tight aggregate credit conditions and high income $\boldsymbol{a}^{tight}_{Z_{high},Z'_{high}}$, the initial forecast error is 2.36 percent which is the lower end of the range of empirical values of [2.35, 4.1] percent as shown in Table 14, which is in turn calculated from data in Table 15. Figure 31 shows that the simulation results are similar if the constant coefficient a^0_0 is adjusted so that the initial forecast error attains a value of 3.9 in upper end of the range, while leaving the slope coefficient unchanged at $a^1_0 = a^{1,tight}_{Z_{high},Z'_{high}}$ (dashed light blue lines). Furthermore, results are similar if the slope coefficient is adjusted and the constant coefficient is unchanged (dotted black lines with circles). And finally, the housing boom dynamics are broadly similar if either coefficient is adjusted so that the initial forecast error is slightly smaller than its empirical value (solid red lines with circles and dashed green lines with small circles).

These adjustments hold the constant gain fixed at $g = 0.3$ annualized as in the main result. Consequently, this results in relatively higher or lower mean squared forecast errors, as shown in Table 19 than the 28.26 observed in the main result that corresponds to the housing boom period as detailed in Table 14 of Appendix D.2. If the learning gains were also adjusted, the results would be even closer to the main simulation.

	Initial lagged beliefs, $\boldsymbol{a}_0$	Initial forecast error	MSE
a^0_0 for high forecast error	$(a^0_{t-1} + 0.017, a^1_{t-1})'$	3.9	43.68
a^0_0 for low forecast error	$(a^0_{t-1} - 0.005, a^1_{t-1})'$	1.8	20.03
a^1_0 for high forecast error	$(a^0_{t-1}, a^1_{t-1} - 0.04)'$	4.1	31.50
a^1_0 for low forecast error	$(a^0_{t-1}, a^1_{t-1} + 0.02)'$	1.0	8.5

Table 19: Lagged beliefs, initial forecast errors, and mean squared forecast errors (MSEs) over the housing boom period, 1999 to 2007.

Note: Initial lagged beliefs $\boldsymbol{a}_0$ can be ad-hoc adjusted by a small factor to set the initial forecast error when learning starts. Without this adjustment, $\boldsymbol{a}_0 = \boldsymbol{a}_{t-1} = \boldsymbol{a}^{tight}_{Z_{high},Z'}$, which are beliefs from the period before learning starts, i.e. the coefficients associated with an economic expansion and tight credit conditions. The MSE from the model is calculated from beliefs formed in 1997 through 2005 and prices from 1999 (first period of learning) through 2007. In the data, the mean-squared forecast error is also formed from beliefs starting in 1997 through those formed through 2005 and prices from 1999 through 2007. The initial forecast error in the model is formed from 2001 prices and 1999 beliefs. 2001 is the first period of adaptive learning corresponding to the loosening of credit conditions and thus the initial forecast error is formed from 2001 prices and 1999 beliefs. Table 15 shows how the ranges for initial forecast errors are taken from the data. These match the model—time 2001 prices with 1999 beliefs of various aggregation methods.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

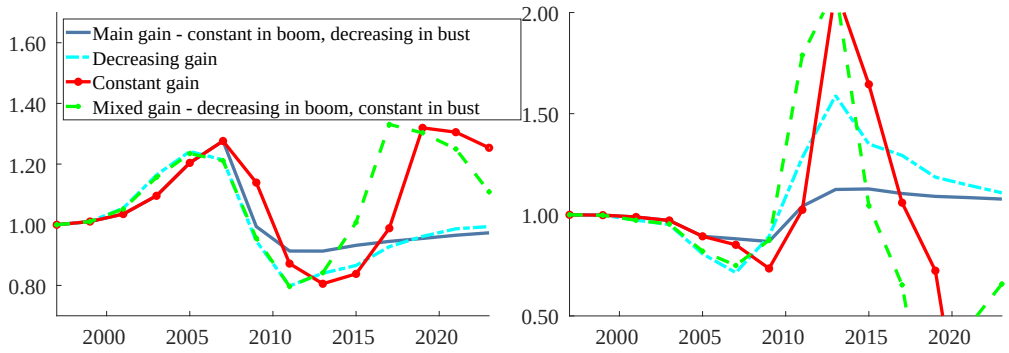
(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

Figure 31: Housing boom simulations from the model under alternative forecast error calibrations.

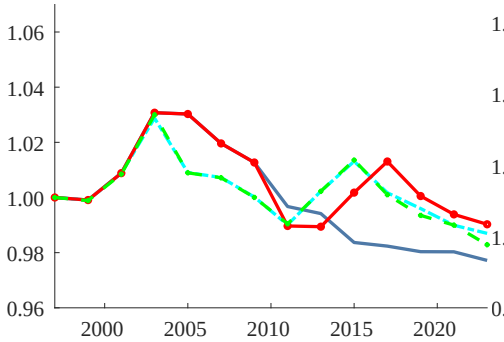
F.4 Mixed and fixed learning gains

Figure 32 shows that housing boom simulations under different types of fixed learning gains are similar to the main result under a mixed learning gain. The mixed learning gain of main result (solid blue lines) is constant, $g_t = g$, when aggregate credit conditions loosen and income expands, but decreasing in all other aggregate states, $g_t = g_{t-1}/(1 + g_{t-1})$. In the decreasing gain simulation (dashed light blue lines), $g_t = 1$ in 2001 at the start of the housing boom and decreases to $g_t = g_{t-1}/(1 + g_{t-1})$ in the following periods. In the constant gain simulation (solid red lines with circles), $g_t = g = 0.508$ in all periods. The simulation titled "Mixed gain - decreasing in boom, constant in bust" (dotted black lines) sets $g_t = 1/t$ in the boom where $t = 1$ when learning starts and sets $g_t = g$ in the bust. This results in an average gain the boom of $\bar{g}_t = 0.45$ and a larger gain in the bust of $g_t = g = 0.508$, which is consistent with the findings of Roschitsch and Twieling (2025) who estimate larger learning gains in the housing bust than in the housing boom in a cross-state specification.

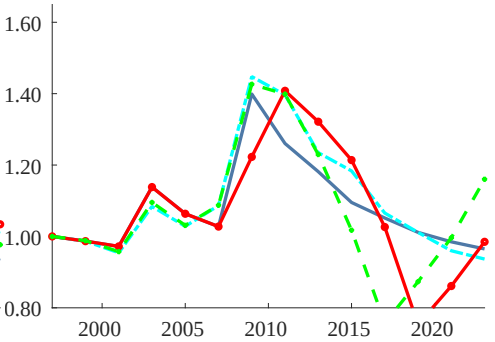


(a) House prices, 1997=1

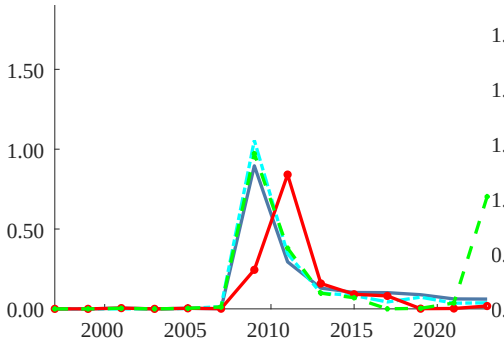
(b) Rent-price ratio, 1997=1



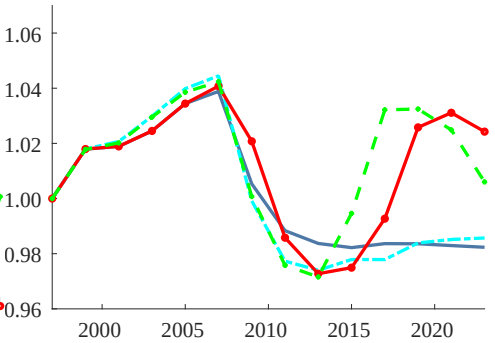
(c) Homeownership rate, 1997=1



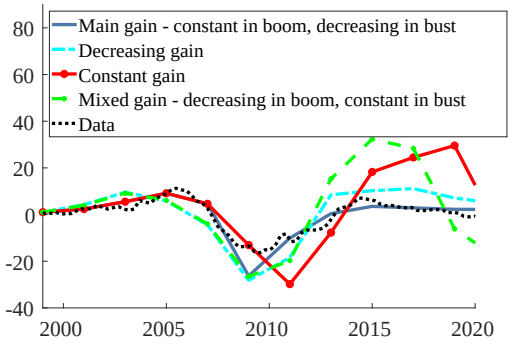
(d) Mortgage leverage, 1997=1



(e) Foreclosure rate, percent



(f) Goods consumption, 1997=1



(g) House price forecast error, percent

Figure 32: Housing boom simulations under fixed rather than mixed learning gains. The house price model's forecast error is $\log p_t - \mathbb{E}_{t-1}[a^0_{\mathcal{Z},t-1} + a^1_{\mathcal{Z},t-1} \log p_{t-1}]$. See Appendix G for sources and definitions of the data series and Appendix D for details on the construction of the proxy for empirical forecast errors from University of Michigan Survey of Consumers.

F.5 Exploring learning mechanisms

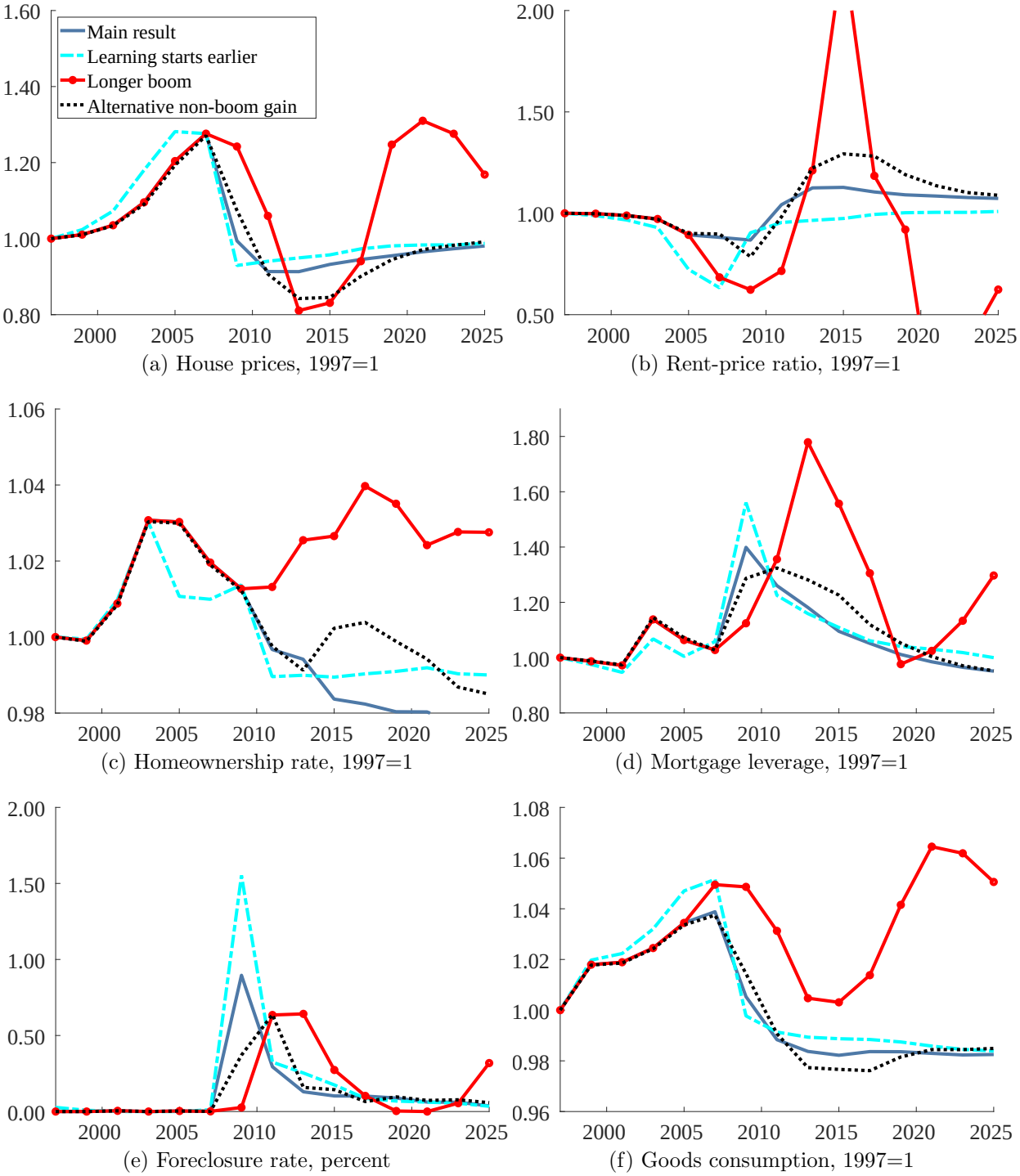
The results shown in Figure 33 explore the properties of adaptive learning.

The simulation titled "Learning starts earlier" (dashed light blue lines), starts adaptive learning at the beginning of the simulation rather than when the economy transitions to the aggregate state characterized by looser credit conditions and high income in 2001 as under the main result (solid blue lines). The results are quite similar with notable differences being a steeper spike in the rent-price ratio, leverage, and foreclosures, shown in Panels 33b, 33d, and 33e, respectively.

In the simulation titled "Longer boom" (solid red lines with circles), the housing boom state, characterized by the aggregate states with high income and looser credit conditions, counterfactually lasts for another 20 years rather than ending in 2009. The simulation paths are exactly the same as those of the main result (solid blue lines) until 2009. Thereafter, house prices, as shown in Panel 33a, contract less suddenly than in the main simulation and reach a deeper trough by 2013. House prices rebound above their pre-boom values around 2018 which actually results in a second housing boom that peaks in late 2022. Although this path tracks the actual rebound in aggregate house prices in the data quite closely, it also results in counterfactual movements in other model prices and quantities shown. For example, the counterfactually large rebound in the price-to-rent ratio is the result of large oscillations in the house price forecast error pushing up rents counterfactually high.

The simulation titled "Alternative non-boom gain" (dotted black lines) initializes the post-boom learning gain at $g_t = g/(1+g)$ as in Marcet and Nicolini (2003) rather than $g_t = g/(1+\tau \times g)$ as in the main simulation. The simulations shown are exactly the same in the housing boom, but the slower decaying of g_t post-boom results in a deeper decline in house prices and other model quantities. Like in the "Learning starts earlier" simulation, the protracted bust in house prices is at the expensive of a counterfactually high rent-price ratio.

Figure 34 shows how the simulations are affected by different decay values, τ, as discussed in the main text. Because the forecast error is so large in the first period of the bust under any specification, the mean-squared forecast error is quite large relative to the data under any specification in the bust. While the mean-squared errors calculated from samples from 2009 onward are similar for the $\tau = 5$ and $\tau = 7$ specifications, they are larger for the $\tau = 1$ and $\tau = 3$, with the $\tau = 1$ being 2 to 8 times larger. $\tau = 1$ corresponds to $g_t = g/(1+g)$ as is commonly found in the literature (Marcet and Nicolini, 2003; Carvalho et al., 2023).



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

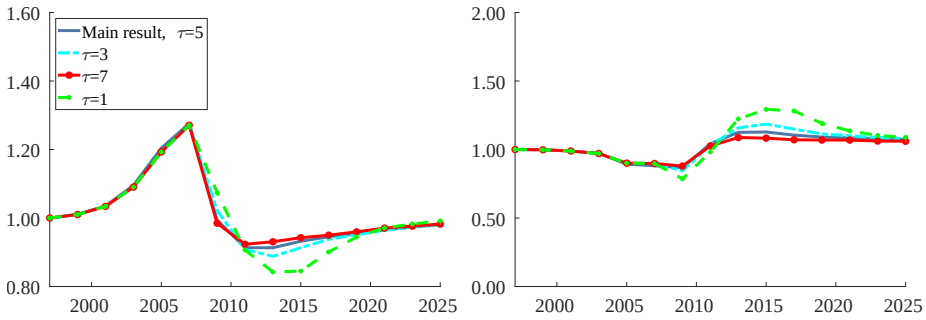
(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

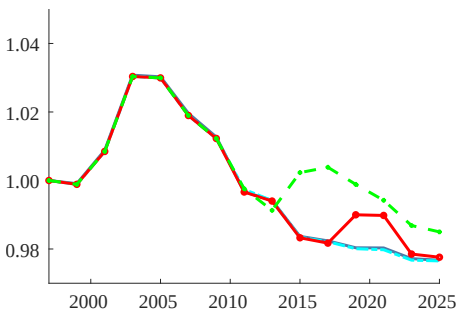
(f) Goods consumption, 1997=1

Figure 33: Housing boom simulations from the model that explore the properties of adaptive learning.

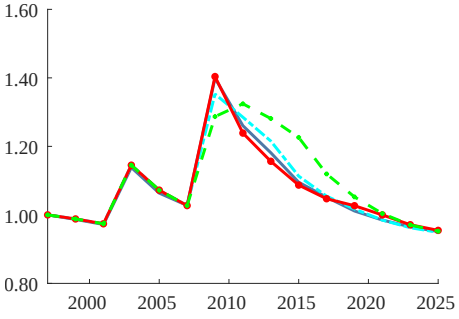


(a) House prices, 1997=1

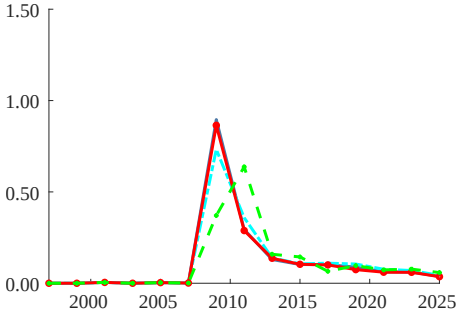
(b) Rent-price ratio, 1997=1



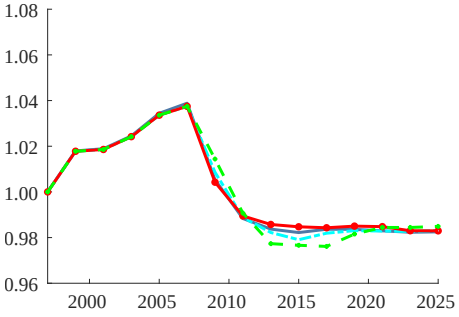
(c) Homeownership rate, 1997=1



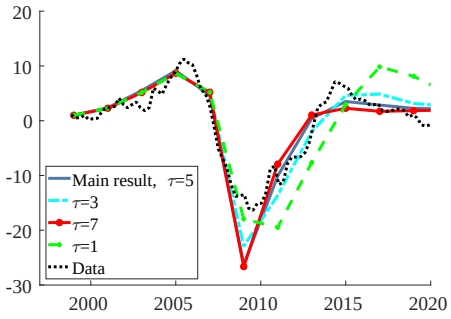
(d) Mortgage leverage, 1997=1



(e) Foreclosure rate, percent



(f) Goods consumption, 1997=1



(g) House price forecast error, percent

Figure 34: Housing boom simulations from the model that show the effects of the decay parameter t in the initial bust gain $g_t = g/(1 + \tau \times g)$. The model's house price forecast error is $\log p_t - \mathbb{E}_{t-1}[a^0_{\mathcal{Z},t-1} + a^1_{\mathcal{Z},t-1} \log p_{t-1}]$. See Appendix G for sources and definitions of the data series and Appendix D for details on the construction of the proxy for empirical forecast errors from University of Michigan Survey of Consumers.

F.6 Alternative learning processes

To show that the results of this paper are robust to different learning processes, instead of updating AR(1) house price forecasting coefficients via the adaptive learning rule in equation (15), Figures 35 and 36 show prices and quantities solved under the Bayesian learning extrapolative house-price growth rate rule of Adam et al. (2025).

In Adam et al. (2025), agents perceive house prices p_t to evolve according to some permanent growth coefficient b_t and some transitory component $\epsilon_t \sim ii\mathcal{N}(0, \sigma_\epsilon^2)$.

$$\frac{p_t}{p_{t-1}} = b_t + \epsilon_t$$

In turn, the growth rate evolves as, $b_t = b_{t-1} + \nu_t, \quad \nu_t \sim ii\mathcal{N}(0, \sigma_\nu^2)$.

Households observe realized house price growth p_t/p_{t-1} and use Bayesian belief updating of observed house price growth to estimate the persistent and transitory components. With conjugate prior beliefs, Adam et al. (2025) define future house price growth as,

$$\gamma_t \equiv \mathbb{E}_t^P \left[\frac{p_{t+1}}{p_t} \right] \tag{26}$$

Which evolves according to the learning rule,

$$\gamma_t - \gamma_{t-1} = \min \left\{ \gamma_{t-1} + \frac{1}{\alpha} \left(\frac{p_{t-1}}{p_{t-2}} - \gamma_{t-1} \right), \bar{\gamma} \right\} \tag{27}$$

They calibrate the Kalman gain, $1/\alpha = 0.03$ at an annualized frequency, directly from the University of Michigan Survey of Consumers rather than encoding the signal-to-noise ratio $\sigma_\nu^2/\sigma_\epsilon^2$ in the Kalman gain.

This paper's learning rule in equation (15), $\boldsymbol{a}_{\mathcal{Z},t} = \boldsymbol{a}_{\mathcal{Z},t-1} + g_t \boldsymbol{x}_{t-2}(\log p_{t-1} - \boldsymbol{x}'_{t-2}\boldsymbol{a}_{t-1})$, can nest that of equation (27) by first altering how house prices enter the model. Setting $a^1_{\mathcal{Z},t}$ to 1 for all realizations in equation (14) below delivers an expression for future house price growth similar to that in equation (26) above.

$$\log p'(p(Z); \mathcal{Z}) = a^0_{\mathcal{Z}} + \underbrace{a^1_{\mathcal{Z}}}_{=1} \log p(Z) \tag{14}$$

$$\log p'(p(Z); \mathcal{Z}) - \log p(Z) = a^0_{\mathcal{Z}} \equiv b_{\mathcal{Z}} \tag{28}$$

Values for $b_{\mathcal{Z}}$ are obtained by re-solving the model under the standard fixed-point method without beliefs so that the growth rate in house prices is sufficient to approximate the distribution of wealth. This yields the following coefficients shown in Table 20 for $b^{tight}_{\mathcal{Z}}$ under tight credit conditions.

Income states	$b_{\mathcal{Z}}^{tight}$
High, High	0.0003
High, Low	-0.0077
Low, High	0.0088
Low, Low	-0.0004

Table 20: Converged house price growth rates under tight aggregate credit conditions, $\log p'(p(Z);\mathcal{Z}) - \log p(Z) = b_{\mathcal{Z}}$

Relative to the AR(1) forecasting rule for house prices in the main text, the growth rate rule has two disadvantages. First, there is no R^2 goodness of fit statistic as one simply takes the sample mean of model-generated house prices rather than running time series regressions. Second, the values for the growth rates in Table 20 are very close to zero because house prices are stationary within each combination of current and future aggregate income states. It may be difficult to find a fixed point with these growth rates so close to zero.

Allowing expected house price growth $b_{\mathcal{Z}}$ to time-vary and evolve endogenously according to the learning rule of Adam et al. (2025) shown in equation (27), yields:

$$b_t = b_{t-1} + g(\log p_{t-1} - \log p_{t-2} - b_{t-1}) \tag{29}$$

While equation (27) requires an upper bound $\bar{\gamma}$ on expected house price growth to prevent agents forming expectations of infinite growth, this is not needed in the setting of this paper for several reasons. First, housing units are fixed in size rather than continuous as in Adam et al. (2025). Once households obtain the maximum sized unit, there is no way for them to increase their share of housing which limits demand for housing units. Second, heterogeneity in ages and incomes means that high house prices may induce younger and worse-off household remain renters for longer than otherwise, which limits the amount of demand for owner-occupied housing.

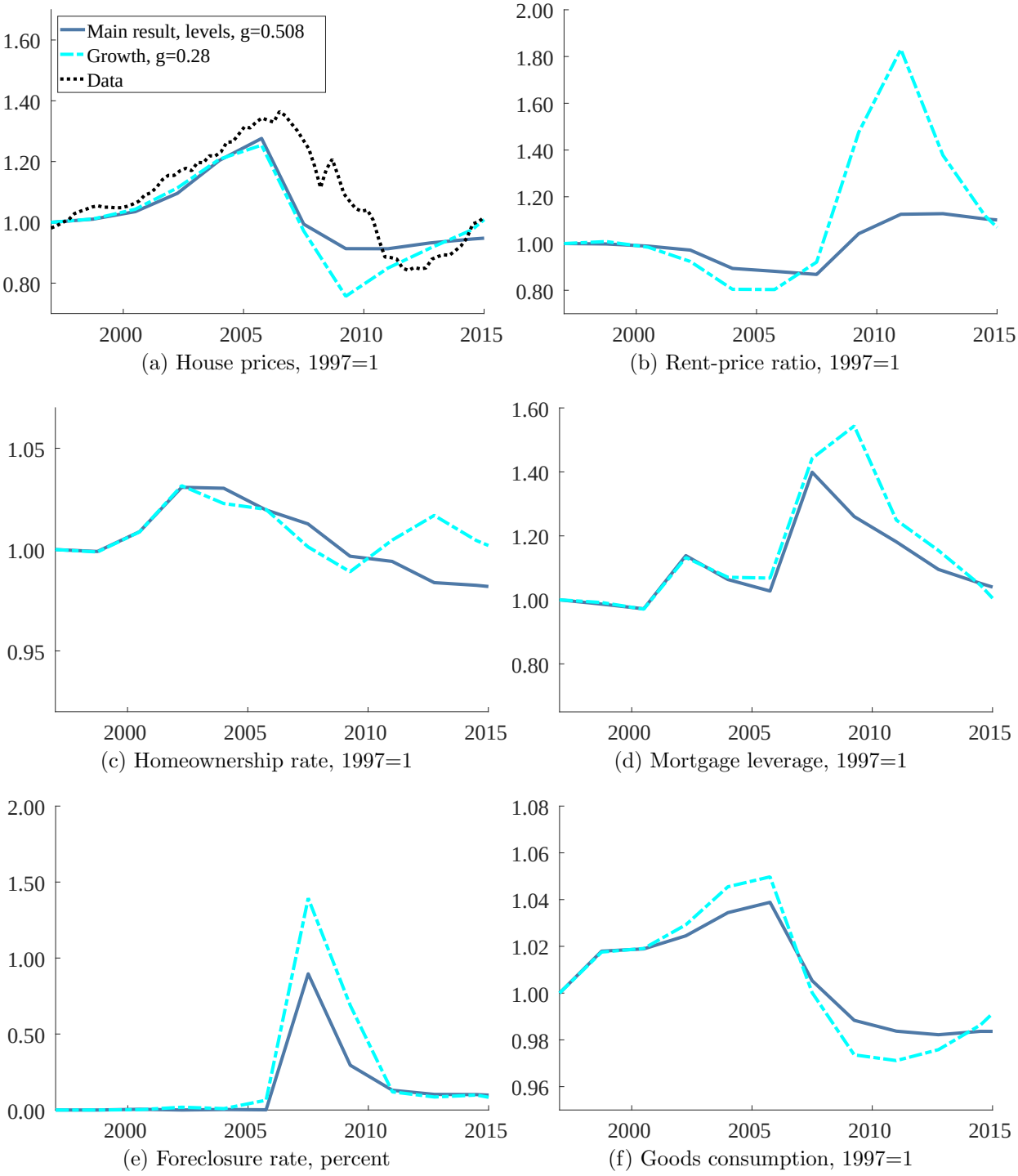
Like Adam et al. (2025), this paper calibrates the constant learning gain g directly from the mean-squared forecast error in the empirical proxy constructed from the University of Michigan Survey of Consumers.

The simulations shown in Figure 35 compares the results under the main specification that relies on the AR(1) forecasting rule with a mixed-learning gain with a constant gain of 0.508 (solid blue line) to a version with a learning rule based on growth rates as in Adam et al. (2025) in equation (29) (dashed cyan lines). In the growth rate simulation, a near identical increase in house prices—and mean-squared forecast error—can be replicated using a lower constant gain of $g = 0.28$, which translates to a gain of 0.15 on an annual basis, well within the range of estimates ranging from 0 to 0.5 elsewhere in the literature, but still

above the 0.03 annualized gain of Adam et al. (2025) calibrated to a non-boom period. The simulation under growth rates shows a deeper bust and a subsequent boom because it only has a constant gain instead of a mixed constant and decreasing gain. In terms of other model prices and quantities, consumption and foreclosures are slightly higher under the growth-rate rule than under the AR(1) rule, but otherwise similar.

Because there is no definitive convention for which forecast error should enter the learning rule, Figure 36 follows Appendix F.2 and shows the robustness of the results under alternative forecast errors.

$$\begin{aligned}
\text{Growth, g=0.28 (dashed cyan lines):} \quad & b_t = b_{t-1} + g(\log p_{t-1} - \log p_{t-2} - b_{t-1}) \\
\text{Growth, alt error 2, g=0.33 (red lines with circle):} \quad & b_t = b_{t-1} + g(\log p_{t-1} - \log p_{t-2} - b_{t-2}) \\
\text{Growth, alt error 1, g=0.28 (dash dot green lines):} \quad & b_t = b_{t-1} + g(\log p_{t-1} - \log p_{t-2} - b_{\mathcal{Z},t-1})
\end{aligned}$$



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

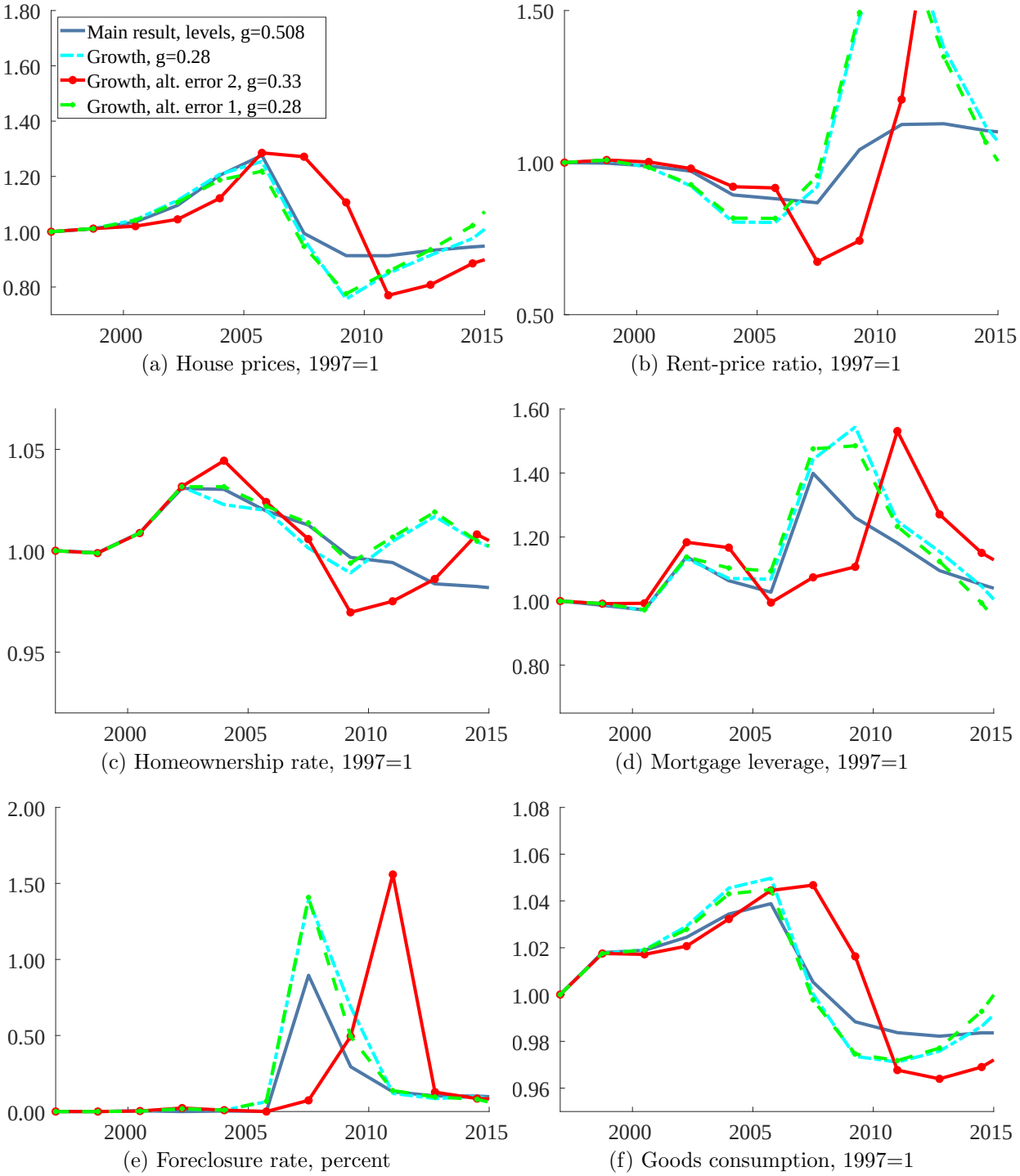
(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

Figure 35: House boom simulations from the model that compare different learning processes.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

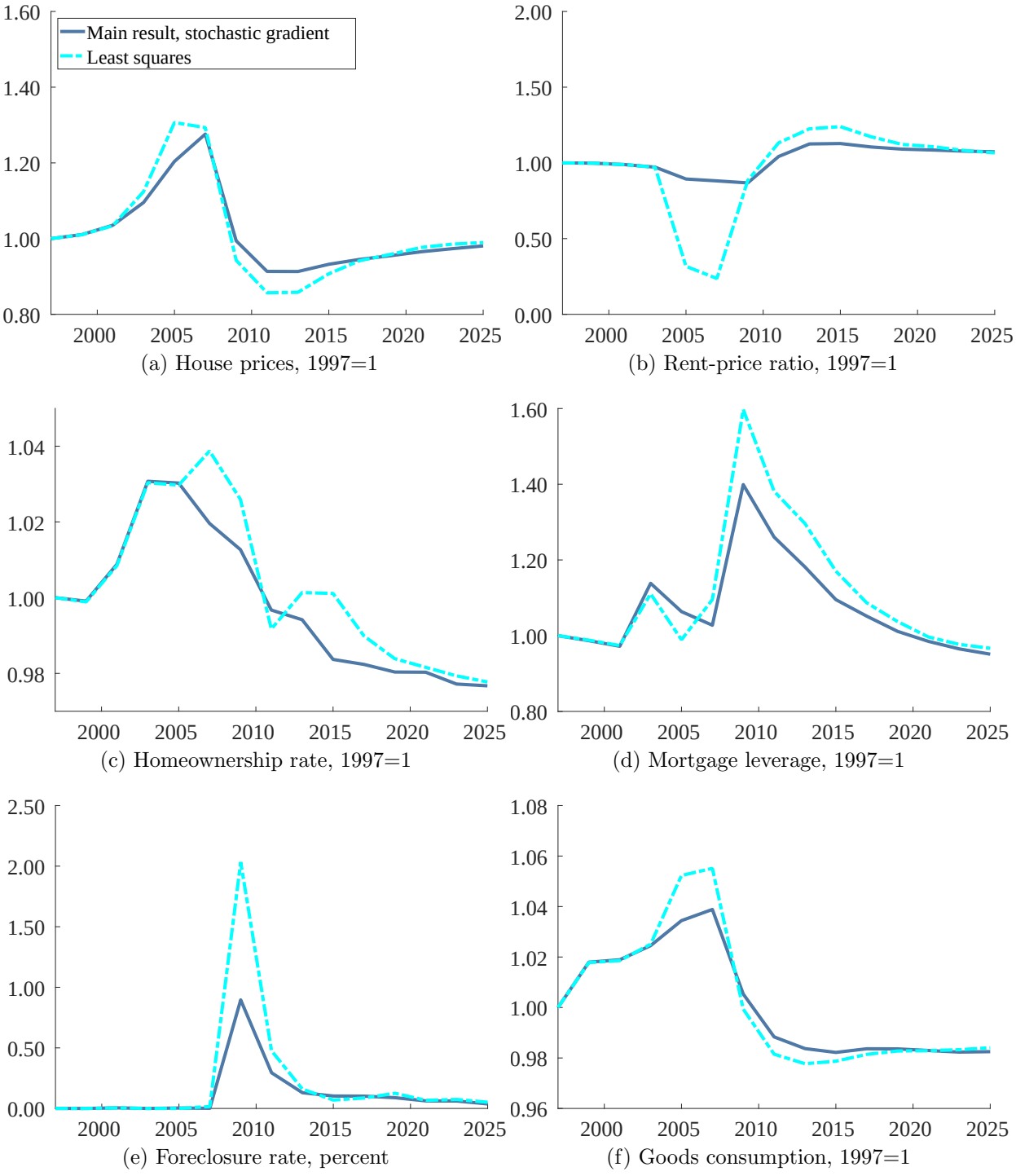
(f) Goods consumption, 1997=1

Figure 36: House boom simulations from the model that compare different learning processes, specifically forecast errors.

Additionally, Figure 37 shows that the simulations are similar if least-squares learning is used instead of stochastic gradient learning for $\boldsymbol{X}_{t-2} = (1, \log p_{t-2})'$,

$$\begin{aligned}\boldsymbol{a}_{\mathcal{Z},t} =& \boldsymbol{a}_{\mathcal{Z},t-1} + g_t R_{\mathcal{Z},t}^{-1} \boldsymbol{x}_{t-2}(\log p_{t-1} - \boldsymbol{x}_{t-2}' \boldsymbol{a}_{t-1}) \\ R_{\mathcal{Z},t} =& R_{\mathcal{Z},t-1} + g_t(\boldsymbol{x}_{t-2}\boldsymbol{x}_{t-2}' - R_{\mathcal{Z},t-1})\end{aligned}$$

Setting $R_{\mathcal{Z},t} = I$ nests the stochastic gradient learning rule used in the main result, as shown by equation (15). At the start of the housing boom when learning starts the initial lagged matrix of second moments, R_{t-1} is diagonal, $R_0 = I$ as agents have no previous experience in this aggregate state. Thereafter, as the economy switches back to the non-boom states in the bust, agents have access to $R_{\mathcal{Z},t-1}$ from all of the previous observations of that state.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

Figure 37: House boom simulations from the model with least-squares learning.

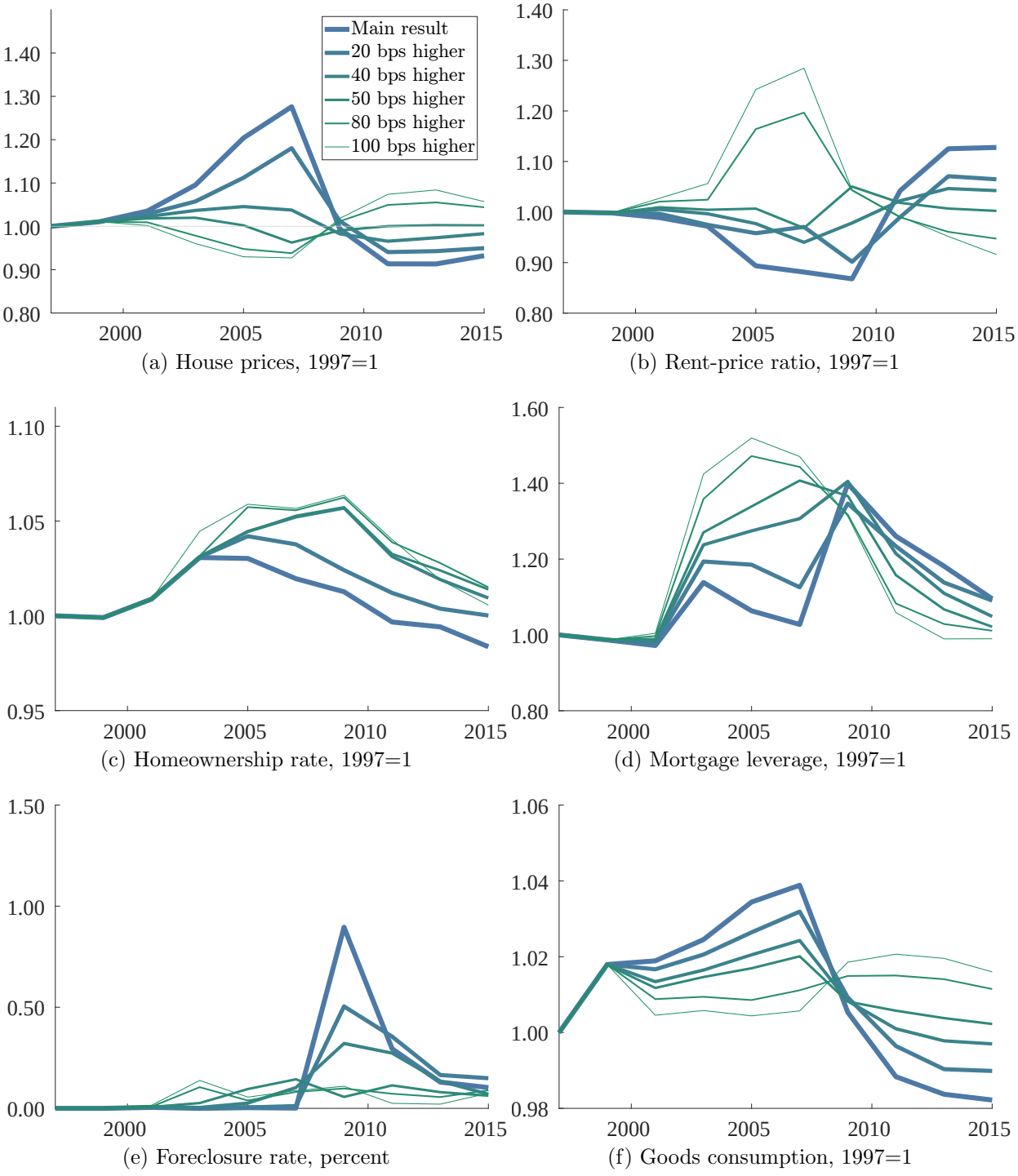
F.7 Effects of Macroprudential Policy

Figure 38 shows how prices and quantities evolve had the central bank used monetary policy to "lean against" rising house prices. This occurs in 2001 when credit conditions loosen and the number of basis points in the legend indicate how much higher interest rates went up relative to the baseline risk-free rate. As interest rates rise, the housing boom is less pronounced and a 50 basis point increase even results in a house price contraction rather than a boom. Consequently, the homeownership rate is actually lowest in the main simulation

Figure 39 shows how house prices and quantities evolve had regulators tightened credit conditions after they loosened in 2001. The legend entries show the years in which they are tightened.

To understand if agents prefer the housing boom economy or that with prudential policy where there is no boom in house prices, I compare the value functions of each agent in the main simulation from their counterparts in the 50 basis point higher interest rate prudential policy counterfactual that has no boom in house prices. Figure 40 plots the evolution of these differences averaged by income and age throughout the boom. For the younger lower-income agents shown in Panel 40a, they initially prefer the boom simulation because they benefit from looser credit conditions, but then prefer the no-boom simulation with higher interest rates until the bust hits in 2009. Thereafter, all of the young agents—except for the highest earners—prefer the boom-bust simulation because housing becomes more affordable as house prices drop in the bust. For middle or higher income younger agents, they tend to prefer the no-boom simulation until 2009 and then the more affordable housing thereafter.

In Panel 40b, the lower- and middle-income older agents typically prefer the boom simulation until house prices contract in 2009. This is because these agents consume more in both housing—due to higher expectations—and in consumption—due to a housing wealth effect from higher prices. By contrast, after 2009 house prices drop in the boom simulation, but level off in the no-boom simulation. As pessimism sets in and the housing wealth effect reverses, these agents consume relatively less in both housing and consumption, which makes them prefer the no-boom simulation after 2009. By contrast, the higher income older agents—like their younger higher-income counterparts—typically prefer the no boom simulation as they prefer less volatility in house prices.



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

Figure 38: Counterfactual housing boom simulations showing the effects of raising interest rates.

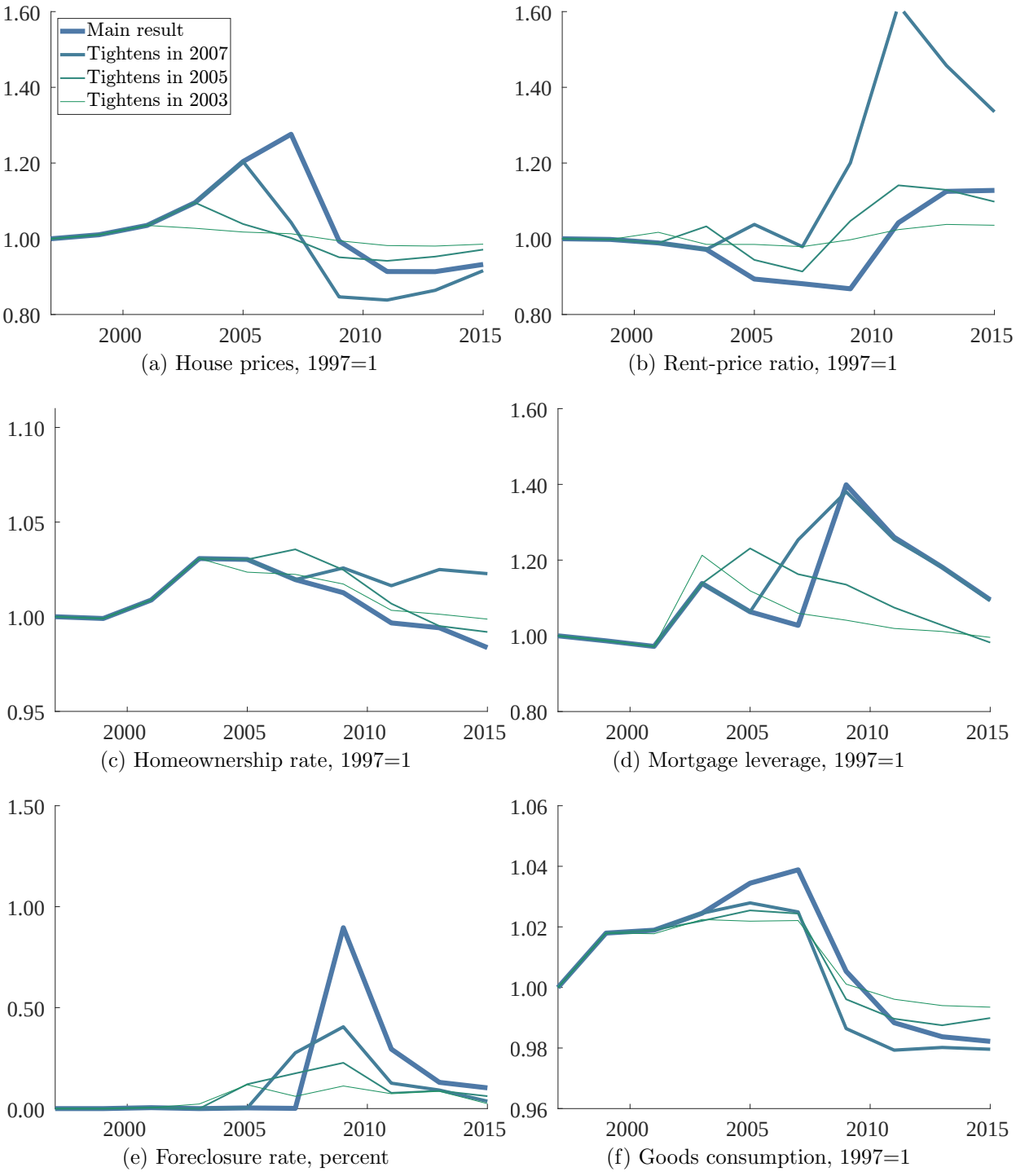
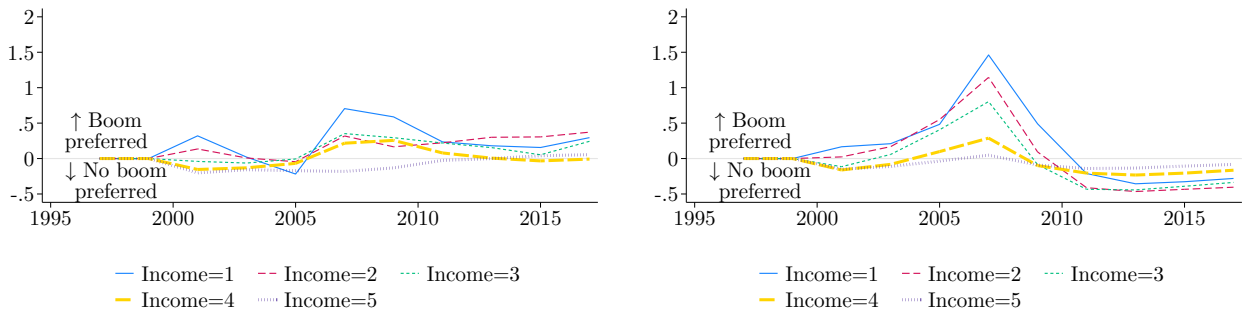


Figure 39: Counterfactual housing boom simulations showing the effects of tightening credit conditions at various years.



(a) Averages for Ages 21 to 29

(b) Averages for Ages 31 to 79

Figure 40: Value functions from the main housing boom simulation less value functions from the macroprudential counterfactual with 50 basis point higher interest rates. Averages of these differences are by incomes and ages

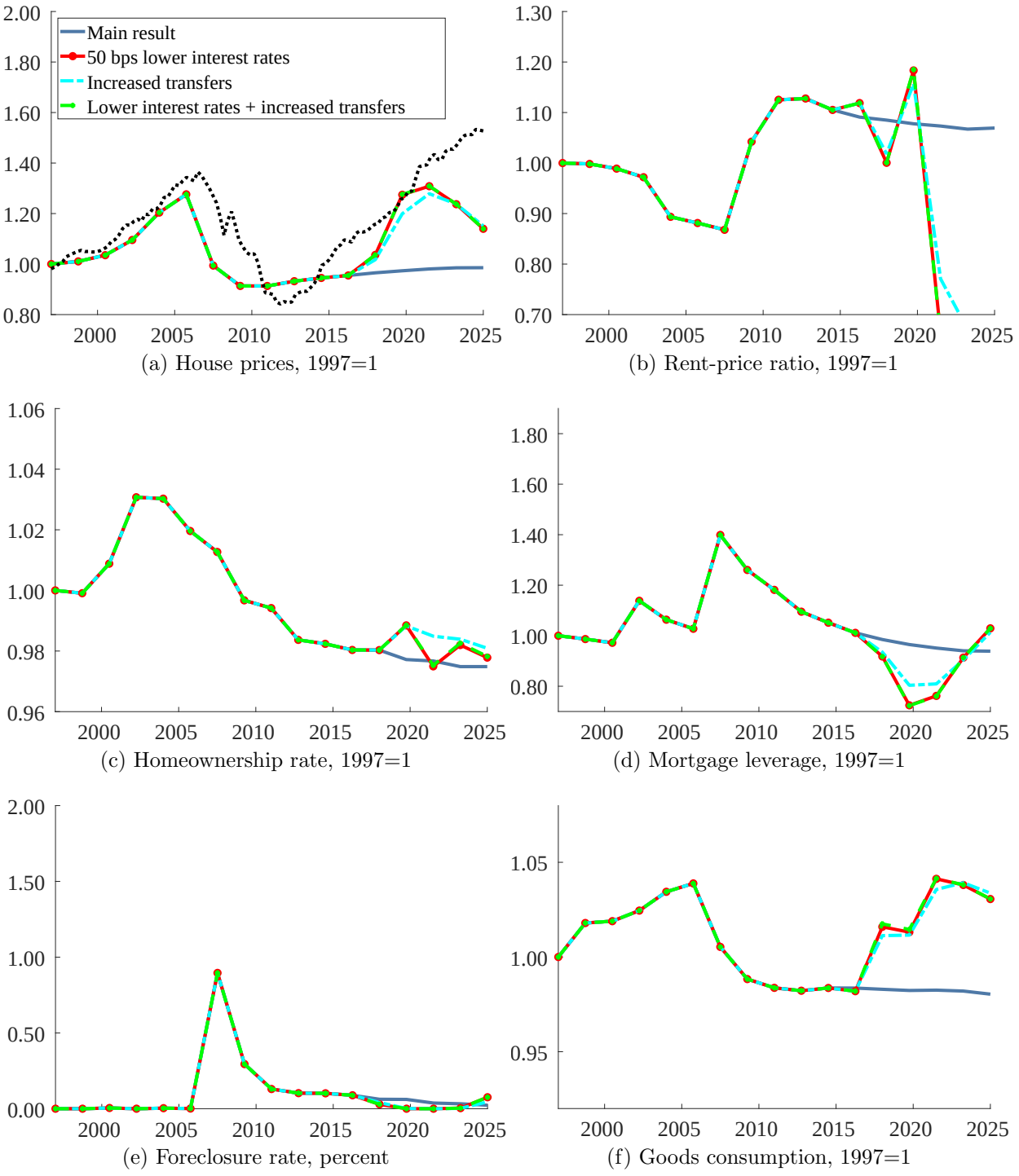
F.8 COVID Boom

Figure 41 shows three simulations in addition to the main result that allow for the arrival of another boom state in 2019, but with tight credit conditions and other changes in fundamentals. Because of the model's timing the COVID boom hits in either 2019 or 2021. In the first simulation, there is a 50 basis point cut in interest rates in 2019, which results in a second boom that matches the rise in house prices in 2020, albeit somewhat more slowly. In the second simulation, there is a fiscal transfer in 2019 as detailed by Carroll et al. (2021, 2023); Crawley et al. (2026) to match those of COVID. The third simulation combines the first two.[59]

All of these COVID simulations can produce the increase in house prices and goods consumption, but do so accompanied by a counterfactual rent-price ratio.

These simulations use the main calibration constant learning gain of $g = 0.3$ on an annual basis. During the COVID boom (2020:Q2 to 2021:Q4), the MSE forecast error from the 75th percentile proxy is 34.44 which is close to the dashed grey line in Panel 4b corresponding to g=0.31. If this were used instead, the boom would be amplified.

[59]Can be found: https://www.bea.gov/help/faq/1409. There was a total of three stimulus payments in COVID. Since the model period is two years, I consider all three. Stimulus 1 pays out \$1,200 per adult and \$500 per dependent for a total of \$2,900 for two adults and one dependent (average household size was 2.6 in 2020) for joint tax filers making under \$150,000. Thereafter, the payments are linearly phased out to \$198,000. For Stimulus 2, the payout was \$1,200 for joint filers and \$600 per dependent for a total of \$1,800 and a phase out that starts at \$150,000 and ends at \$174,000. For stimulus 3, the payment was \$1,400 per individual so \$4,200 for a household of three. The start of the phase out was \$150,000 and end at \$160,000. All values are then scaled to \$1998 dollars (divided by 1.66).



(a) House prices, 1997=1

(b) Rent-price ratio, 1997=1

(c) Homeownership rate, 1997=1

(d) Mortgage leverage, 1997=1

(e) Foreclosure rate, percent

(f) Goods consumption, 1997=1

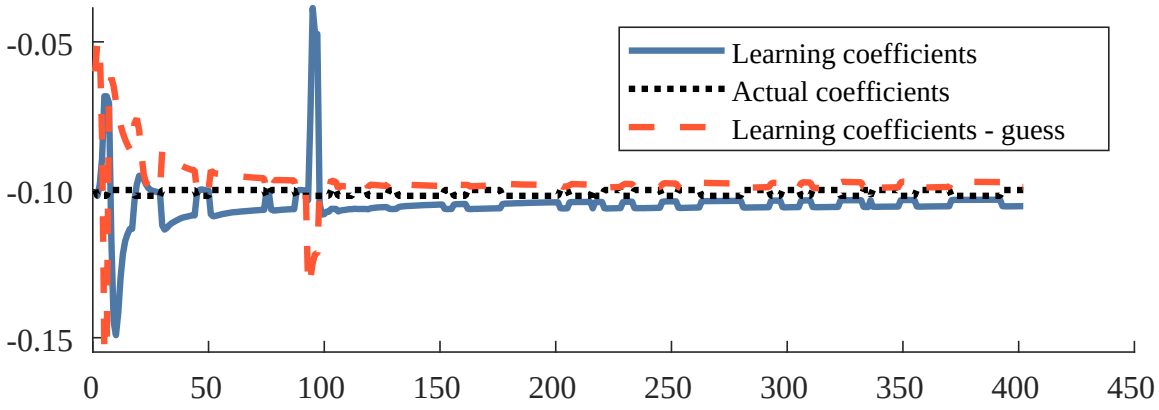
Figure 41: Counterfactual housing boom simulations from the model to study the COVID housing boom in the 2020s. Dashed line in panel (a) is house price data.

F.9 Convergence of Learning Coefficients

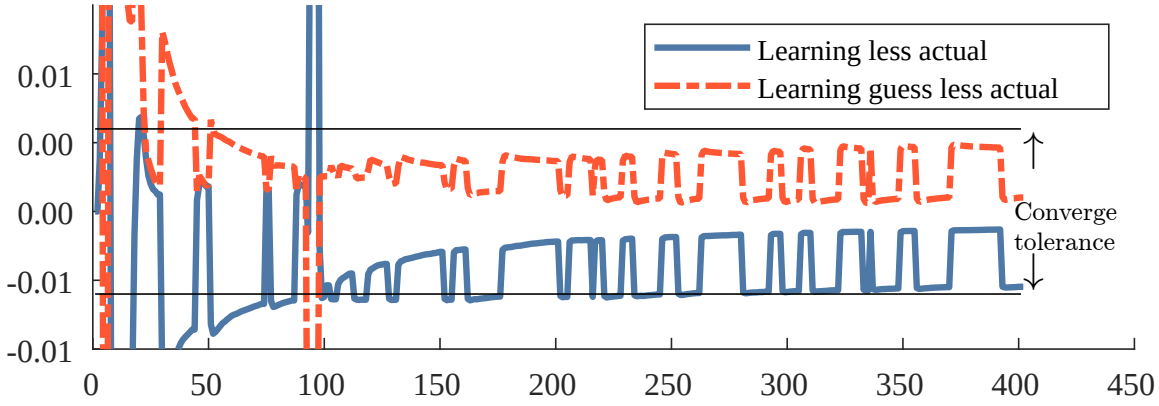
Figures 42 and 43 show that coefficients under adaptive learning settle at an ergodic distribution near their known counterparts. A counterfactual return to the housing boom state 100 periods (200 years) after the 2000s boom shows that coefficients adjust in response to forecast errors but revert back to an ergodic distribution in the non-boom state. Although there are observable differences between the learning coefficients and their known counterparts in Panels 42a and 43a, Panels 42b and 43b show that these differences are well within the convergence tolerance used to solve for the fixed coefficients.

In the case of the main result shown by the "learning coefficients" simulation, the model starts from converged near-rational coefficients for the pre-boom aggregate state characterized by tight credit conditions, $\boldsymbol{a}^{tight}_{Z_{high},Z'_{low}} = \{-0.111, 0.838\}$ and $\boldsymbol{a}^{tight}_{Z_{high},Z'_{high}} = \{-0.102, 0.838\}$. The simulated coefficients depart from these non-boom states during the housing boom because of the constant gain learning that puts relatively more weight on forecast errors. Nonetheless, the coefficients eventually resettle within the convergence tolerance as learning continues and the economy returns to the non-boom state.

The simulation titled "Learning coefficients guess" instead solves the model at some guess, $\boldsymbol{a}^{tight}_{Z_{high},Z'_{low}} = \{-0.06, 0.82\}$ and $\boldsymbol{a}^{tight}_{Z_{high},Z'_{high}} = \{-0.05, 0.8\}$. While these coefficients deliver a different path than that of the main simulation, they nonetheless settle within the same convergence tolerance, which implies that learning can eventually recover the near-rational coefficients. As a result, researchers could potentially skip solving for the fixed point coefficients entirely, solve the model at some guess for coefficients, and then recover reasonable values of the actual coefficients via learning. Compute time is 10 or more hours to solve for four pairs of non-boom coefficients, $\boldsymbol{a}^{tight}_{Z,Z'}$ via fixed point iteration. By contrast, the one-step method only takes one hour. A potential caveat of using the one-step method to solve for near-rational coefficients is the guess must be within the range of coefficients the model is actually solved at. If this is poorly informed, then a lot of belief grid points N may be needed, which could be computationally costly because the model must be solved $N^{\#Z'}$ times.

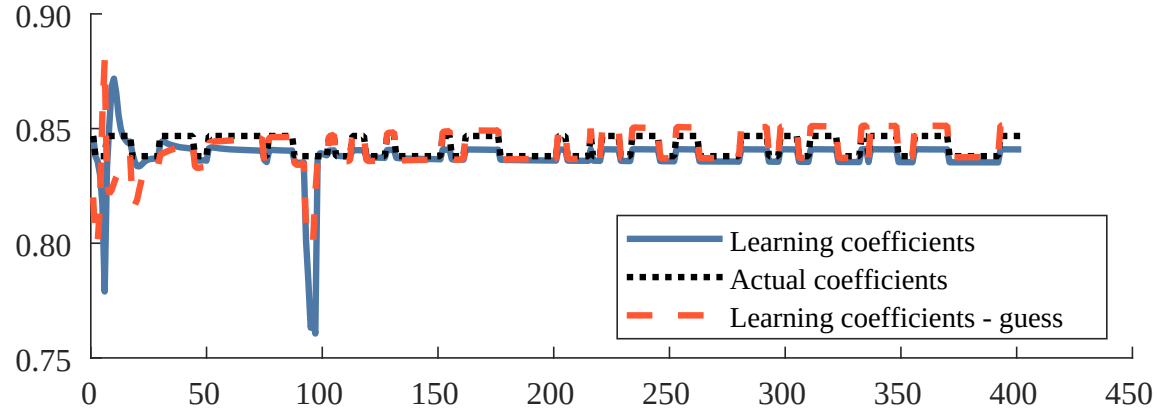


(a) Evolution of a_t^0 coefficient

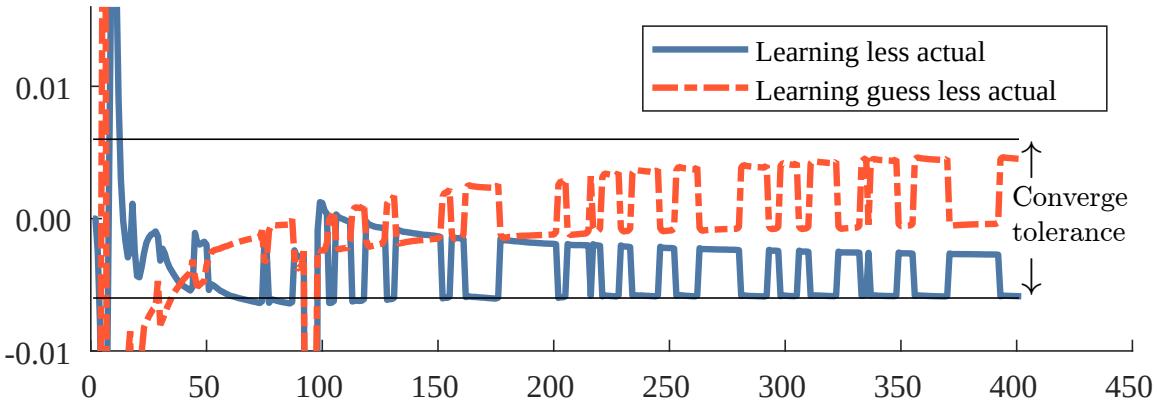


(b) Difference between learning and known coefficients

Figure 42: Time path of beliefs for 400 periods. Period 0 is the start of the housing boom near period 100 there is another counterfactual unexpected loosening of credit conditions.



(a) Evolution of a_t^1 coefficient



(b) Difference between learning and known coefficients

Figure 43: Time path of beliefs for 400 periods. Period 0 is the start of the housing boom and near period 100 there is another counterfactual unexpected loosening of credit conditions.

G Appendix: Data Definitions

The aggregate data definitions follow those in Appendix E.1 of Kaplan et al. (2020). Consumption and house price data have been detrended using a linear time trend estimated from 1975 to 1997.

- **Consumption**: Quarterly nominal nondurable expenditures (line 8 of NIPA Table 2.3.5 Personal Consumption Expenditures by Major Type of Product) divided by the price index for nondurable consumption (line 8 of NIPA Table 2.3.4. Price Indexes for Personal Consumption Expenditures by Major Type of Product).
- **Homeownership**: Census Bureau U.S homeownership rate (Table 14) and by the age of household head (Table 19).
- **House Prices**: In the main results , it is the house price index for the entire United States (Federal Housing Finance Agency (`FRED: USSTHPI`)) divided by the price index for nondurable consumption (line 8 of NIPA Table 2.3.4. Price Indexes for Personal Consumption Expenditures by Major Type of Product). For Appendix D.3, the series follows Adam et al. (2025) and is the seasonally adjusted Case-Shiller house price index (`FRED: CSUSHPISA`).
- **Rent-Price Ratio**: In the main results, the series is rents (Bureau of Labor Statistics Price Index for Rent of Primary Residences) divided by the FHFA house price index described above. For Appendix D.3, the series follows Adam et al. (2025) and is the OECD series price-rent ratio for the United States.
- **Foreclosures**: Number of consumers with new foreclosures and bankruptcies (Federal Reserve Bank of New York Quarterly Report on Household Debt and Credit) divided by the civilian noninstitutional population (`FRED: CNP16OV`).
- **Leverage**: Flow of Funds Table B.101 Balance Sheet of Households and Nonprofit Organizations. Home mortgage liabilities (FL163165505) divided by the sum of household owner occupied housing at market value (LM155035015) and nonprofit organization real estate at Market Value (LM165035005).
- **Labor Productivity**: U.S. Total Labor Productivity (`FRED: ULQELP01USQ661S`)
- **Mortgage and Treasury Interest Rates**: 30-year fixed rate mortgage (`FRED: GAGE30US`) and 10-year Treasury at a constant maturity (`FRED: GS10`)
- **House price expectations**: University of Michigan Surveys of Consumers. The construction of the empirical proxy and data definitions are discussed in detail in Appendix D. New York Fed Survey of Consumer Expectations.

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